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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 2

FURTHER COMPLEX NUMBERS

SUGGESTED TIME: 14 PERIODS

IN THIS CHAPTER

This chapter deepens the study of complex numbers: their forms, De Moivre's theorem and its applications, loci in the Argand diagram, and transformations of the complex plane. It culminates in similitudes, modulus inequalities and the classification of rigid motions by orientation.

- By the end of this chapter I can convert a complex number between Cartesian, trigonometric and exponential forms and use each form appropriately.
- By the end of this chapter I can state and apply De Moivre's theorem to compute powers and nth roots of complex numbers.
- By the end of this chapter I can use De Moivre's theorem to express $\cos(n\theta)$, $\sin(n\theta)$ in terms of $\cos\theta$ and $\sin\theta$ and to linearise powers of cosine and sine.
- By the end of this chapter I can interpret and sketch loci defined by modulus and argument conditions in the Argand diagram.
- By the end of this chapter I can describe elementary transformations from the z-plane to the w-plane and find images of points and curves.
- By the end of this chapter I can represent reflections, rotations, enlargements, stretches and shears by 2×2 matrices and compose them.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Forms of a Complex Number and De Moivre's Theorem
2. Applications of De Moivre's Theorem
3. Loci in the Argand Diagram
4. Transformations from the z-plane to the w-plane
5. Plane Transformations and Matrix Representation
6. Similitudes: Complex Form, Matrix Interpretation and Applications
7. Modulus Inequalities and Rigid Motions
8. Integration activity
9. Methods and worked examples
10. Exercises
11. Solutions to the exercises
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CHAPTER OBJECTIVES

- By the end of this chapter I can convert a complex number between Cartesian, trigonometric and exponential forms and use each form appropriately.
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- By the end of this chapter I can use De Moivre's theorem to express $\cos(n\theta)$, $\sin(n\theta)$ in terms of $\cos\theta$ and $\sin\theta$ and to linearise powers of cosine and sine.
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- By the end of this chapter I can describe elementary transformations from the z-plane to the w-plane and find images of points and curves.
- By the end of this chapter I can represent reflections, rotations, enlargements, stretches and shears by 2×2 matrices and compose them.
- By the end of this chapter I can identify a similitude, write its complex form $z' = az + b$ or $z' = a\bar{z} + b$, and determine its characteristic elements.
- By the end of this chapter I can solve and interpret modulus inequalities in the complex plane and apply them to geometric problems.
- By the end of this chapter I can classify rigid motions as orientation-preserving (translations, rotations) or orientation-reversing (reflections, glide reflections).

PREREQUISITES

- Basic algebra of complex numbers from Lower Sixth: modulus, argument, conjugate, operations in Cartesian form
- Trigonometric identities: addition formulae, double-angle formulae, solving trigonometric equations
- Equations of lines and circles in the Cartesian plane; distance between two points
- Basic 2×2 matrix operations: addition, multiplication, determinant, inverse
- Solving polynomial equations, including factorisation and the quadratic formula with complex roots

1 FORMS OF A COMPLEX NUMBER AND DE MOIVRE'S THEOREM

When an engineer at the Douala seaport rotates a crane arm through an angle, or when MTN engineers analyse alternating-current signals across the Littoral region, they are secretly multiplying complex numbers. A single complex number can describe a position on a map of Yaoundé, a rotation of a satellite dish, and a stretch of a bridge cable — all at once. How can one algebraic object encode turning, scaling and sliding on the plane? This chapter reveals how complex numbers, through De Moivre's theorem and the Argand diagram, become a complete language for plane geometry and transformations. Our first task is to master the three ways of writing a complex number — Cartesian, trigonometric and exponential — and to discover the astonishing rule that powers of complex numbers are computed by simply multiplying angles.

1.1 Review of the Cartesian Form

DEFINITION **Complex number, Cartesian form**

A **complex number** is an expression $z = x + iy$ where $x, y \in \mathbb{R}$ and i is a number satisfying $i^2 = -1$. We call $x = \mathrm{Re}(z)$ the **real part** and $y = \mathrm{Im}(z)$ the **imaginary part** of z. The **conjugate** of z is $\bar{z} = x - iy$, and its **modulus** is

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}.$$

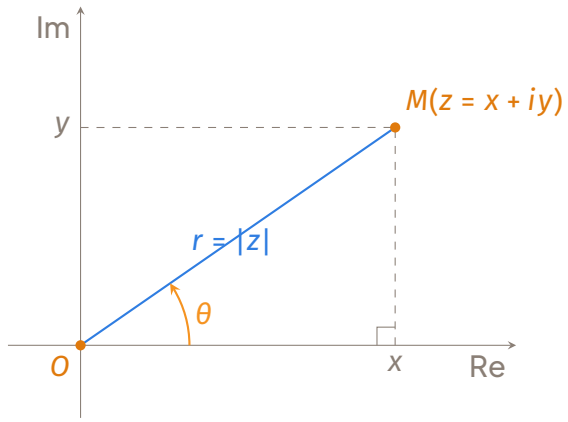
Recall the essential algebraic facts, all proved by direct expansion: $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, $z + \bar{z} = 2\,\mathrm{Re}(z)$, $z - \bar{z} = 2i\,\mathrm{Im}(z)$, and $z\bar{z} = |z|^2$. The modulus measures a distance: geometrically, $|z|$ is the distance from the origin to the point representing z. Note also that $|\bar{z}| = |z|$ and that $|z| = 0$ if and only if $z = 0$.

1.2 The Argand Diagram: Affixes and Geometry

The plane equipped with an orthonormal frame $(O; \vec{u}, \vec{v})$ is called the **Argand diagram** (or complex plane). To each complex number $z = x + iy$ we associate the point $M(x, y)$, called the image of z; conversely z is called the **affix** of M, written z_M. Similarly, the vector $\vec{w} = a\vec{u} + b\vec{v}$ has affix $z_{\vec{w}} = a + ib$. This bijection between $\mathbb{C}$ and the plane is the key that unlocks all the geometry of this chapter.

Two geometric facts make the Argand diagram powerful:

- **Addition.** If A and B have affixes z_A and z_B, then $z_A + z_B$ is the affix of $\vec{OA} + \vec{OB}$: addition of complex numbers is addition of vectors (parallelogram rule).
- **Subtraction.** The vector $\vec{AB}$ has affix $z_B - z_A$, and the distance $AB = |z_B - z_A|$. The modulus of a difference is a distance between two points.



The Argand diagram: the affix $z = x + iy$, its modulus $r = |z|$ and its argument $\theta = \arg(z)$ form a right triangle with $x = r\cos\theta$, $y = r\sin\theta$.

1.3 Argument of a Non-zero Complex Number

The modulus tells us how far M is from O; to locate M completely we also need a direction.

DEFINITION **Modulus and principal argument**

Let $z \neq 0$ with image M. An **argument** of z is any measure θ of the oriented angle $(\vec{u}, \vec{OM})$. The unique argument lying in $(-\pi, \pi]$ is called the **principal argument**, denoted $\operatorname{Arg}(z)$. Every argument has the form

$$\arg(z) = \operatorname{Arg}(z) + 2k\pi, \qquad k \in \mathbb{Z}.$$

If $z = x + iy$ with $|z| = r$, then $\theta = \operatorname{Arg}(z)$ satisfies

$$\cos\theta = \frac{x}{r}, \qquad \sin\theta = \frac{y}{r}, \qquad \tan\theta = \frac{y}{x} \ (x \neq 0).$$

The pair $(\cos\theta, \sin\theta)$ fixes θ uniquely in $(-\pi, \pi]$; the tangent alone does not, since $\tan\theta = \tan(\theta + \pi)$. Always use the signs of x and y to choose the correct quadrant. In particular, a positive real number has argument 0, a negative real number has principal argument π, and a pure imaginary iy has principal argument $\frac{\pi}{2}$ if $y > 0$ and $-\frac{\pi}{2}$ if $y < 0$.

WARNING **The argument is defined only up to $2k\pi$**

A very common mistake is to write "$\arg(z) = \frac{\pi}{3}$" as if the argument were a single number. In fact $\arg(z)$ is a family of values differing by multiples of 2π. In every solution, state clearly which determination you use: either the principal argument $\operatorname{Arg}(z) \in (-\pi, \pi]$, or write $\arg(z) \equiv \theta \pmod{2\pi}$. Also remember: the number 0 has no argument — the direction of the zero vector is undefined.

1.4 Trigonometric and Exponential Forms

Putting modulus and argument together gives the second great way of writing a complex number.

DEFINITION **Trigonometric (polar) form**

Every non-zero complex number can be written

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z| > 0,\ \theta = \arg(z).$$

This is the **trigonometric form** (or polar form) of z.

METHOD **Converting from Cartesian to trigonometric form**

Given $z = x + iy \neq 0$:

1. Compute the modulus $r = \sqrt{x^2 + y^2}$.
2. Factorise: $z = r\left(\frac{x}{r} + i\frac{y}{r}\right)$.
3. Find θ such that $\cos\theta = \frac{x}{r}$ and $\sin\theta = \frac{y}{r}$, using the signs of x and y to fix the quadrant.
4. Conclude $z = r(\cos\theta + i\sin\theta)$, stating $\operatorname{Arg}(z) = \theta$.

EXEMPLE **From Cartesian to polar form**

Convert $z = -\sqrt{3} + i$ to trigonometric form. We have $r = \sqrt{3+1} = 2$, so

$$z = 2\left(-\frac{\sqrt{3}}{2} + i\,\frac{1}{2}\right).$$

We need $\cos\theta = -\frac{\sqrt{3}}{2}$ and $\sin\theta = \frac{1}{2}$: this is $\theta = \frac{5\pi}{6}$ (second quadrant, since $x < 0$, $y > 0$). Hence

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right), \qquad \operatorname{Arg}(z) = \frac{5\pi}{6}.$$

Euler's brilliant idea was to compress $\cos\theta + i\sin\theta$ into a genuine exponential. One defines, for real θ,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

so that every non-zero complex number has the **exponential form** $z = re^{i\theta}$. This definition is consistent because $e^{i\theta}$ obeys the usual exponential law $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$ — which is exactly the addition formula for sines and cosines, as the proof of the product rule below shows. From $e^{i\theta} =$

$\cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$, adding and subtracting gives **Euler's formulae**:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These will be indispensable when we linearise trigonometric powers in the next section. Note also the useful special values $e^{i\pi} = -1$, $e^{i\pi/2} = i$ and $e^{-i\pi/2} = -i$.

1.5 Multiplication and Division in Polar Form

Here is the heart of the matter: polar form turns multiplication into something geometrically transparent.

PROPERTY **Modulus and argument of a product and quotient**

For non-zero complex numbers $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$:

$$|z_1 z_2| = |z_1|\,|z_2|, \qquad \arg(z_1 z_2) \equiv \arg(z_1) + \arg(z_2) \pmod{2\pi},$$

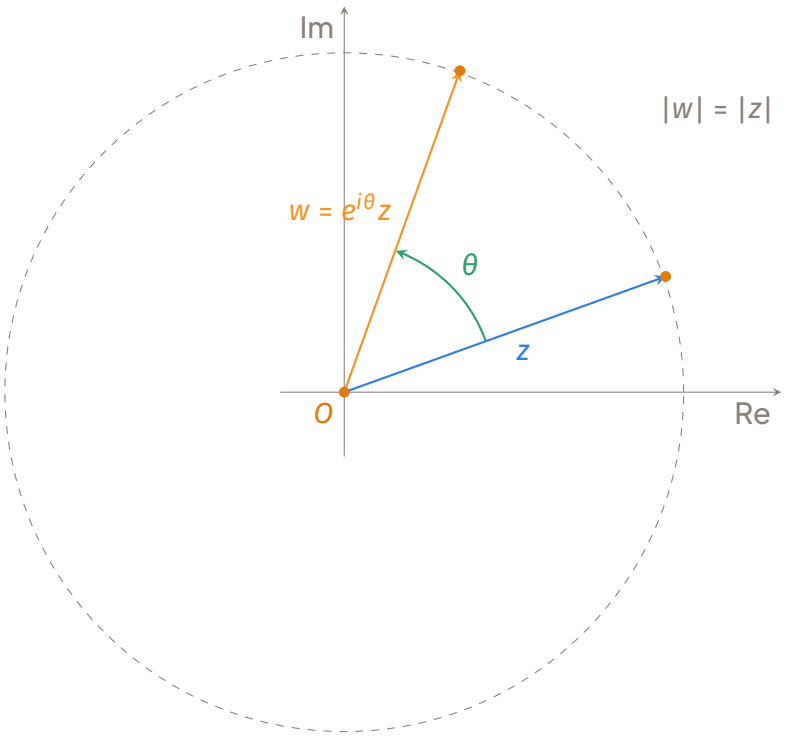
$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \qquad \arg\left(\frac{z_1}{z_2}\right) \equiv \arg(z_1) - \arg(z_2) \pmod{2\pi}.$$

Proof (examinable). Write $z_1 z_2 = r_1 r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$. Expanding and using the addition formulae $\cos(\theta_1+\theta_2) = \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2$ and $\sin(\theta_1+\theta_2) = \sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2$, we obtain

$$z_1 z_2 = r_1 r_2\big(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big),$$

which is the trigonometric form of $z_1 z_2$ with modulus $r_1 r_2$ and argument $\theta_1 + \theta_2$. For the quotient, apply the product rule to $z_1 = \frac{z_1}{z_2}\cdot z_2$: this gives $|z_1| = \left|\frac{z_1}{z_2}\right||z_2|$ and $\arg(z_1) \equiv \arg\left(\frac{z_1}{z_2}\right) + \arg(z_2) \pmod{2\pi}$, from which both claims follow. ∎

Geometrically, multiplying by $z_2 = r_2 e^{i\theta_2}$ rotates the plane about O through the angle θ_2 and scales distances by the factor r_2. In particular, multiplication by $e^{i\theta}$ (modulus 1) is a pure rotation of angle θ about the origin — this is exactly the crane arm of our opening hook. Since $\bar{z} = re^{-i\theta}$, we also read off $\arg(\bar{z}) \equiv -\arg(z) \pmod{2\pi}$ and $\arg\left(\frac{1}{z}\right) \equiv -\arg(z) \pmod{2\pi}$.



Multiplication by $e^{i\theta}$ rotates the image of z about O through the angle θ; the modulus is preserved, so the point moves along its circle centred at O.

1.6 De Moivre's Theorem

What happens when we multiply a complex number by itself repeatedly? The product rule says the modulus is raised to a power and the arguments accumulate. This observation becomes one of the most useful theorems of the whole course.

PROPERTY **De Moivre's theorem — proof examinable**

For every real number θ and every integer $n \in \mathbb{Z}$,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta),$$

equivalently $(e^{i\theta})^n = e^{in\theta}$.

Proof. Case $n \geq 0$: induction. Let $P(n)$ be the statement $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$. Base case. For $n = 0$: $(\cos\theta + i\sin\theta)^0 = 1 = \cos 0 + i\sin 0$, so $P(0)$ holds.

Inductive step. Assume $P(n)$ true for some $n \geq 0$. Then

$$\begin{aligned}(\cos\theta + i\sin\theta)^{n+1} &= (\cos\theta + i\sin\theta)^n(\cos\theta + i\sin\theta)\\ &= \big(\cos(n\theta) + i\sin(n\theta)\big)(\cos\theta + i\sin\theta)\\ &= \cos(n\theta + \theta) + i\sin(n\theta + \theta) \quad \text{(product rule)}\\ &= \cos\big((n+1)\theta\big) + i\sin\big((n+1)\theta\big),\end{aligned}$$

so $P(n+1)$ holds. By induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Case $n < 0$. Write $n = -m$ with $m \in \mathbb{N}^*$. Since $\cos\theta + i\sin\theta \neq 0$ (its modulus is 1),

$$(\cos\theta + i\sin\theta)^{-m} = \frac{1}{(\cos\theta + i\sin\theta)^m} = \frac{1}{\cos(m\theta) + i\sin(m\theta)}.$$

Multiplying numerator and denominator by the conjugate $\cos(m\theta) - i\sin(m\theta)$, whose product with the denominator is $\cos^2(m\theta) + \sin^2(m\theta) = 1$, gives

$$(\cos\theta + i\sin\theta)^{-m} = \cos(m\theta) - i\sin(m\theta) = \cos(-m\theta) + i\sin(-m\theta) = \cos(n\theta) + i\sin(n\theta),$$

using the parity of cosine and sine. Hence the theorem holds for all $n \in \mathbb{Z}$. ■

WARNING **De Moivre is stated for integer exponents**

The identity $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ is proved here for $n \in \mathbb{Z}$. For fractional exponents the formula becomes ambiguous: a complex number has n distinct n-th roots, so $\big(\cos\theta + i\sin\theta\big)^{1/n}$ takes n different values, not one. Writing $\cos(\theta/n) + i\sin(\theta/n)$ gives only one of them. We shall exploit this properly when studying roots of unity; for now, never apply De Moivre blindly to fractions.

1.7 Computing Powers Efficiently

De Moivre's theorem converts the painful task of expanding $(x + iy)^n$ into a one-line computation: raise the modulus to the power n and multiply the argument by n. Explicitly, if $z = re^{i\theta}$ then

$$z^n = r^n e^{in\theta} = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

WORKED EXAMPLE 1 **Powers via polar form**

Compute $(1+i)^{12}$ and $(\sqrt{3}+i)^9$, giving answers in Cartesian form.

SOLUTION

For $1 + i$: $r = \sqrt{2}$, and $\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$ gives $\operatorname{Arg}(1+i) = \frac{\pi}{4}$. Hence $1 + i = \sqrt{2}\,e^{i\pi/4}$ and by De Moivre,

$$(1+i)^{12} = (\sqrt{2})^{12}\,e^{i\cdot 12\pi/4} = 2^6 e^{3i\pi} = 64\big(\cos 3\pi + i\sin 3\pi\big) = -64,$$

since $\cos 3\pi = \cos\pi = -1$ and $\sin 3\pi = 0$.

For $\sqrt{3}+i$: $r=\sqrt{3+1}=2$, $\cos\theta=\frac{\sqrt{3}}{2}$, $\sin\theta=\frac{1}{2}$, so $\operatorname{Arg}=\frac{\pi}{6}$. Thus

$$(\sqrt{3}+i)^9 = 2^9 e^{i\cdot 9\pi/6} = 512\,e^{3i\pi/2} = 512\left(\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}\right) = -512\,i.$$

Notice how reducing the final angle modulo 2π (here $3\pi\equiv\pi$, while $\frac{3\pi}{2}$ is already principal) makes the Cartesian answer immediate.

EXEMPLE **A negative power**

Compute $(1+i)^{-8}$. With $1+i=\sqrt{2}\,e^{i\pi/4}$:

$$(1+i)^{-8} = (\sqrt{2})^{-8}e^{-2i\pi} = \frac{1}{16}\big(\cos(-2\pi)+i\sin(-2\pi)\big) = \frac{1}{16}.$$

KEY POINTS **Key points of this section**

- Three forms of $z\neq 0$: Cartesian $x+iy$, trigonometric $r(\cos\theta+i\sin\theta)$, exponential $re^{i\theta}$, with $r=|z|$, $\theta=\arg(z)$.
- The principal argument $\operatorname{Arg}(z)\in(-\pi,\pi]$ is unique; general arguments differ by $2k\pi$. The number 0 has no argument.
- Moduli multiply, arguments add: $|z_1z_2|=|z_1||z_2|$, $\arg(z_1z_2)\equiv\arg(z_1)+\arg(z_2)\pmod{2\pi}$. Multiplication by $e^{i\theta}$ is a rotation of angle θ about O.
- Euler's formulae: $\cos\theta=\frac{e^{i\theta}+e^{-i\theta}}{2}$, $\sin\theta=\frac{e^{i\theta}-e^{-i\theta}}{2i}$.
- De Moivre: $(\cos\theta+i\sin\theta)^n=\cos(n\theta)+i\sin(n\theta)$ for all $n\in\mathbb{Z}$ — proved by induction for $n\geq 0$, extended to negatives via the conjugate.
- To compute z^n: convert to polar form, then $z^n=r^n\big(\cos(n\theta)+i\sin(n\theta)\big)$.

EXERCISE 1 **Practice**

1. Write in trigonometric and exponential form: (a) $-1+i\sqrt{3}$; (b) $-4i$; (c) $3-3i$.
2. Given $z_1=2e^{i\pi/6}$ and $z_2=3e^{-i\pi/4}$, find the modulus and principal argument of z_1z_2 and of $\dfrac{z_1}{z_2}$.
3. Compute $(1-i)^{10}$ and $(\sqrt{3}-i)^{-6}$ in Cartesian form.

4. Show that $(1+i)^n + (1-i)^n$ is real for every $n \in \mathbb{N}$, and find its value in terms of n.

SOLUTION **Exercise 1**

1. (a) $r = 2$, $\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $-1 + i\sqrt{3} = 2e^{2i\pi/3}$. (b) $-4i = 4e^{-i\pi/2}$. (c) $r = 3\sqrt{2}$, $\theta = -\frac{\pi}{4}$, so $3 - 3i = 3\sqrt{2}\,e^{-i\pi/4}$.
2. $z_1 z_2 = 6e^{-i\pi/12}$: modulus 6, Arg $= -\frac{\pi}{12}$. $\frac{z_1}{z_2} = \frac{2}{3}e^{5i\pi/12}$: modulus $\frac{2}{3}$, Arg $= \frac{5\pi}{12}$.
3. $1 - i = \sqrt{2}\,e^{-i\pi/4}$, so $(1-i)^{10} = 2^5 e^{-10i\pi/4} = 32e^{-i\pi/2} = -32i$. Also $\sqrt{3} - i = 2e^{-i\pi/6}$, so $(\sqrt{3}-i)^{-6} = 2^{-6}e^{i\pi} = -\frac{1}{64}$.
4. $1 + i = \sqrt{2}\,e^{i\pi/4}$ and $1 - i = \sqrt{2}\,e^{-i\pi/4}$. By De Moivre, $(1+i)^n + (1-i)^n = 2^{n/2}\left(e^{in\pi/4} + e^{-in\pi/4}\right) = 2^{n/2} \cdot 2\cos\frac{n\pi}{4} = 2^{(n+2)/2}\cos\frac{n\pi}{4} \in \mathbb{R}$ by Euler's formula.

DID YOU KNOW? **Abraham de Moivre**

Abraham de Moivre (1667–1754) was a French mathematician who fled to England and became a friend of Isaac Newton. His famous formula, published in the early eighteenth century, later became the bridge that led Euler to the identity $e^{i\pi} + 1 = 0$, often called the most beautiful equation in mathematics because it links the five fundamental constants $0, 1, i, e, \pi$. Today, the same mathematics you have just learnt drives the signal processing inside every mobile phone transmitting across Douala and Yaoundé.

2 APPLICATIONS OF DE MOIVRE'S THEOREM

In the previous section we established De Moivre's theorem, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, and its exponential form $(re^{i\theta})^n = r^n e^{in\theta}$. That single identity is a remarkably powerful machine. In this section we put it to work in four directions: extracting all the nth roots of a complex number, expanding $\cos n\theta$ and $\sin n\theta$ as polynomials, linearising powers $\cos^n\theta$ and $\sin^n\theta$ (a gift for integration), and summing trigonometric series. These are among the most frequently examined techniques in the GCE Advanced Level Further Mathematics paper.

2.1 The nth Roots of a Complex Number

Motivation. In $\mathbb{R}$, the equation $x^2 = 9$ has two solutions, $x = \pm 3$, and $x^3 = 8$ has only one, $x = 2$. In $\mathbb{C}$ the picture is far richer: the fundamental theorem of algebra guarantees that $z^n = w$ (with $w \neq 0$) has exactly n solutions. De Moivre's theorem lets us find them all explicitly.

DEFINITION **nth roots of a complex number**

Let $w \in \mathbb{C}$, $w \neq 0$, and let $n \in \mathbb{N}$, $n \geq 2$. An **nth root** of w is any complex number z such that $z^n = w$. A solution of $z^n = 1$ is called an **nth root of unity**.

PROPERTY **Formula for the nth roots**

Let $w = re^{i\theta}$ with $r > 0$. Then $z^n = w$ has exactly n distinct solutions, given by

$$z_k = r^{1/n}\, e^{i(\theta+2k\pi)/n}, \qquad k = 0, 1, 2, \ldots, n-1.$$

Derivation (examinable). Write the unknown z in polar form, $z = \rho e^{i\varphi}$ with $\rho > 0$. By De Moivre, $z^n = \rho^n e^{in\varphi}$. The equation $z^n = re^{i\theta}$ then forces

$$\rho^n = r \quad \text{and} \quad n\varphi = \theta + 2k\pi,\ k \in \mathbb{Z},$$

because two complex numbers in polar form are equal exactly when their moduli are equal and their arguments differ by a multiple of 2π. Since $\rho > 0$ is real, $\rho = r^{1/n}$ (the ordinary positive real nth root), and $\varphi = \dfrac{\theta + 2k\pi}{n}$. Taking $k = 0, 1, \ldots, n-1$ gives n distinct values of φ in an interval of length 2π; any other integer k merely repeats one of these (adding 2π to an argument changes nothing). Hence there are exactly n distinct roots. ■

PROPERTY **Geometric distribution of the nth roots**

The n roots of $z^n = w$ all have the same modulus $r^{1/n}$ and their arguments differ successively by $\dfrac{2\pi}{n}$. Hence their images in the Argand diagram are the vertices of a **regular n-gon** inscribed in the circle of centre O and radius $r^{1/n}$.

WARNING **There are exactly n roots — not one!**

A very common error is to give only the "obvious" root, for instance writing $z^3 = 8 \Rightarrow z = 2$ and stopping. In $\mathbb{C}$, $z^3 = 8$ has three solutions: 2, $2e^{2i\pi/3} = -1 + i\sqrt{3}$ and $2e^{-2i\pi/3} = -1 - i\sqrt{3}$. Always count your answers: for $z^n = w$ you must list n distinct roots.

METHOD **Solving $z^n = w$ step by step**

1. Write w in polar form: $w = re^{i\theta}$ (find $r = |w|$ and an argument θ).
2. The common modulus of all roots is $r^{1/n}$.

3. The arguments are $\dfrac{\theta + 2k\pi}{n}$ for $k = 0, 1, \dots, n-1$.
4. List the n roots $z_k = r^{1/n} e^{i(\theta+2k\pi)/n}$, and if required convert each to Cartesian form $a + ib$.
5. Check: you must have exactly n roots, equally spaced by $\dfrac{2\pi}{n}$ around the circle of radius $r^{1/n}$.

WORKED EXAMPLE 2 **Solving $z^4 = -16$**

Solve $z^4 = -16$ in $\mathbb{C}$, giving the roots in Cartesian form.

SOLUTION

Solution. Write -16 in polar form: $|-16| = 16$ and $\arg(-16) = \pi$, so $-16 = 16e^{i\pi}$. The common modulus of the roots is $16^{1/4} = 2$, and the arguments are

$$\varphi_k = \frac{\pi + 2k\pi}{4}, \qquad k = 0, 1, 2, 3,$$

giving $\varphi_0 = \dfrac{\pi}{4}, \varphi_1 = \dfrac{3\pi}{4}, \varphi_2 = \dfrac{5\pi}{4}, \varphi_3 = \dfrac{7\pi}{4}$. Hence

$$z_0 = 2e^{i\pi/4} = \sqrt{2} + i\sqrt{2}, \qquad z_1 = 2e^{3i\pi/4} = -\sqrt{2} + i\sqrt{2},$$
$$z_2 = 2e^{5i\pi/4} = -\sqrt{2} - i\sqrt{2}, \qquad z_3 = 2e^{7i\pi/4} = \sqrt{2} - i\sqrt{2}.$$

The four roots are the vertices of a square inscribed in the circle of radius 2.

WORKED EXAMPLE 3 **Solving $z^3 = 8i$**

Solve $z^3 = 8i$ in $\mathbb{C}$.

SOLUTION

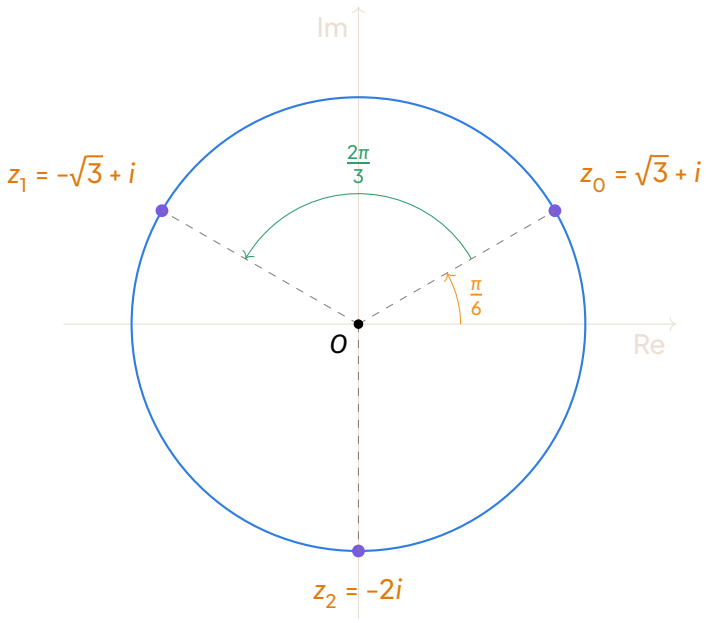
Solution. We have $|8i| = 8$ and $\arg(8i) = \dfrac{\pi}{2}$, so $8i = 8e^{i\pi/2}$. The cube roots have modulus $8^{1/3} = 2$ and arguments

$$\varphi_k = \frac{\frac{\pi}{2} + 2k\pi}{3} = \frac{\pi}{6} + \frac{2k\pi}{3}, \qquad k = 0, 1, 2,$$

that is $\dfrac{\pi}{6}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$. Therefore

$$z_0 = 2e^{i\pi/6} = \sqrt{3} + i, \qquad z_1 = 2e^{5i\pi/6} = -\sqrt{3} + i, \qquad z_2 = 2e^{3i\pi/2} = -2i.$$

The three roots are spaced at 120° on the circle of radius 2, as shown below.



The three cube roots of $8i$: equally spaced at 120° on the circle of radius 2.

WARNING **Degrees and radians: do not mix them**

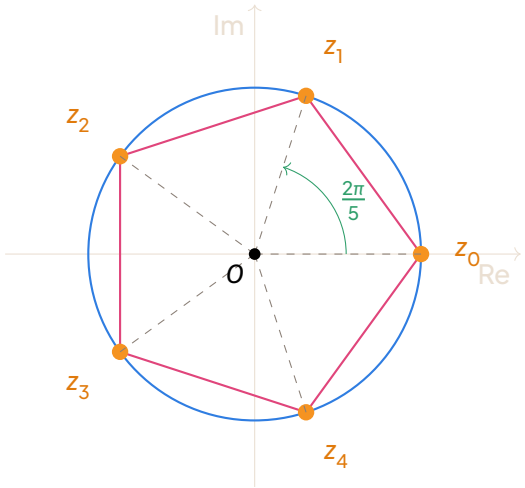
When listing the arguments $\frac{\theta + 2k\pi}{n}$, work entirely in radians (or entirely in degrees, but radians are the GCE standard). A classic blunder is to write $\theta = \frac{\pi}{2}$ and then compute $\frac{\pi/2+360k}{3}$. Mixing the two systems produces wrong angles and wrong roots. Convert first, then compute.

2.2 Roots of Unity

The equation $z^n = 1$ is the special case $w = 1 = 1 \cdot e^{i\cdot 0}$. Its solutions, the **nth roots of unity**, are

$$z_k = e^{2ik\pi/n}, \qquad k = 0, 1, \dots, n-1,$$

the vertices of a regular n-gon inscribed in the unit circle, one vertex being $z_0 = 1$.



The five fifth roots of unity: vertices of a regular pentagon on the unit circle.

PROPERTY **Sum and product of the nth roots of unity**

Let $z_0, z_1, \dots, z_{n-1}$ be the nth roots of unity, with $n \geq 2$. Then

$$z_0 + z_1 + \cdots + z_{n-1} = 0 \qquad \text{and} \qquad z_0 z_1 \cdots z_{n-1} = (-1)^{n-1}.$$

Justification. The roots are the solutions of $z^n - 1 = 0$. In a polynomial $z^n + 0 \cdot z^{n-1} + \cdots - 1$, the sum of the roots is minus the coefficient of z^{n-1}, which is 0; the product is $(-1)^n \times$ (constant term) $= (-1)^n(-1) = (-1)^{n-1}$. Alternatively, writing $\omega = e^{2i\pi/n}$, the roots are $1, \omega, \omega^2, \dots, \omega^{n-1}$, a geometric progression whose sum is $\dfrac{\omega^n - 1}{\omega - 1} = 0$ since $\omega^n = 1$. ■

KEY POINTS **Key facts about roots of unity**

- Setting $\omega = e^{2i\pi/n}$, the roots are $1, \omega, \omega^2, \dots, \omega^{n-1}$ and $\omega^n = 1$.
- Their sum is 0 (for $n \geq 2$); geometrically, the polygon is "balanced" about O.
- Non-real roots occur in conjugate pairs, since the coefficients of $z^n - 1$ are real.

2.3 Expanding $\cos n\theta$ and $\sin n\theta$

De Moivre gives $\cos n\theta + i \sin n\theta = (\cos\theta + i \sin\theta)^n$. Expanding the right-hand side with the **binomial theorem** and then equating real and imaginary parts produces polynomial identities.

EXEMPLE **The triple-angle formulae**

With $c = \cos\theta$, $s = \sin\theta$:

$$\cos 3\theta + i\sin 3\theta = (c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2s - s^3).$$

Equating real and imaginary parts:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta,$$

$$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta,$$

where we used $\sin^2\theta = 1 - \cos^2\theta$ and $\cos^2\theta = 1 - \sin^2\theta$.

METHOD **Expanding** $\cos n\theta$ **and** $\sin n\theta$

1. Write $\cos n\theta + i\sin n\theta = (c + is)^n$ (De Moivre).
2. Expand $(c + is)^n$ by the binomial theorem; use $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, …
3. Equate real parts to get $\cos n\theta$, imaginary parts to get $\sin n\theta$.
4. If a polynomial in $\cos\theta$ alone (or $\sin\theta$ alone) is wanted, eliminate the other function via $c^2 + s^2 = 1$.

Expressing $\tan n\theta$ **in terms of** $t = \tan\theta$. Divide the imaginary part by the real part and then divide numerator and denominator by $\cos^n\theta$. For example, with $t = \tan\theta$:

$$\tan 3\theta = \frac{3c^2s - s^3}{c^3 - 3cs^2} = \frac{3t - t^3}{1 - 3t^2}.$$

This is the pattern of the classical "tan $n\theta$" formulae, and the same technique works for any n.

2.4 Linearisation of $\cos^n\theta$ and $\sin^n\theta$

The reverse problem is just as important, especially for integration: express $\cos^n\theta$ or $\sin^n\theta$ as a sum of cosines or sines of multiple angles. The tool is now Euler's formulae.

METHOD **Linearising** $\cos^n\theta$ **using Euler's formulae**

1. Write $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$, so $\cos^n\theta = \dfrac{1}{2^n}\left(e^{i\theta} + e^{-i\theta}\right)^n$.
2. Expand with the binomial theorem.
3. Group symmetric terms $e^{ik\theta} + e^{-ik\theta} = 2\cos k\theta$.

4. Simplify to obtain a sum of cosines of multiple angles. (For $\sin^n\theta$, start from $\sin\theta = \dfrac{e^{i\theta}-e^{-i\theta}}{2i}$; sines of multiple angles appear when n is odd.)

WORKED EXAMPLE 4 **Linearising $\cos^4\theta$**

Express $\cos^4\theta$ as a sum of cosines of multiple angles. Hence evaluate $\displaystyle\int_0^{\pi/2}\cos^4\theta\,d\theta$.

SOLUTION

Solution. By Euler's formula and the binomial theorem,

$$\begin{aligned}\cos^4\theta &= \frac{1}{16}\left(e^{i\theta}+e^{-i\theta}\right)^4\\ &= \frac{1}{16}\left(e^{4i\theta}+4e^{2i\theta}+6+4e^{-2i\theta}+e^{-4i\theta}\right)\\ &= \frac{1}{16}(2\cos 4\theta+8\cos 2\theta+6) = \frac{1}{8}\cos 4\theta+\frac{1}{2}\cos 2\theta+\frac{3}{8}.\end{aligned}$$

Then

$$\int_0^{\pi/2}\cos^4\theta\,d\theta = \left[\frac{1}{32}\sin 4\theta+\frac{1}{4}\sin 2\theta+\frac{3\theta}{8}\right]_0^{\pi/2} = 0+0+\frac{3\pi}{16} = \frac{3\pi}{16}.$$

Without linearisation this integral would be painful; with it, it is immediate.

DID YOU KNOW? **Why examiners love linearisation**

Integrals such as $\int\cos^6 x\,dx$ or $\int\sin^3 x\cos^2 x\,dx$ appear regularly in GCE Further Mathematics and in first-year university courses (including at the University of Buea and Yaoundé I). Linearisation via Euler's formulae converts them into sums of easy integrals $\int\cos kx\,dx$. It is one of the highest-value techniques of this chapter.

2.5 Summing Trigonometric Series

A final beautiful application: sums such as

$$C=\sum_{k=0}^{n-1}\cos(\theta+k\alpha),\qquad S=\sum_{k=0}^{n-1}\sin(\theta+k\alpha)$$

are handled by recognising them as the real and imaginary parts of a **geometric series** of complex exponentials.

EXEMPLE **The standard summation**

Consider $C + iS = \sum_{k=0}^{n-1} e^{i(\theta+k\alpha)} = e^{i\theta} \sum_{k=0}^{n-1} \left(e^{i\alpha}\right)^k$. For $\alpha \not\equiv 0 \pmod{2\pi}$, the geometric sum gives

$$C + iS = e^{i\theta} \cdot \frac{e^{in\alpha} - 1}{e^{i\alpha} - 1}.$$

Using the half-angle trick, $e^{ix} - 1 = e^{ix/2} \cdot 2i \sin\frac{x}{2}$:

$$C + iS = e^{i\theta} \cdot \frac{e^{in\alpha/2} \cdot 2i \sin\frac{n\alpha}{2}}{e^{i\alpha/2} \cdot 2i \sin\frac{\alpha}{2}} = \frac{\sin\frac{n\alpha}{2}}{\sin\frac{\alpha}{2}} e^{i\left(\theta + \frac{(n-1)\alpha}{2}\right)}.$$

Taking real and imaginary parts:

$$\boxed{C = \frac{\sin\frac{n\alpha}{2}}{\sin\frac{\alpha}{2}} \cos\left(\theta + \frac{(n-1)\alpha}{2}\right), \qquad S = \frac{\sin\frac{n\alpha}{2}}{\sin\frac{\alpha}{2}} \sin\left(\theta + \frac{(n-1)\alpha}{2}\right).}$$

Note the elegance: the sum behaves like a single cosine (or sine) taken at the average angle $\theta + \frac{(n-1)\alpha}{2}$, scaled by the factor $\frac{\sin(n\alpha/2)}{\sin(\alpha/2)}$.

WARNING **Check the ratio before summing**

The geometric-series formula requires the common ratio $e^{i\alpha} \neq 1$, i.e. α not a multiple of 2π. If $\alpha = 2m\pi$, every term equals $\cos\theta$ (or $\sin\theta$) and the sums are simply $C = n\cos\theta, S = n\sin\theta$. Always treat this degenerate case separately.

EXERCISE 2 **Practice**

1. Solve $z^5 = 32$ in $\mathbb{C}$ and represent the roots on an Argand diagram.
2. Show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.
3. Linearise $\sin^3\theta$ and hence find $\int_0^{\pi} \sin^3\theta\, d\theta$.
4. Using $\omega = e^{2i\pi/3}$, verify directly that $1 + \omega + \omega^2 = 0$.
5. Sum $\sum_{k=0}^{n-1} \cos k\theta$ for $\theta \neq 2m\pi$.

SOLUTION **Exercises 1–5**

1. $32 = 32e^{i\cdot 0}$, so $z_k = 2e^{2ik\pi/5}$, $k = 0, \dots, 4$: the fifth roots are 2, $2e^{2i\pi/5}$, $2e^{4i\pi/5}$, $2e^{6i\pi/5}$, $2e^{8i\pi/5}$, vertices of a regular pentagon on the circle of radius 2. 2. Expand $(c + is)^5$: real part $c^5 - 10c^3s^2 + 5cs^4$; replace $s^2 = 1 - c^2$: $c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 = 16c^5 - 20c^3 + 5c$. 3. $\sin^3\theta = \frac{1}{(2i)^3}(e^{i\theta} - e^{-i\theta})^3 = \frac{3\sin\theta - \sin 3\theta}{4}$; the integral is $\left[-\frac{3\cos\theta}{4} + \frac{\cos 3\theta}{12}\right]_0^{\pi} = \frac{4}{3}$. 4. $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, $\omega^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$; sum $= 1 - 1 + 0i = 0$. 5. With θ playing the role of α and initial angle 0: $\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}} \cos\frac{(n-1)\theta}{2}$.

KEY POINTS **Section summary**

- $z^n = w = re^{i\theta}$ has exactly n roots: $z_k = r^{1/n}e^{i(\theta+2k\pi)/n}$, $k = 0, \dots, n-1$ — vertices of a regular n-gon.
- The nth roots of unity are $1, \omega, \dots, \omega^{n-1}$ with $\omega = e^{2i\pi/n}$; their sum is 0.
- De Moivre + binomial theorem ⇒ polynomial forms of $\cos n\theta$, $\sin n\theta$, $\tan n\theta$.
- Euler's formulae + binomial theorem ⇒ linearisation of $\cos^n\theta$, $\sin^n\theta$ (ideal for integration).
- Trigonometric sums are real/imaginary parts of geometric series of exponentials.

3 LOCI IN THE ARGAND DIAGRAM

In the previous section we exploited De Moivre's theorem to extract roots, linearise trigonometric expressions and sum series. In every one of those applications the complex number z was treated as an algebraic object. We now return to the geometric point of view: a complex number is also a point of the plane. The central question of this section is the following: if a complex number z is required to satisfy a condition involving a modulus or an argument, what is the set of points $M(z)$ that satisfy it? Such a set is called a **locus**. Locus questions are among the most frequently examined items in the GCE Advanced Level paper, and they are also the gateway to the transformations of the z-plane that we study in the next section.

3.1 The Affine Plane and Affixes

Recall that to every complex number $z = x + iy$ (with $x, y \in \mathbb{R}$) we associate the point $M(x, y)$ of the plane equipped with an orthonormal frame $(O; \vec{u}, \vec{v})$. This plane, viewed as the set of affixes of points, is called the **affine complex plane** (or Argand plane).

DEFINITION **Affix of a point and of a vector**

Let the plane be referred to an orthonormal frame $(O;\vec{u},\vec{v})$.

- The point $M(x,y)$ is said to have **affix** $z_M = x + iy$. We write $M(z_M)$.
- The vector $\vec{w}\begin{pmatrix}a\\b\end{pmatrix}$ has affix $z_{\vec{w}} = a + ib$.
- If $A(z_A)$ and $B(z_B)$ are two points, then the vector $\vec{AB}$ has affix

$$z_{\vec{AB}} = z_B - z_A.$$

- The midpoint I of the segment $[AB]$ has affix

$$z_I = \frac{z_A + z_B}{2}.$$

These formulas are immediate consequences of the coordinate formulas $\vec{AB}(x_B - x_A,\ y_B - y_A)$ and $I\left(\frac{x_A+x_B}{2}, \frac{y_A+y_B}{2}\right)$, but they are far more powerful in complex form because they combine with the modulus and argument.

PROPERTY **Modulus and argument of a difference**

Let $A(z_A)$ and $B(z_B)$. Then

$$AB = |z_B - z_A| \qquad \text{and} \qquad (\vec{u},\vec{AB}) \equiv \arg(z_B - z_A) \pmod{2\pi}.$$

This single property is the key to the whole section: **a condition on $|z-a|$ is a condition on a distance, and a condition on $\arg(z-a)$ is a condition on a direction.**

3.2 Circles: the locus $|z-a| = r$

PROPERTY **Circle**

Let $A(a)$ be a fixed point and $r > 0$. The locus of points $M(z)$ such that

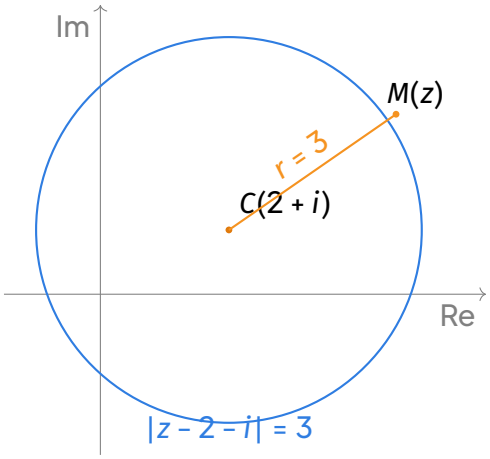
$$|z - a| = r$$

is the **circle** of centre A and radius r. The condition $|z-a| \le r$ describes the closed disc, and $|z-a| < r$ the open disc.

The proof is one line: $|z-a| = |z_M - z_A| = AM$, so the condition says exactly $AM = r$.

EXEMPLE

The equation $|z - 2 - i| = 3$ represents the circle of centre $C(2 + i)$, i.e. $C(2, 1)$, and radius 3. Writing $z = x + iy$, we recover the Cartesian equation $(x - 2)^2 + (y - 1)^2 = 9$.



The locus $|z - 2 - i| = 3$: circle of centre $C(2, 1)$ and radius 3.

3.3 The Perpendicular Bisector: $|z - a| = |z - b|$

Suppose a condition says that $M(z)$ is as far from A as from B. Geometric intuition (think of two radio transmitters in Yaounde and Douala: the set of positions receiving both equally strongly) tells us the answer is a straight line.

PROPERTY **Perpendicular bisector**

Let $A(z_A)$ and $B(z_B)$ be distinct points. The locus of points $M(z)$ such that

$$|z - z_A| = |z - z_B|$$

is the **perpendicular bisector** of the segment $[AB]$: the line through the midpoint $I\left(\frac{z_A + z_B}{2}\right)$ perpendicular to (AB).

Proof (examinable). The condition $|z - z_A| = |z - z_B|$ means $MA = MB$. Squaring with $z = x + iy$, $z_A = x_A + iy_A$, $z_B = x_B + iy_B$:

$$(x - x_A)^2 + (y - y_A)^2 = (x - x_B)^2 + (y - y_B)^2,$$

which simplifies to

$$2(x_B - x_A)x + 2(y_B - y_A)y = x_B^2 + y_B^2 - x_A^2 - y_A^2,$$

the equation of a line whose normal vector is $\vec{AB}(x_B - x_A, y_B - y_A)$ and which passes through I (sub-

stitute the coordinates of I: both sides agree). Hence it is the perpendicular bisector of $[AB]$. ■

WORKED EXAMPLE 5 **A perpendicular bisector locus**

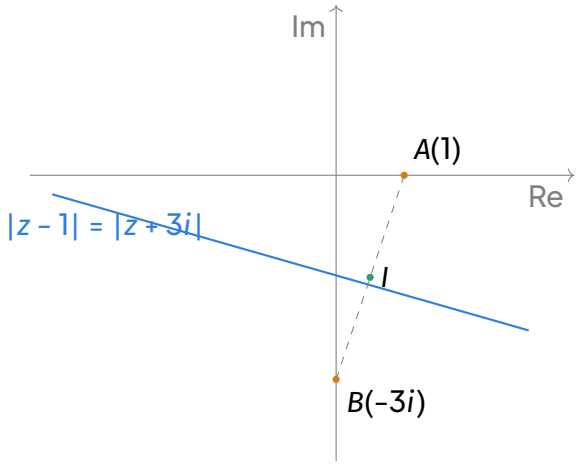
Describe and sketch the locus of $M(z)$ such that $|z - 1| = |z + 3i|$.

SOLUTION

The condition is $|z - 1| = |z - (-3i)|$, i.e. $MA = MB$ with $A(1, 0)$ and $B(0, -3)$. The locus is the perpendicular bisector of $[AB]$. The midpoint is $I\left(\frac{1}{2}, -\frac{3}{2}\right)$. Since $\vec{AB}(-1, -3)$, a point $M(x, y)$ of the bisector satisfies $\vec{IM} \cdot \vec{AB} = 0$:

$$-\left(x - \tfrac{1}{2}\right) - 3\left(y + \tfrac{3}{2}\right) = 0 \quad \Longleftrightarrow \quad x + 3y + 4 = 0.$$

The locus is the line $x + 3y + 4 = 0$, passing through $I(0.5, -1.5)$, perpendicular to (AB).



The perpendicular bisector of $[AB]$ with $A(1)$, $B(-3i)$.

3.4 Circles of Apollonius: $|z - a| = k|z - b|$

What happens when the distances are in a fixed ratio $k \neq 1$? The locus is no longer a line but a circle — the **circle of Apollonius**. This is a GCE-useful enrichment: the question appears regularly as a structured exercise.

PROPERTY **Circle of Apollonius**

Let $A(z_A)$, $B(z_B)$ distinct and $k > 0$, $k \neq 1$. The locus of $M(z)$ such that

$$|z - z_A| = k\,|z - z_B|$$

is a circle whose diameter is the segment joining the two points of the line (AB) that divide $[AB]$ in the ratio k (one internally, one externally). Its centre lies on the line (AB).

Derivation (examinable in outline). Squaring $MA^2 = k^2 MB^2$ with $z = x + iy$ gives

$$(x - x_A)^2 + (y - y_A)^2 = k^2\left[(x - x_B)^2 + (y - y_B)^2\right].$$

Expanding and grouping terms:

$$(1 - k^2)(x^2 + y^2) - 2(x_A - k^2 x_B)x - 2(y_A - k^2 y_B)y + (x_A^2 + y_A^2) - k^2(x_B^2 + y_B^2) = 0.$$

Since $k \neq 1$, we divide by $1 - k^2$ to obtain an equation of the form $x^2 + y^2 + ux + vy + w = 0$, which is a circle (provided the radius squared is positive, which it is for distinct A, B). The centre is $\left(\frac{x_A - k^2 x_B}{1-k^2}, \frac{y_A - k^2 y_B}{1-k^2}\right)$, which lies on line AB. ■

EXEMPLE

The locus $|z - 4| = 2|z - 1|$: squaring with $z = x + iy$,

$$(x - 4)^2 + y^2 = 4\left[(x - 1)^2 + y^2\right] \iff 3x^2 + 3y^2 - 12 = 0 \iff x^2 + y^2 = 4.$$

The locus is the circle centred at O of radius 2. (Here $A(4, 0)$, $B(1, 0)$, $k = 2$; the internal division point is 2, the external is -2; the circle has diameter $[-2, 2]$.)

3.5 Argument Loci: Half-Lines and Circles

PROPERTY **Half-line**

Let $A(a)$ and $\theta \in \mathbb{R}$. The locus of points $M(z)$, $M \neq A$, such that

$$\arg(z - a) \equiv \theta \pmod{2\pi}$$

is the **half-line** starting at A (the point A itself excluded) making the angle θ with the positive real axis.

WARNING **The half-line excludes its origin**

A very common mistake is to draw the whole ray including A, or even the full line. But $\arg(z - a)$ is not defined when $z = a$, since $\arg(0)$ does not exist. Mark A with an open (hollow) dot. Similarly, $\arg(z - a) = \theta$ is only half of the line through A in direction θ: the opposite half corresponds to $\arg(z - a) = \theta + \pi$.

For loci involving two fixed points, the key tool is the argument of a quotient.

PROPERTY **Angle between vectors via a quotient of affixes**

Let $A(z_A)$, $B(z_B)$, $C(z_C)$ be distinct points. Then

$$\arg\left(\frac{z_C - z_A}{z_C - z_B}\right) \equiv (\vec{CB}, \vec{CA}) \pmod{2\pi},$$

i.e. this argument measures the angle $B\hat{C}A$ at the vertex C. In particular, with $M(z)$ a variable point,

$$\arg\left(\frac{z - z_A}{z - z_B}\right) \equiv (\vec{MB}, \vec{MA}) \pmod{2\pi}.$$

Proof. $\arg\left(\frac{z - z_A}{z - z_B}\right) = \arg(z - z_A) - \arg(z - z_B) = (\vec{u}, \vec{MA}) - (\vec{u}, \vec{MB}) = (\vec{MB}, \vec{MA})$ by the Chasles relation on angles. ■

PROPERTY **Thales' circle and arcs of circles**

Let $A(z_A)$, $B(z_B)$ be distinct.

- The locus of $M(z)$, $M \notin \{A, B\}$, such that

$$\arg\left(\frac{z - z_A}{z - z_B}\right) \equiv \pm\frac{\pi}{2} \pmod{\pi}$$

  is the **circle of diameter** $[AB]$ (Thales' circle), with A and B excluded.

- More generally, for a constant α with $0 < \alpha < \pi$, the condition

$$\arg\left(\frac{z - z_A}{z - z_B}\right) \equiv \alpha \pmod{2\pi}$$

  describes one arc of the circle through A and B (the arc on which the segment $[AB]$ subtends an angle α). The condition $\equiv \alpha \pmod{\pi}$ gives the whole circle (both arcs) minus A and B.

Indeed, $(\vec{MB}, \vec{MA}) = \pm\frac{\pi}{2}$ means $MA \perp MB$, and the set of points from which a segment $[AB]$ is seen at a right angle is exactly the circle of diameter $[AB]$. The inscribed angle theorem guarantees that a constant angle $\alpha \pmod{2\pi}$ corresponds to one arc.

EXEMPLE

The locus $\arg\left(\frac{z - 2}{z + 2}\right) = \frac{\pi}{2}$ is the circle of diameter $[AB]$ with $A(2, 0)$, $B(-2, 0)$: the circle cen-

tred at O of radius 2, with the points $(\pm 2, 0)$ excluded.

3.6 Regions Defined by Inequalities

Inequalities define regions of the plane rather than curves. The standard vocabulary:

KEY POINTS **Standard regions**

- $|z - a| \le r$: **closed disc**, centre $A(a)$, radius r (boundary circle included: solid line).
- $|z - a| < r$: **open disc** (boundary excluded: dashed line).
- $r_1 \le |z| \le r_2$: **annulus** centred at O, between the circles of radii r_1 and r_2.
- $\alpha \le \arg(z) \le \beta$: **sector** (wedge) of the plane between the half-lines of arguments α and β, with O excluded.

WARNING **Open versus closed boundaries**

Do not confuse $|z - a| < r$ (open disc: shade the interior and draw the boundary circle dashed) with $|z - a| \le r$ (closed disc: boundary solid). The same convention applies to strict inequalities on arguments. In the GCE, an incorrectly drawn boundary costs marks even when the region is correctly identified.

METHOD **Identifying and sketching a locus**

1. Translate the condition geometrically: $|z - a|$ is the distance AM; $\arg(z - a)$ is the direction of $\vec{AM}$; a quotient $\frac{z-a}{z-b}$ encodes the angle at M.
2. Identify the type: modulus equal to a constant → circle; two equal moduli → perpendicular bisector; ratio of moduli → Apollonius circle; constant argument → half-line; right-angle argument → Thales' circle; constant argument (mod 2π) of a quotient → arc of circle; inequality → region.
3. Determine the characteristic elements: centre, radius, midpoint, angle, division points for Apollonius.
4. Sketch on an Argand diagram: mark fixed points, draw the locus, respect boundary conventions (solid/dashed, hollow dots for excluded points), and shade the required region for inequalities — shade a test point (e.g. $z = 0$) to check which side of a boundary is wanted.

5. Verify algebraically with $z = x + iy$ if in doubt.

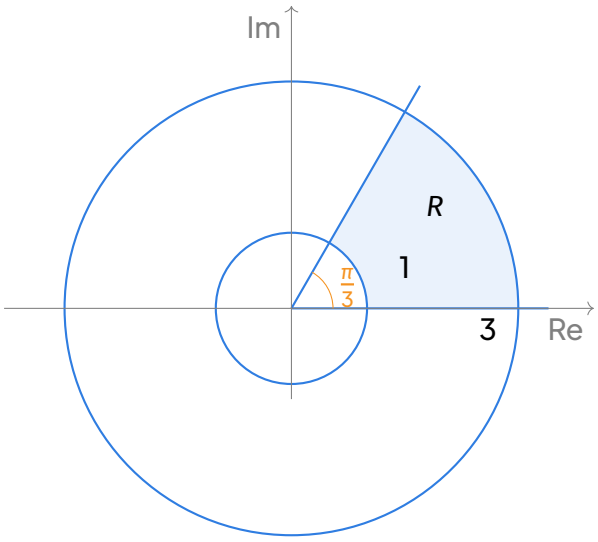
WORKED EXAMPLE 6 **Intersection of loci: annulus and sector**

Shade the region of the Argand diagram defined by

$$1 \le |z| \le 3 \qquad \text{and} \qquad 0 \le \arg(z) \le \frac{\pi}{3}.$$

SOLUTION

The first condition is the closed annulus between the circles centred at O of radii 1 and 3 (both boundaries included, drawn solid). The second is the sector between the positive real axis and the half-line from O at angle $\frac{\pi}{3}$. The required region is the intersection: the portion of the annulus lying inside the sector. Both boundary circles are solid; both bounding half-lines are solid; the origin is excluded (as $\arg(0)$ is undefined) but this does not affect the shaded region.



The region R: $1 \le |z| \le 3$ and $0 \le \arg(z) \le \frac{\pi}{3}$ (annular sector).

EXERCISE 3

On the same Argand diagram, sketch and shade the set of points $M(z)$ satisfying simultaneously $|z - 2i| \le 2$ and $|z| > 1$. State precisely which parts of the boundary are included.

SOLUTION **Exercise 1**

$|z - 2i| \leq 2$ is the closed disc of centre $(0, 2)$ and radius 2 (it passes through O); $|z| > 1$ is the exterior of the open disc of radius 1. The required set is the part of the first disc lying outside or on the unit circle: the circle of centre $(0, 2)$ is drawn solid, the unit circle dashed (strict inequality), and the region between them inside the disc is shaded.

DID YOU KNOW? **Why loci matter beyond the exam**

Locus conditions are the language of many technologies. The set of points where a mobile phone signal from two relay antennas arrives with equal strength is a perpendicular bisector; points where the ratio of signal powers is constant lie on Apollonius circles — a fact used in positioning and in the design of cellular networks. The same mathematics will reappear in the next section, where we ask: what becomes of these circles and lines when the plane is transformed by a mapping $z \mapsto w = f(z)$?

4 TRANSFORMATIONS FROM THE Z-PLANE TO THE W-PLANE

In the previous section we studied loci: given a condition on z (for example $|z - 2i| = 3$), we described the set of points of the Argand plane satisfying it. We now change our point of view. Instead of asking where a point may lie, we ask where it goes: we feed each point z of the plane into a rule, and a new point w comes out. The rule is a **transformation** of the plane, and complex numbers make such transformations astonishingly easy to handle, because a single formula like $w = (1 + i)z + 3$ encodes a rotation, an enlargement and a translation all at once.

DEFINITION **Transformation from the z-plane to the w-plane**

A **transformation from the z-plane to the w-plane** is a rule T which associates to each complex number $z = x + iy$ (in its domain) a unique complex number

$$w = T(z) = u + iv,$$

where u and v are real functions of x and y. We picture z in one Argand diagram (the **z-plane**, with axes x and y) and its image w in a second Argand diagram (the **w-plane**, with axes u and v). If C is a curve in the z-plane, its **image** is the set $T(C) = \{T(z) : z \in C\}$ in the w-plane.

The central question of this section is: given a curve (line, circle, ray) in the z-plane and a map T, what is the equation of the image curve in the w-plane?

4.1 The three elementary maps

Translation: $w = z + b$.

Let $b = p + iq$ be fixed. Writing $z = x + iy$, we get $w = (x + p) + i(y + q)$: every point is shifted by the vector $\begin{pmatrix} p \\ q \end{pmatrix}$. Geometrically, $w - z = b$ means the displacement from z to w is the same everywhere. Translations preserve distances, angles, shapes and sizes.

Rotation composed with enlargement: $w = az$, with $a = re^{i\alpha}$ ($r > 0$).

This is where the exponential form shines. If $z = \rho e^{i\theta}$, then

$$w = az = r\rho\, e^{i(\theta+\alpha)}.$$

So $|w| = r|z|$ and $\arg w = \arg z + \alpha$: the map enlarges distances from the origin by the factor $r = |a|$ and rotates the plane through the angle $\alpha = \arg a$ about O. This is exactly a **direct similitude** centred at O, which we shall study formally later in this chapter. (If a is real and positive, it is a pure enlargement; if $|a| = 1$, it is a pure rotation.)

Reflection in the real axis: $w = \bar{z}$.

If $z = \rho e^{i\theta}$ then $\bar{z} = \rho e^{-i\theta}$: the modulus is kept and the argument is negated. Every point is mirrored across the x-axis. Note that this map reverses orientation: a triangle traversed anticlockwise is sent to a triangle traversed clockwise.

EXEMPLE **Image of a point under $w = (1 + i)z + 3$**

Take $z_0 = 2 - i$. Since $1 + i = \sqrt{2}\, e^{i\pi/4}$, the map is: enlarge by $\sqrt{2}$, rotate by $\frac{\pi}{4}$, then translate by 3. Directly:

$$w_0 = (1 + i)(2 - i) + 3 = (2 - i + 2i - i^2) + 3 = (3 + i) + 3 = 6 + i.$$

Check with the geometric description: $|z_0| = \sqrt{5}$, so $|w_0 - 3| = \sqrt{2} \cdot \sqrt{5} = \sqrt{10}$, and indeed $|3 + i| = \sqrt{10}$. ✓

4.2 Image of a line and of a circle under $w = az + b$

Because $w = az + b$ is a similitude (rotation + enlargement + translation), it preserves shapes: **lines map to lines and circles map to circles**. Algebraically this is easy to see because the map is invertible: $z = \dfrac{w - b}{a}$ (provided $a \neq 0$).

WORKED EXAMPLE 7 **Image of the circle $|z| = 2$ under $w = (1 + i)z + 3$**

Find the image of the circle $|z| = 2$ under the map $w = (1 + i)z + 3$, and describe it geometrically.

SOLUTION

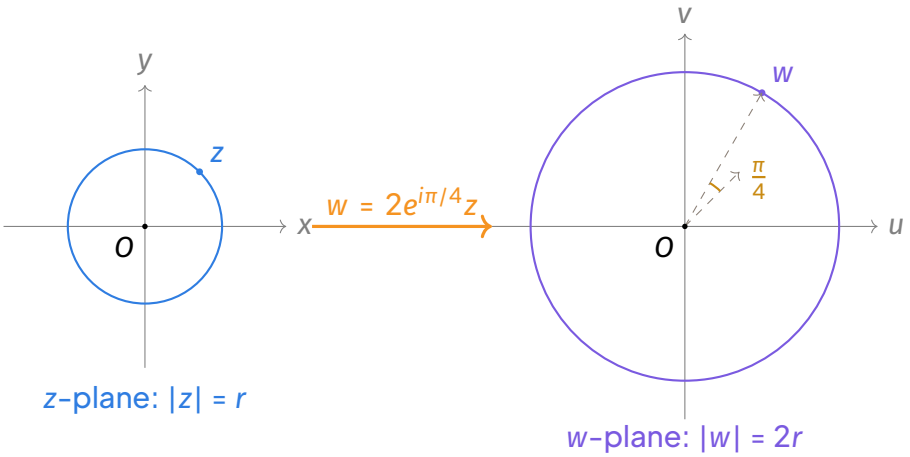
Step 1: invert the map. $z = \dfrac{w-3}{1+i}$.
Step 2: substitute into the locus condition.

$$|z| = 2 \iff \left|\frac{w-3}{1+i}\right| = 2 \iff \frac{|w-3|}{|1+i|} = 2 \iff |w-3| = 2\sqrt{2}.$$

Step 3: interpret. The image is the circle of centre $3 + 0i$ (the image of the centre 0) and radius $2\sqrt{2} = 2|1+i|$ (the original radius multiplied by the scale factor $\sqrt{2}$). Geometrically: enlarge by $\sqrt{2}$ (radius $2 \to 2\sqrt{2}$), rotate by $\pi/4$ (centre 0 unchanged), translate by 3 (centre $0 \to 3$).

EXEMPLE **Image of a line under $w = az + b$**

Find the image of the line $\mathrm{Re}(z) = 1$ under $w = 2iz + 1$. Invert: $z = \dfrac{w-1}{2i}$. Write $w = u + iv$: $z = \dfrac{(u-1)+iv}{2i} = \dfrac{v}{2} - i\,\dfrac{u-1}{2}$. Hence $\mathrm{Re}(z) = \dfrac{v}{2} = 1$, i.e. $v = 2$: the image is the horizontal line $\mathrm{Im}(w) = 2$. Geometrically, the vertical line was rotated by $\pi/2$ (becoming horizontal), stretched by 2, then shifted up by 1.



Under $w = 2e^{i\pi/4}z$, the circle $|z| = r$ is enlarged by the factor 2 and rotated through $\dfrac{\pi}{4}$; since a circle centred at O is invariant under rotation about O, only the enlargement is visible: the image is $|w| = 2r$.

4.3 The reciprocal map $w = \dfrac{1}{z}$

The map $z \mapsto \dfrac{1}{z}$ is defined on $\mathbb{C} \setminus \{0\}$ and is called **inversion** (composed with reflection in the real axis). Its behaviour is completely described by the modulus–argument analysis.

PROPERTY **Modulus and argument under $w = 1/z$**

For $z \neq 0$, writing $z = \rho e^{i\theta}$,

$$w = \frac{1}{z} = \frac{1}{\rho} e^{-i\theta}, \qquad \text{so} \qquad |w| = \frac{1}{|z|} \quad \text{and} \quad \arg w = -\arg z.$$

In particular: points far from O map close to O and vice versa; the upper half-plane maps to the lower half-plane; and the unit circle is invariant ($|z| = 1 \Rightarrow |w| = 1$), each point $e^{i\theta}$ being sent to $e^{-i\theta}$.

Image of the line $\mathrm{Re}(z) = c$ ($c \neq 0$).

Let $z = c + iy$ with $y \in \mathbb{R}$, and write $w = u + iv = \dfrac{1}{c + iy}$. Then

$$u = \frac{c}{c^2 + y^2}, \qquad v = \frac{-y}{c^2 + y^2}.$$

Compute:

$$u^2 + v^2 = \frac{c^2 + y^2}{(c^2 + y^2)^2} = \frac{1}{c^2 + y^2} = \frac{u}{c}.$$

Hence $u^2 + v^2 - \dfrac{u}{c} = 0$, i.e.

$$\left(u - \frac{1}{2c}\right)^2 + v^2 = \left(\frac{1}{2c}\right)^2 :$$

a **circle through the origin**, of centre $\dfrac{1}{2c}$ on the real axis and radius $\dfrac{1}{2|c|}$. The origin itself is excluded (it would correspond to $z = \infty$).

WARNING **$w = 1/z$ is undefined at $z = 0$**

The reciprocal map is only defined for $z \neq 0$. Consequently: a line through the origin maps to a line through the origin (minus O), since $\arg w = -\arg z$ keeps the points on a ray; but a line not through the origin maps to a circle through the origin (with O excluded). Never write "the image of a line is a line" without checking whether the line passes through O. Similarly, a circle through O maps to a line, and a circle not through O maps to a circle.

WORKED EXAMPLE 8 **Image of $\mathrm{Re}(z) = 1$ under $w = 1/z$**

Find the image of the line $\mathrm{Re}(z) = 1$ under $w = \dfrac{1}{z}$.

SOLUTION

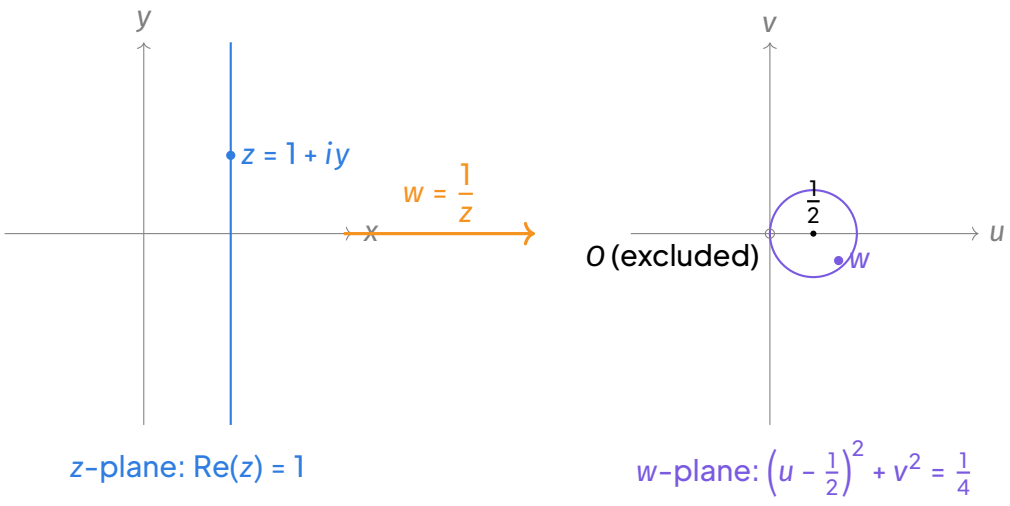
Parametrise: $z = 1 + iy$, $y \in \mathbb{R}$. Then with $w = u + iv$:

$$w = \frac{1}{1+iy} = \frac{1-iy}{1+y^2} \implies u = \frac{1}{1+y^2}, \quad v = \frac{-y}{1+y^2}.$$

Eliminate y:

$$u^2 + v^2 = \frac{1+y^2}{(1+y^2)^2} = \frac{1}{1+y^2} = u.$$

So $u^2 - u + v^2 = 0$, i.e. $\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}$. The image is the circle of centre $\dfrac{1}{2}$ and radius $\dfrac{1}{2}$, passing through the origin (the origin itself excluded, since $u = \frac{1}{1+y^2} > 0$ for all finite y).



The line $\mathrm{Re}(z) = 1$ (which avoids the origin) is mapped by $w = 1/z$ to the circle of centre $\frac{1}{2}$ and radius $\frac{1}{2}$, passing through O. The point $z = 1 + iy$ shown maps to w in the lower half-plane, since $\arg w = -\arg z$.

4.4 The map $w = z^2$

If $z = \rho e^{i\theta}$, then $w = \rho^2 e^{2i\theta}$: **moduli are squared, arguments are doubled**.

- **Unit circle:** $|z| = 1 \Rightarrow |w| = 1$; the image is again the unit circle, but described twice as θ runs from 0 to 2π (the map is two-to-one: z and $-z$ have the same image).
- **Rays from O:** the ray $\arg z = \theta_0$ maps to the ray $\arg w = 2\theta_0$. In particular the positive real axis

is fixed (as a set), and the positive imaginary axis ($\theta_0 = \pi/2$) maps to the negative real axis.

- **Circle** $|z| = r$: maps to $|w| = r^2$, described twice.

WARNING **Not every map sends circles to circles**

The maps $w = az + b$, $w = \bar{z}$ and $w = 1/z$ send every line or circle to a line or circle. This is false for $w = z^2$ in general. For instance, the line $\mathrm{Re}(z) = 1$ maps under $w = z^2$ to a parabola: with $z = 1 + iy$, $w = (1 - y^2) + 2iy$, so $u = 1 - y^2$, $v = 2y$, and eliminating y gives $u = 1 - \frac{v^2}{4}$ — certainly not a circle. Always compute; never assume.

4.5 General method: parametrise, substitute, eliminate

METHOD **Finding the image of a curve under a complex map**

1. Parametrise the given locus in the z-plane (e.g. $z = c + iy$ for a vertical line, $z = z_0 + re^{it}$ for a circle), or write its equation in terms of z and $\bar{z}$.
2. Substitute into $w = T(z)$, or invert the map and substitute $z = T^{-1}(w)$ into the locus condition — whichever is simpler.
3. Eliminate the parameter (or simplify the condition) to obtain a relation between $u = \mathrm{Re}(w)$ and $v = \mathrm{Im}(w)$.
4. Identify the image curve and state any excluded points (e.g. the origin for $w = 1/z$).

4.6 Möbius maps: a powerful class of transformations

DEFINITION **Linear fractional (Möbius) map**

A **Möbius map** is a transformation of the form

$$w = \frac{az + b}{cz + d}, \qquad ad - bc \neq 0,$$

defined for $z \neq -\frac{d}{c}$ (when $c \neq 0$). The condition $ad - bc \neq 0$ guarantees the map is not constant.

Every Möbius map is a composition of translations, inversions $z \mapsto 1/z$ and maps $z \mapsto az$: indeed, for $c \neq 0$,

$$\frac{az + b}{cz + d} = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}.$$

Since each ingredient sends lines and circles to lines or circles, so does every Möbius map. This

is a powerful exam shortcut: to find the image of a circle, it is enough to find the images of three of its points.

EXEMPLE **A Möbius image**

Under $w = \dfrac{z}{z+1}$, find the image of the circle $|z| = 1$ (note $z = -1$ is on the circle, so expect a line). Invert: $z = \dfrac{w}{1-w}$. Then $|z| = 1 \iff |w| = |1-w| \iff |w| = |w-1|$: the set of points equidistant from 0 and 1, i.e. the vertical line $\mathrm{Re}(w) = \dfrac{1}{2}$.

DID YOU KNOW? **Möbius transformations and the Riemann sphere**

Möbius maps are the only conformal (angle-preserving) automorphisms of the Riemann sphere $\mathbb{C} \cup \{\infty\}$. They appear in physics (relativity, fluid dynamics), in cartography (stereographic projection), and in the study of fractals. The fact that they map "circles to circles" (where lines are circles through ∞) makes them indispensable in geometry.

EXERCISE 4 **Practice**

1. Find the image of the line $\mathrm{Im}(z) = 2$ under $w = (1-i)z + 2i$.
2. Find the image of the circle $|z-1| = 1$ under $w = \dfrac{1}{z}$. Why is the image a line?
3. Under $w = z^2$, find the image of the ray $\arg z = \dfrac{\pi}{6}$ and of the circle $|z| = 3$.
4. Find the image of the line $\mathrm{Re}(z) = \dfrac{1}{2}$ under $w = \dfrac{1}{z}$, giving the centre and radius of the image circle.

SOLUTION **Practice**

1. Invert: $z = \dfrac{w-2i}{1-i}$. With $w = u+iv$: $z = \dfrac{u+i(v-2)}{1-i} = \dfrac{(u+i(v-2))(1+i)}{2} = \dfrac{u-(v-2)}{2} + i\,\dfrac{u+v-2}{2}$. Then $\mathrm{Im}(z) = 2 \iff u+v-2 = 4 \iff v = 6-u$: the line $u+v=6$.

2. $|z-1| = 1 \iff \left|\dfrac{1}{w} - 1\right| = 1 \iff |1-w| = |w|$: the perpendicular bisector of 0 and 1, i.e. $\mathrm{Re}(w) = \frac{1}{2}$. The image is a line because the original circle passes through the origin.

3. The ray $\arg z = \pi/6$ maps to the ray $\arg w = \pi/3$ (origin excluded). The circle $|z| = 3$ maps to $|w| = 9$, described twice.

4. With $z = \frac{1}{2} + iy$: $u = \dfrac{1/2}{1/4+y^2}$, $v = \dfrac{-y}{1/4+y^2}$, and $u^2+v^2 = \dfrac{1}{1/4+y^2} = 2u$. Hence $(u-1)^2+v^2 = 1$:

circle of centre 1 and radius 1, through O (excluded).

KEY POINTS **Key points**

- $w = z + b$: translation; $w = az$ with $a = re^{i\alpha}$: enlargement of ratio r composed with rotation of angle α; $w = \bar{z}$: reflection in the real axis.
- Under $w = az + b$: lines → lines, circles → circles (radius multiplied by $|a|$).
- Under $w = \frac{1}{z}$: $|w| = \frac{1}{|z|}$, $\arg w = -\arg z$; the unit circle is invariant; a line avoiding O maps to a circle through O.
- Under $w = z^2$: moduli squared, arguments doubled; circles and lines are not preserved in general.
- Möbius maps $w = \frac{az + b}{cz + d}$ send lines and circles to lines or circles.
- Method: **parametrise** → **substitute** → **eliminate** → identify, stating excluded points.

These transformations are not only geometric objects in their own right: in the next sections we shall see that the maps $w = az + b$ are exactly the direct similitudes, and that their action on points (x, y) can be written with 2×2 matrices — a bridge between complex numbers and the matrix transformations of the plane.

5 PLANE TRANSFORMATIONS AND MATRIX REPRESENTATION

In the previous section we studied how a complex function $w = f(z)$ maps points from the z-plane to the w-plane. Many of those mappings — translations, rotations, dilations, and even the Joukowski map — are built from elementary geometric operations on the plane. In this section we place those operations on a firm algebraic footing: every **linear transformation** of the plane can be represented by a 2×2 matrix acting on column vectors $\begin{pmatrix} x \\ y \end{pmatrix}$. This matrix viewpoint not only unifies the treatment of reflections, rotations, stretches and shears, but also gives us powerful tools (determinant, inverse, composition) that are indispensable in the GCE Advanced Level examination.

5.1 Linear Transformations of the Plane and Matrix Representation

DEFINITION **Linear plane transformation**

A transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is linear if for all vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ and all scalars $\lambda \in \mathbb{R}$:

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \qquad T(\lambda\mathbf{u}) = \lambda T(\mathbf{u}).$$

An immediate consequence is $T(\mathbf{0}) = \mathbf{0}$ (the origin is always fixed). Every linear transformation can be written as

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix},$$

where the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is called the **matrix of the transformation** (relative to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$). The columns of M are precisely the images of the basis vectors:

$$M\mathbf{e}_1 = \begin{pmatrix} a \\ c \end{pmatrix} = T(1, 0), \qquad M\mathbf{e}_2 = \begin{pmatrix} b \\ d \end{pmatrix} = T(0, 1).$$

EXEMPLE **Finding the matrix of a linear transformation**

Let T be the transformation that maps (x, y) to $(2x - y,\ x + 3y)$. Write T in matrix form.
The coefficients of x and y in the first component give the first row $(2, -1)$; the second component gives the second row $(1, 3)$. Hence

$$M = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}.$$

Check: $M\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}$.

WARNING **Translation is not linear**

A translation $T(x, y) = (x + a,\ y + b)$ with $(a, b) \neq (0, 0)$ does not satisfy $T(\mathbf{0}) = \mathbf{0}$, hence it is not linear and cannot be represented by a 2×2 matrix acting on $\begin{pmatrix} x \\ y \end{pmatrix}$ alone. (In homogeneous coordinates one uses a 3×3 matrix, but that is beyond the current syllabus.)

DID YOU KNOW? **Link with complex numbers**

A complex number $z = x + iy$ can be identified with the vector $\begin{pmatrix} x \\ y \end{pmatrix}$. Multiplication by a fixed complex number $w = a + ib$ corresponds to the linear transformation

$$(x + iy) \mapsto (a + ib)(x + iy) = (ax - by) + i(bx + ay) \quad \longleftrightarrow \quad \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

Matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ represent **similitudes** (rotation combined with enlargement), a key concept we will develop in the next section.

5.2 Reflections

A reflection is a linear transformation that flips the plane across a line through the origin. The most common reflection matrices are:

- Reflection in the x-axis: $M_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
- Reflection in the y-axis: $M_y = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- Reflection in the line $y = x$: $M_{y=x} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- Reflection in the line $y = (\tan\theta)\,x$ (making an angle θ with the positive x-axis):

$$M_\theta = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

EXPERIMENT **Deriving the general reflection matrix**

To reflect in the line $L_\theta : y = (\tan\theta)x$:

1. Rotate by $-\theta$ to align L_θ with the x-axis: $R(-\theta)$.
2. Reflect in the x-axis: M_x.
3. Rotate back by θ: $R(\theta)$.

The composite matrix is $M_\theta = R(\theta)M_xR(-\theta)$. Using $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $M_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$,

compute:

$$M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

This derivation is examinable and explains the double-angle formulas.

PROPERTY **Properties of reflection matrices**

For any reflection matrix M:

1. $M^2 = I$ (applying the reflection twice returns the original point).
2. $\det M = -1$ (the orientation is reversed; area is preserved).
3. M is symmetric ($M^T = M$) and orthogonal ($M^T M = I$).

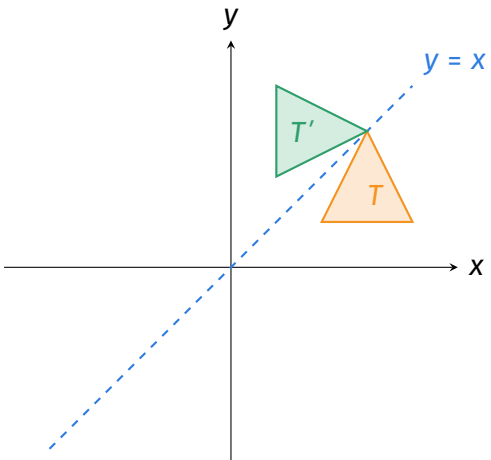
These properties are examinable; you may be asked to verify $M^2 = I$ or to find θ given a matrix of the form $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ with $a^2 + b^2 = 1$.

EXEMPLE **Verifying $M^2 = I$ for reflection in $y = x$**

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$M^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Also $\det M = -1$, confirming the orientation reversal.



Matrix of reflection in $y = x$: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Reflection of a triangle in the line $y = x$. The matrix swaps the coordinates.

5.3 Rotations about the Origin

A rotation through an angle θ (anticlockwise) about the origin is a linear transformation with matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

PROPERTY **Properties of rotation matrices**

For any rotation matrix $R(\theta)$:

1. $\det R(\theta) = \cos^2\theta + \sin^2\theta = 1$ (area and orientation are preserved).
2. $R(\theta)^{-1} = R(-\theta) = R(\theta)^T$ (the inverse is the transpose).
3. $R(\alpha)R(\beta) = R(\alpha + \beta)$ (rotations add angles).

EXEMPLE **Rotation by 60°**

$\theta = 60^\circ = \frac{\pi}{3}$, $\cos\frac{\pi}{3} = \frac{1}{2}$, $\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

$$R\left(\frac{\pi}{3}\right) = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

The image of $(1, 0)$ is $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$; the image of $(0, 1)$ is $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

5.4 Enlargements (Dilations) and Stretches

An **enlargement** (or dilation) centred at the origin with scale factor k multiplies every vector by k. Its matrix is $kI = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$.

A **stretch** parallel to the coordinate axes uses different scale factors a (in the x-direction) and b (in the y-direction):

$$S = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

If $a, b > 0$ the transformation preserves orientation ($\det S = ab > 0$); if one factor is negative, a reflection is combined with the stretch. If $a = b$, the stretch reduces to an enlargement.

EXEMPLE **Stretch with $a = 2$, $b = 1$**

$S = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. The unit square $[0,1] \times [0,1]$ becomes a 2×1 rectangle. Area is multiplied by $\det S = 2$.

5.5 Shears

A **shear** parallel to the x-axis with factor k fixes the x-axis and shifts points horizontally by an amount proportional to their y-coordinate:

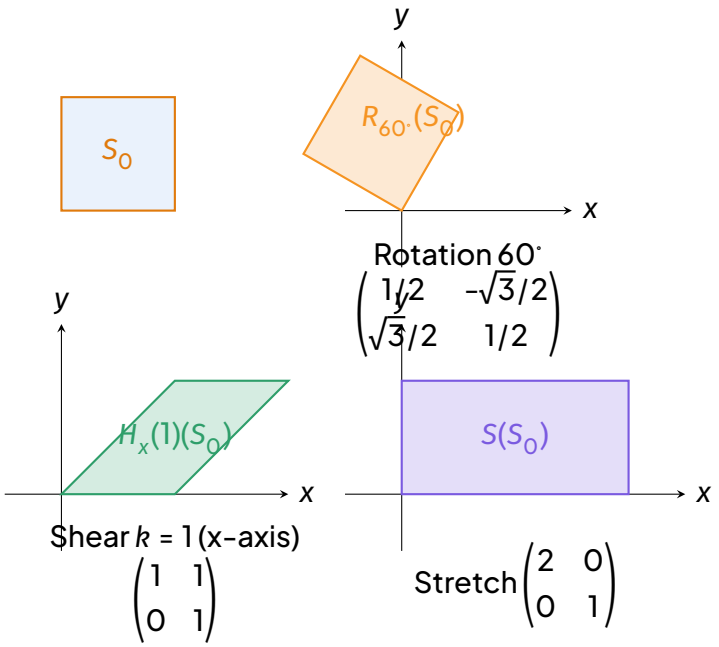
$$H_x(k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + ky \\ y \end{pmatrix}.$$

A shear parallel to the y-axis with factor k fixes the y-axis and shifts points vertically by an amount proportional to their x-coordinate:

$$H_y(k) = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}, \qquad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ kx + y \end{pmatrix}.$$

PROPERTY **Effect of a shear**

- $\det H_x(k) = \det H_y(k) = 1$; area is preserved.
- The unit square becomes a parallelogram of the same area.
- A shear is not an isometry (distances and angles change).



Images of the unit square under a rotation of 60°, a shear with $k = 1$ (parallel to x-axis), and a stretch diag(2, 1).

5.6 Composite Transformations

If a transformation T_1 with matrix A is followed by T_2 with matrix B, the combined transformation $T = T_2 \circ T_1$ has matrix BA (note the reverse order: the matrix of the first transformation appears on the right).

METHOD **Matrix of a composite transformation**

To find the matrix of the transformation "apply T_1 then T_2":

1. Write the matrices M_1 (for T_1) and M_2 (for T_2).
2. Multiply in the order $M = M_2 M_1$ (rightmost applied first).
3. The resulting matrix M represents the composite transformation.

WARNING **Order of multiplication**

A very common mistake is to multiply in the order $M_1 M_2$. Remember: if you apply T_1 then T_2, the matrix is $M_2 M_1$. In function notation $(T_2 \circ T_1)(\mathbf{x}) = T_2(T_1(\mathbf{x})) = M_2(M_1\mathbf{x}) = (M_2 M_1)\mathbf{x}$.

EXEMPLE **Rotation then shear**

Rotate by 90° ($R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$), then apply shear $H_x(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Composite matrix:

$$M = H_x(1)\,R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

If we reversed the order (shear then rotation) we would get $R\,H_x(1) = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \neq M$, showing non-commutativity.

EXEMPLE **Reflection then rotation**

Reflect in the line $y = x$ ($M_{y=x} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$), then rotate by 90° ($R_{90} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$). Composite matrix:

$$M = R_{90}M_{y=x} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = M_y.$$

The composite is a reflection in the y-axis. This illustrates that the product of a reflection (det -1) and a rotation (det 1) is a reflection (det -1).

5.7 Determinant and Area Scale Factor

PROPERTY **Determinant as signed area scale factor**

Let T be a linear transformation with matrix M. For any region R in the plane with area A, the area of the image $T(R)$ is $|\det M| \cdot A$. The sign of $\det M$ indicates whether orientation is preserved (+) or reversed (−).

KEY POINTS **Key facts about determinants**

- $\det(kI) = k^2$ (enlargement by k multiplies area by k^2).
- $\det\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} = ab$ (stretch).

- $\det H_x(k) = \det H_y(k) = 1$ (shear preserves area).
- $\det R(\theta) = 1$ (rotation preserves area and orientation).
- $\det M_{\text{reflection}} = -1$ (reflection preserves area but reverses orientation).
- If $\det M = 0$, the transformation is **singular**: it collapses the plane onto a line or a point (e.g., projection onto an axis). Such a transformation has no inverse.

5.8 Inverse Transformations and Identifying a Transformation from its Matrix

If $\det M \neq 0$, the transformation is invertible and its inverse has matrix $M^{-1} = \frac{1}{\det M}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ for $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

METHOD **Identifying a transformation from its matrix (exam technique)**

Given a 2×2 matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

1. Compute $\det M$.
2. If $\det M = 0$: singular transformation (projection or collapse).
3. If $\det M > 0$ and $M^TM = (\det M)I$ (i.e. columns are orthogonal and have same length $\sqrt{\det M}$): it is a similitude (rotation + enlargement). The scale factor is $\sqrt{\det M}$, and the rotation angle θ satisfies $\cos\theta = \frac{a}{\sqrt{\det M}}$, $\sin\theta = \frac{c}{\sqrt{\det M}}$.
4. If $\det M = 1$ and $M^TM = I$: it is a rotation (find θ from $\cos\theta = a$, $\sin\theta = c$).
5. If $\det M = -1$ and $M^TM = I$: it is a reflection (find the line of fixed points: solve $M\mathbf{x} = \mathbf{x}$).
6. If M is diagonal $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$: it is a stretch (or enlargement if $a = b$).
7. If $M = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$: it is a shear.
8. If M is a product of the above, decompose it (e.g., by recognising standard forms or using polar decomposition) — but in exams you usually recognise standard forms directly.

WORKED EXAMPLE 9 **Identifying a transformation**

The matrix $M = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ represents a single geometric transformation. Identify it fully.

SOLUTION

Solution:

1. $\det M = (0)(0) - (-1)(-1) = -1$. Orientation is reversed.
2. $M^T M = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. So M is orthogonal.
3. An orthogonal matrix with determinant -1 is a reflection.
4. The line of reflection is the set of fixed points: $M\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow x = -y$.

   Hence the line is $y = -x$ (or $\theta = -45^\circ$).

Answer: Reflection in the line $y = -x$.

WORKED EXAMPLE 10 **Identifying a similitude**

Identify the transformation represented by $M = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$.

SOLUTION

Solution:

1. $\det M = 4 - (-4) = 8 > 0$.
2. $M^T M = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = 8I = (\det M)I$.
3. Columns are orthogonal and have length $\sqrt{8} = 2\sqrt{2}$.
4. Scale factor $k = \sqrt{\det M} = 2\sqrt{2}$.
5. Rotation angle: $\cos\theta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$, $\sin\theta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$.

Answer: Enlargement scale factor $2\sqrt{2}$ followed by rotation 45° anticlockwise (or equivalently, multiplication by $2\sqrt{2}e^{i\pi/4} = 2 + 2i$ in the complex plane).

EXERCISE 5 **Practice**

1. Find the matrix of the transformation that reflects in the line $y = \sqrt{3}\,x$ (i.e. $\theta = 60^\circ$). Verify that $M^2 = I$ and $\det M = -1$.

2. A triangle has vertices $A(1,2)$, $B(3,1)$, $C(2,4)$. It undergoes a shear $H_y(2)$ followed by a rotation of -90°. Find the coordinates of the image vertices.

3. The matrix $M = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$ represents a reflection combined with an enlargement. Determine the scale factor of the enlargement and the line of reflection.

4. Show that the transformation with matrix $\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$ is singular. Describe its geometric effect.

5. Find the matrix of the composite transformation: stretch $S = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ followed by rotation $R(90^\circ)$. Is the result a similitude? Justify.

SOLUTION **Exercise 1**

The line makes angle $\theta = 60^\circ$, so $2\theta = 120^\circ$. $\cos 120^\circ = -\tfrac{1}{2}$, $\sin 120^\circ = \tfrac{\sqrt{3}}{2}$. $M = \begin{pmatrix} -\tfrac{1}{2} & \tfrac{\sqrt{3}}{2} \\ \tfrac{\sqrt{3}}{2} & \tfrac{1}{2} \end{pmatrix}$.

$M^2 = \begin{pmatrix} \tfrac{1}{4}+\tfrac{3}{4} & -\tfrac{\sqrt{3}}{4}+\tfrac{\sqrt{3}}{4} \\ -\tfrac{\sqrt{3}}{4}+\tfrac{\sqrt{3}}{4} & \tfrac{3}{4}+\tfrac{1}{4} \end{pmatrix} = I$, $\det M = -\tfrac{1}{4} - \tfrac{3}{4} = -1$.

SOLUTION **Exercise 2**

$H_y(2) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, $R(-90^\circ) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Composite $M = R(-90^\circ)H_y(2) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$.

Apply to each vertex: $A' = M\binom{1}{2} = \binom{2\cdot1+2}{-1} = \binom{4}{-1}$, $B' = M\binom{3}{1} = \binom{6+1}{-3} = \binom{7}{-3}$, $C' = M\binom{2}{4} = \binom{4+4}{-2} = \binom{8}{-2}$.

SOLUTION **Exercise 3**

$\det M = -9 - 16 = -25 = -5^2$. The matrix is 5 times an orthogonal matrix: $M = 5\begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix}$.

The orthogonal part has determinant -1, so it is a reflection. The line of reflection satisfies $\begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \frac{3}{5}x + \frac{4}{5}y = x \Rightarrow 4y = 2x \Rightarrow y = \frac{1}{2}x$. Enlargement scale factor = 5.

SOLUTION **Exercise 4**

$\det = 4 - 4 = 0$, singular. The matrix has rank 1; its columns are $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, which are linearly dependent (second is half the first). The transformation maps the whole plane onto the line spanned by $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (equation $y = 2x$). It is a projection onto that line combined with a scaling (the image of (x, y) is $(2x + y)\begin{pmatrix} 1 \\ 2 \end{pmatrix}$).

SOLUTION **Exercise 5**

$R(90^\circ) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Composite $M = R(90^\circ)S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix}$. $\det M = 0 - (-6) = 6 > 0$. $M^TM = \begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}\begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \neq 6I$. The columns are orthogonal but have different lengths (2 and 3). Hence $M^TM \neq (\det M)I$. Answer: The result is not a similitude; it is a rotation combined with a stretch (different scale factors in perpendicular directions), which distorts angles.

6 SIMILITUDES: COMPLEX FORM, MATRIX INTERPRETATION AND APPLICATIONS

In the previous section we studied plane transformations through matrices: rotations, reflections, enlargements, stretches and shears, all described by 2 × 2 matrices acting on position vectors. We now single out a particularly beautiful family of transformations, the **similitudes**, which combine a rotation, an enlargement and possibly a translation or reflection into a single geometric object. The extraordinary fact is that complex numbers describe these transformations with a single formula: $z' = az + b$ (or $z' = a\bar{z} + b$). This is one of the most elegant meetings of algebra and geometry in the whole Further Mathematics syllabus, and it is a favourite of GCE Advanced Level examiners.

6.1 Definition and First Examples

EXPERIMENT **Growing a map without distorting it**

Imagine a photocopier set to 150%. The map of Cameroon it produces is bigger, but every angle is unchanged: the angle between two roads is the same on the copy as on the original, and the ratio of two distances is preserved. A transformation with this behaviour — all distances multiplied by the same positive constant — is exactly what we call a similitude.

DEFINITION **Similitude and ratio**

A **similitude** of the plane is a transformation s for which there exists a fixed real number $k > 0$, called the **ratio** of the similitude, such that for all points M, N with images $M' = s(M)$, $N' = s(N)$,

$$M'N' = k \cdot MN.$$

If $k = 1$ the similitude is an **isometry** (it preserves distances). If $k > 1$ it enlarges; if $0 < k < 1$ it contracts.

Rotations, translations and reflections (ratio 1), and homotheties (enlargements) of ratio k, are all similitudes. The new phenomenon is that we may combine a rotation and an enlargement about the same centre: this is still a similitude, and it is the typical case.

6.2 Complex Form of a Similitude

PROPERTY **Complex form of similitudes**

Every similitude of the plane has one of the following two complex forms:

- Direct similitude (preserves orientation): $z' = az + b$, with $a, b \in \mathbb{C}$, $a \neq 0$;
- Indirect (opposite) similitude (reverses orientation): $z' = a\bar{z} + b$, with $a, b \in \mathbb{C}$, $a \neq 0$.

Conversely, every map of one of these forms is a similitude of ratio $k = |a|$. (The derivation below is examinable in spirit: you must be able to justify that $z' = az+b$ multiplies distances by $|a|$.)

Justification. Let z_1, z_2 be the affixes of M, N and z_1', z_2' those of their images.

- For a direct similitude $z' = az + b$:

$$|z_1' - z_2'| = |(az_1 + b) - (az_2 + b)| = |a(z_1 - z_2)| = |a|\,|z_1 - z_2|,$$

so $M'N' = |a| \cdot MN$.

- For an indirect similitude $z' = a\bar{z} + b$:

$$|z_1' - z_2'| = |a\bar{z}_1 + b - (a\bar{z}_2 + b)| = |a(\bar{z}_1 - \bar{z}_2)| = |a|\,|\bar{z}_1 - \bar{z}_2| = |a|\,|z_1 - z_2|,$$

since conjugation preserves moduli ($|\bar{z}| = |z|$).

In both cases all distances are multiplied by $k = |a|$. ■

KEY POINTS **Reading a direct similitude**

For the direct similitude $z' = az + b$, write $a = re^{i\theta}$ $(r > 0)$. Then:

- the **ratio** is $k = |a| = r$;
- the **angle** of the similitude is $\theta = \arg(a)$ (defined modulo 2π);
- geometrically, s is the composite of the homothety of ratio r and the rotation of angle θ, about the same centre.

6.3 Classification of Direct Similitudes

The nature of $z' = az + b$ depends entirely on a:

Condition on a	Transformation	Fixed points
$a = 1, b \neq 0$	Translation by vector of affix b	none
$a = 1, b = 0$	Identity	all points
$\|a\| = 1, a \neq 1$	Rotation of angle $\arg a$	one (the centre)
$a \in \mathbb{R}, a > 0, a \neq 1$	Homothety (enlargement) of ratio a	one (the centre)
general $a \neq 1$	Rotation of angle $\arg a$ **and** homothety of ratio $\|a\|$ about the same centre	one (the centre)

METHOD **Centre, ratio and angle of $z' = az + b$ $(a \neq 1)$**

1. Ratio: compute $k = |a|$.
2. Angle: compute $\theta = \arg a$ (modulo 2π).
3. Centre: the centre Ω is the unique **fixed point**; solve

$$z = az + b \iff z_0 = \frac{b}{1 - a}.$$

4. Conclusion: state "s is the similitude of centre $\Omega(z_0)$, ratio k and angle θ", i.e. the composite of the homothety $h(\Omega, k)$ and the rotation $r(\Omega, \theta)$ (these two commute).

WARNING **The centre is not the origin!**

The most frequent error is to say that $z' = az+b$ is "a rotation about the origin". The rotation part acts about the centre Ω, the fixed point obtained by solving $z = az+b$. Only when $b = 0$ is the centre the origin. Always compute $z_0 = \dfrac{b}{1-a}$ first.

WORKED EXAMPLE 11 **Identifying a similitude**

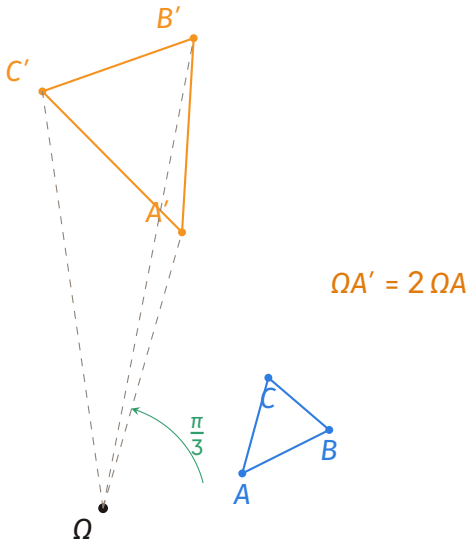
Characterise the transformation $s : z' = (1+i)z + (2-i)$.

SOLUTION

Solution. Here $a = 1+i$ and $b = 2-i$.
Ratio: $k = |a| = \sqrt{1^2+1^2} = \sqrt{2}$.
Angle: $\theta = \arg(1+i) = \dfrac{\pi}{4}$.
Centre: solve $z = (1+i)z + 2 - i$:

$$z - (1+i)z = 2 - i \iff -iz = 2 - i \iff z = \frac{2-i}{-i} = \frac{(2-i)\,i}{-i\cdot i} = \frac{2i+1}{1} = 1+2i.$$

Conclusion: s is the direct similitude of centre $\Omega(1+2i)$, ratio $\sqrt{2}$ and angle $\dfrac{\pi}{4}$: the composite of the rotation of centre Ω and angle $\pi/4$ with the homothety of centre Ω and ratio $\sqrt{2}$.



Direct similitude of centre Ω, ratio 2 and angle $\pi/3$: triangle ABC is mapped to the similar triangle $A'B'C'$, twice as large and rotated by 60° about Ω.

6.4 Indirect Similitudes

An **indirect similitude** has the form $z' = a\bar{z} + b$. The conjugation $z \mapsto \bar{z}$ is the reflection in the real axis; composing it with a direct similitude reverses orientation.

EXEMPLE **Special cases of indirect similitudes**

- $z' = \bar{z}$: reflection in the real axis.
- $z' = e^{i\theta}\bar{z}$: reflection in the line through O making angle $\theta/2$ with the real axis.
- $z' = \bar{z} + b$ with b real and non-zero: **glide reflection** — reflection in the real axis followed by a translation parallel to it.
- $z' = a\bar{z} + b$ with $|a| \neq 1$: a reflection (or glide reflection) combined with an enlargement of ratio $|a|$.

WARNING **Direct or indirect?**

Do not confuse the two families. Direct similitudes $z' = az + b$ preserve orientation: a triangle labelled anticlockwise keeps its anticlockwise labelling. Indirect similitudes $z' = a\bar{z} + b$ reverse orientation, because of the conjugation. In an examination, the presence of $\bar{z}$ in the formula is the unmistakable signature of an indirect similitude.

6.5 Matrix Interpretation

Let $a = p + iq = re^{i\theta}$ and write $z = x + iy$, $z' = x' + iy'$. Expanding $z' = az + b$ with $b = b_1 + ib_2$:

$$x' + iy' = (p + iq)(x + iy) + b_1 + ib_2 = (px - qy + b_1) + i(qx + py + b_2).$$

Hence, in matrix form,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

PROPERTY **Similarity matrices**

A matrix of the form

$$M = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}, \qquad (p, q) \neq (0, 0),$$

is called a **similarity matrix**. Writing $r = \sqrt{p^2 + q^2}$ and $\theta = \arg(p + iq)$, it factorises as

$$M = r \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = rR(\theta),$$

i.e. a rotation matrix scaled by r. Thus $z' = az + b$ corresponds exactly to the matrix map $\vec{x} \mapsto rR(\theta)\vec{x}+\vec{b}$: a rotation, an enlargement of ratio r, then a translation. Note $\det M = p^2+q^2 = r^2 > 0$, confirming that direct similitudes preserve orientation.
For an indirect similitude $z' = a\bar{z} + b$ with $a = p + iq$, we have

$$x' + iy' = (p + iq)(x - iy) + b_1 + ib_2 = (px + qy + b_1) + i(qx - py + b_2),$$

giving the matrix form

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

The determinant is $-p^2 - q^2 = -r^2 < 0$, confirming orientation reversal.

EXEMPLE **From complex form to matrix**

For $s : z' = (1 + i)z + (2 - i)$, we have $p = q = 1$, so

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \sqrt{2}\, R\left(\tfrac{\pi}{4}\right)\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix},$$

which matches the ratio $\sqrt{2}$ and angle $\pi/4$ found earlier.

6.6 Geometric Properties of Similitudes

PROPERTY **Conservation properties**

Every similitude of ratio k:

1. multiplies all distances by k and all areas by k^2;
2. preserves angles (in size; direct similitudes also preserve their orientation);
3. preserves ratios of distances and division of a segment (in particular it maps the midpoint of $[MN]$ to the midpoint of $[M'N']$);
4. maps lines to lines and circles to circles (a line may be viewed as a circle of infinite radius);
5. maps any figure to a similar figure: triangles are mapped to similar triangles.

(Proofs of (1)–(3) follow directly from $|z_1' - z_2'| = k|z_1 - z_2|$ and are examinable.)

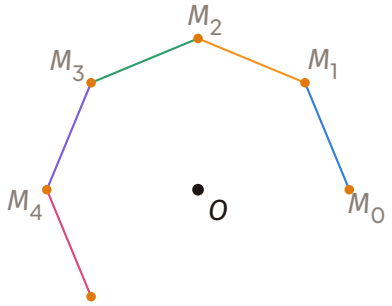
Proof that midpoints are preserved. If M, N have affixes z_1, z_2, the midpoint I has affix $\frac{z_1+z_2}{2}$. Its

image under $z' = az + b$ has affix

$$a\cdot\frac{z_1+z_2}{2}+b=\frac{(az_1+b)+(az_2+b)}{2}=\frac{z_1'+z_2'}{2},$$

which is the affix of the midpoint of $[M'N']$. ■

Proof that circles map to circles. A circle with centre z_c and radius R has equation $|z-z_c|=R$. Under $z'=az+b$ ($a\neq 0$), the image satisfies $|z'-(az_c+b)|=|a|\,|z-z_c|=|a|R$, which is a circle of centre az_c+b and radius $|a|R$. For $z'=a\bar{z}+b$, the equation becomes $|z'-(a\bar{z}_c+b)|=|a|R$, also a circle. ■



Spiral similarity: successive images of the segment $[OM_0]$ under $z'=\frac{1+i}{\sqrt{2}}z$ (ratio 1, angle $\pi/4$) trace out a rotational chain; with ratio $k\neq 1$ the points would spiral inwards ($k<1$) or outwards ($k>1$) — hence the name spiral similarity.

6.7 Composition of Similitudes and Applications

PROPERTY **Composition of similitudes**

The composite of two similitudes of ratios k_1 and k_2 is a similitude of ratio k_1k_2. For direct similitudes $s_1: z'=a_1z+b_1$ and $s_2: z'=a_2z+b_2$,

$$s_2\circ s_1:\ z'=a_2(a_1z+b_1)+b_2=a_1a_2\,z+(a_2b_1+b_2),$$

so the angles add: $\arg(a_1a_2)=\arg a_1+\arg a_2$. In particular, the composite of two rotations is a rotation (angles add) or a translation (angles sum to 0 modulo 2π), and the composite of two homotheties is a homothety or a translation.

WORKED EXAMPLE 12 **Composing two similitudes**

Let $s_1: z'=2iz+1$ and $s_2: z'=(1-i)z-2$. Determine the nature and characteristic elements of $s=s_2\circ s_1$.

SOLUTION

Solution. Compute the composite:

$$s(z) = (1-i)(2iz+1) - 2 = 2i(1-i)z + (1-i) - 2 = (2i+2)z - 1 - i.$$

So $s: z' = (2+2i)z - 1 - i$, a direct similitude with $a = 2+2i$.
Ratio: $k = |2+2i| = 2\sqrt{2}$ (check: $|2i| \cdot |1-i| = 2\sqrt{2}$ ✓).
Angle: $\theta = \arg(2+2i) = \frac{\pi}{4}$ (check: $\arg(2i) + \arg(1-i) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$ ✓).
Centre: $z_0 = \frac{b}{1-a} = \frac{-1-i}{1-2-2i} = \frac{-1-i}{-1-2i} = \frac{(-1-i)(-1+2i)}{(-1)^2+2^2} = \frac{1-2i+i+2}{5} = \frac{3-i}{5}$.
Conclusion: s is the direct similitude of centre $\Omega\left(\frac{3-i}{5}\right)$, ratio $2\sqrt{2}$ and angle $\frac{\pi}{4}$.

WORKED EXAMPLE 13 **Writing the complex expression from geometric data**

Find the complex form of the direct similitude s of centre $\Omega(2+i)$, ratio $\frac{1}{2}$ and angle $-\frac{\pi}{2}$, and determine the image of $A(3+3i)$.

SOLUTION

Solution. We need a with $|a| = \frac{1}{2}$ and $\arg a = -\frac{\pi}{2}$: $a = \frac{1}{2}e^{-i\pi/2} = -\frac{i}{2}$.
The centre is the fixed point: $z_0 = \frac{b}{1-a}$, so $b = z_0(1-a) = (2+i)\left(1+\frac{i}{2}\right) = 2+i+i+\frac{i^2}{2} = \frac{3}{2}+2i$.
Hence $s: z' = -\frac{i}{2}z + \frac{3}{2} + 2i$.
Image of A: $z'_A = -\frac{i}{2}(3+3i) + \frac{3}{2} + 2i = -\frac{3i}{2} + \frac{3}{2} + \frac{3}{2} + 2i = 3 + \frac{i}{2}$. So $A'\left(3+\frac{i}{2}\right)$.

EXEMPLE **Proving triangles similar**

To prove that triangles ABC and DEF are directly similar, exhibit a direct similitude mapping one to the other: find a, b with $z_D = az_A + b$ and $z_E = az_B + b$ (two equations, two unknowns), then check that $z_F = az_C + b$. Since similitudes preserve angles and ratios of distances, the triangles are similar with ratio $|a|$.

WARNING **Examination traps**

- Solving $z = az + b$ when $a = 1$: there is no fixed point (unless $b = 0$) — the map is a translation, not a similitude with centre.

- The ratio is $|a|$, never a itself: for $a = -3$ the ratio is 3 (and the angle is π).
- When composing, the order matters for the centre (though not for the ratio and angle): $s_2 \circ s_1$ means apply s_1 first.
- For indirect similitudes $z' = a\bar{z} + b$, fixed points satisfy $z = a\bar{z} + b$, which generally gives a whole line of fixed points (reflection) or none (glide reflection) when $|a| = 1$.

DID YOU KNOW? **Why "spiral similarity"?**

A direct similitude with $k \neq 1$ and $\theta \neq 0$ sends any point M to M' such that triangle $\Omega MM'$ always has the same shape: the angle at Ω is θ and $\frac{\Omega M'}{\Omega M} = k$. Iterating the map makes the images of a point spiral around Ω along a logarithmic spiral — the curve seen in snail shells and sunflower heads. Similitudes are also the mathematical engine behind fractals: each piece of a fractal is the image of the whole under a collection of contracting similitudes.

EXERCISE 6 **Practice**

1. Characterise (centre, ratio, angle) each similitude: (a) $z' = 3z - 4 + 2i$; (b) $z' = iz + 1 + i$; (c) $z' = (-1 + i\sqrt{3})z$.
2. Write the complex form of the similitude of centre $\Omega(1 - i)$, ratio 2, angle $\frac{\pi}{3}$, and find the image of $B(2 + i)$.
3. Let $s_1 : z' = (1 + i)z$ and $s_2 : z' = \frac{1 - i}{2}z + 3$. Determine the nature of $s_2 \circ s_1$.
4. Show that $t : z' = 2\bar{z} + 3i$ is an indirect similitude and give its ratio.

SOLUTION **Exercise 1**

(a) $a = 3$ real positive: homothety of ratio 3, centre $z_0 = \frac{-4 + 2i}{1 - 3} = 2 - i$. (b) $a = i$: $|a| = 1$, $\arg a = \pi/2$: rotation of angle $\pi/2$, centre $z_0 = \frac{1 + i}{1 - i} = \frac{(1 + i)^2}{2} = i$. (c) $a = -1 + i\sqrt{3} = 2e^{2i\pi/3}$, $b = 0$: centre O, ratio 2, angle $\frac{2\pi}{3}$.

SOLUTION **Exercise 2**

$a = 2e^{i\pi/3} = 1 + i\sqrt{3}$; $b = z_0(1 - a) = (1 - i)(-i\sqrt{3}) = -\sqrt{3} - i\sqrt{3}$. So $z' = (1 + i\sqrt{3})z - \sqrt{3}(1 + i)$. Image of B: $z'_B = (1 + i\sqrt{3})(2 + i) - \sqrt{3} - i\sqrt{3} = 2 + i + 2i\sqrt{3} - \sqrt{3} - \sqrt{3} - i\sqrt{3} = (2 - 2\sqrt{3}) + i(1 + \sqrt{3})$.

SOLUTION **Exercise 3**

$s_2 \circ s_1 : z' = \dfrac{1 - i}{2}(1 + i)z + 3 = \dfrac{2}{2}z + 3 = z + 3$. Since $a = 1$, this is the translation by the vector of affix 3 (the rotation angles $\pi/4$ and $-\pi/4$ cancel, and the ratios $\sqrt{2}$ and $1/\sqrt{2}$ multiply to 1).

SOLUTION **Exercise 4**

The formula contains $\bar{z}$, so t is an indirect similitude; its ratio is $k = |a| = |2| = 2$. It is the composite of a reflection-type map with an enlargement of ratio 2.

KEY POINTS **Key points of the section**

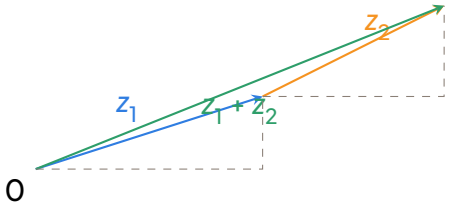
- A similitude multiplies all distances by a fixed ratio $k > 0$.
- Direct: $z' = az + b$; indirect: $z' = a\bar{z} + b$; in both cases $k = |a|$.
- For $a \neq 1$, the centre is the fixed point $z_0 = \dfrac{b}{1 - a}$; the angle is $\arg a$.
- Matrix form: $\begin{pmatrix} p & -q \\ q & p \end{pmatrix} = rR(\theta)$ plus a translation for direct; $\begin{pmatrix} p & q \\ q & -p \end{pmatrix}$ for indirect.
- Similitudes preserve angles, ratios, midpoints; they send lines to lines and circles to circles.
- Ratios multiply and angles add under composition.

7 MODULUS INEQUALITIES AND RIGID MOTIONS

In the previous section we studied similitudes, which combine a rotation, a scaling and a translation. If we remove the scaling factor (i.e. set the ratio to 1) we obtain the rigid motions or isometries of the plane — transformations that preserve distances. Before classifying them we first need a solid tool for estimating and comparing moduli: the triangle inequality and its

consequences. These inequalities are also the key to describing regions in the Argand diagram defined by conditions such as $|z - a| < r$ or $|z - a| \geq |z - b|$.

7.1 The Triangle Inequality and its Consequences



Geometric proof of the triangle inequality: the sum of two sides of a triangle is at least the third side.

PROPERTY **Triangle Inequality**

For any two complex numbers z_1 and z_2,

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

Equality holds if and only if z_1 and z_2 have the same argument (i.e. they point in the same direction), or one of them is zero.

Proof (examinable). Write $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$ with $r_1, r_2 \geq 0$. Then

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1} + \overline{z_2}) = |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1\overline{z_2}) = r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2).$$

Since $\cos(\theta_1 - \theta_2) \leq 1$, we obtain $|z_1 + z_2|^2 \leq (r_1 + r_2)^2$, hence the inequality. Equality requires $\cos(\theta_1 - \theta_2) = 1$, i.e. $\theta_1 = \theta_2 \pmod{2\pi}$.

PROPERTY **Reverse Triangle Inequality**

For any $z_1, z_2 \in \mathbb{C}$,

$$\big||z_1| - |z_2|\big| \leq |z_1 - z_2|.$$

Proof. Apply the triangle inequality to $z_1 = (z_1 - z_2) + z_2$ and to $z_2 = (z_2 - z_1) + z_1$:

$$|z_1| = |(z_1 - z_2) + z_2| \leq |z_1 - z_2| + |z_2| \implies |z_1| - |z_2| \leq |z_1 - z_2|.$$

Swapping z_1 and z_2 gives $|z_2| - |z_1| \leq |z_1 - z_2|$. Combining both yields the result.

KEY POINTS **Key Uses of the Triangle Inequalities**

- Bounding a sum: $|z_1 + z_2| \le |z_1| + |z_2|$ (upper bound).
- Bounding a difference from below: $|z_1 - z_2| \ge \big||z_1| - |z_2|\big|$ (lower bound).
- Estimating moduli of polynomials: if $|z| = R$ then $|a_n z^n + \cdots + a_0| \le |a_n|R^n + \cdots + |a_0|$ (useful for locating roots — a GCE-useful extra).
- Error regions: a GPS receiver in Douala gives a position z_0 with accuracy r; the true position lies in the disc $|z - z_0| \le r$.

WARNING **Common Mistake**

$|z_1 + z_2| = |z_1| + |z_2|$ is not true in general. It holds only when z_1 and z_2 have exactly the same argument (or one is zero). Always check the arguments before claiming equality.

7.2 Solving Modulus Inequalities in the Argand Diagram

The inequalities $|z - a| < r$, $|z - a| \ge r$, $|z - a| \le |z - b|$, and double inequalities $r_1 < |z - a| \le r_2$ define discs, half-planes, annuli and their combinations. The boundary is always a circle or a line (the perpendicular bisector of a and b for $|z - a| = |z - b|$).

METHOD **Sketching the Solution Region of a Modulus Inequality**

1. Identify the geometric locus of the corresponding equality (circle, line, or ray).
2. Choose a test point not on the boundary (often the origin, if it is not on the boundary) and substitute it into the inequality.
3. Shade the side containing the test point if it satisfies the inequality; otherwise shade the opposite side.
4. For a system of inequalities, the solution region is the intersection of the individual shaded regions.
5. Indicate whether boundaries are included (solid line) or excluded (dashed line).

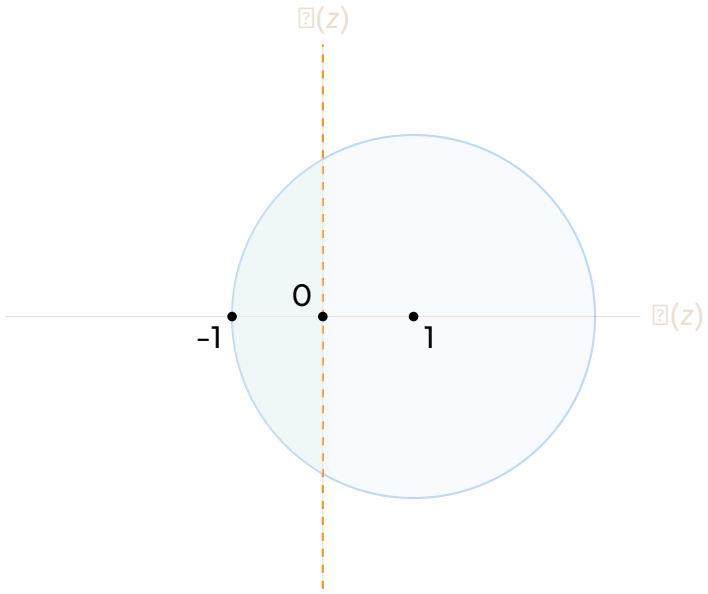
WORKED EXAMPLE 14 **System of Modulus Inequalities**

Sketch the region defined by

$$|z-1| \le 2 \quad \text{and} \quad |z-1| \ge |z+1|.$$

Solution.

- $|z-1| \le 2$ is the closed disc centred at 1 with radius 2. Boundary: solid circle.
- $|z-1| \ge |z+1|$ means the distance to 1 is at least the distance to −1. The boundary $|z-1| = |z+1|$ is the perpendicular bisector of 1 and −1, i.e. the imaginary axis $\Re(z) = 0$. Test $z = 2$: $|2-1| = 1$, $|2+1| = 3$, so $1 \ge 3$ is false. Hence the region is the left half-plane $\Re(z) \le 0$ (including the axis).
- The intersection is the part of the disc $|z-1| \le 2$ that lies on or to the left of the imaginary axis.



Solution region of $|z-1| \le 2$ and $|z-1| \ge |z+1|$ (shaded).

7.3 Rigid Motions (Isometries) of the Plane

DEFINITION **Rigid Motion / Isometry**

A transformation $T : \mathbb{C} \to \mathbb{C}$ is a rigid motion (or isometry) if it preserves distances:

$$|T(z_1) - T(z_2)| = |z_1 - z_2| \qquad \forall\, z_1, z_2 \in \mathbb{C}.$$

Every isometry of the plane can be written in one of two complex forms:

PROPERTY **Complex Form of Plane Isometries**

Let T be an isometry of $\mathbb{C}$. Then there exist $\theta \in \mathbb{R}$ and $b \in \mathbb{C}$ such that either

$$\text{(direct)} \quad T(z) = e^{i\theta}z + b \qquad \text{or} \qquad \text{(indirect)} \quad T(z) = e^{i\theta}\overline{z} + b.$$

Proof sketch. An isometry fixes the origin after a translation, so we may assume $T(0) = 0$. Then $|T(z)| = |z|$ and $|T(z) - T(w)| = |z - w|$. Using the polarization identity one shows T is $\mathbb{R}$-linear and preserves the inner product, hence its matrix is orthogonal. In complex notation orthogonal matrices correspond to multiplication by $e^{i\theta}$ (determinant +1, orientation preserving) or multiplication by $e^{i\theta}$ followed by conjugation (determinant -1, orientation reversing). Adding back the translation gives the stated forms.

DEFINITION **Orientation**

A direct isometry (determinant +1) preserves orientation; an indirect isometry (determinant -1) reverses orientation. Equivalently, a direct isometry maps a positively oriented triangle to a positively oriented triangle.

7.4 Classification of Plane Isometries

PROPERTY **Classification of Plane Isometries**

Every isometry of the plane is exactly one of the following four types:

1. Translation: $z \mapsto z + b$ ($b \neq 0$). Direct, no fixed points.
2. Rotation: $z \mapsto e^{i\theta}z + b$ with $\theta \not\equiv 0 \pmod{2\pi}$. Direct, exactly one fixed point (the centre).
3. Reflection: $z \mapsto e^{i\theta}\overline{z} + b$ where the translation component b is orthogonal to the direction $e^{i\theta/2}$ (i.e. b has no component parallel to the mirror axis). Indirect, fixed points form a line (the mirror axis).
4. Glide Reflection: $z \mapsto e^{i\theta}\overline{z} + b$ where the translation component b has a non-zero component parallel to the mirror axis. Indirect, no fixed points.

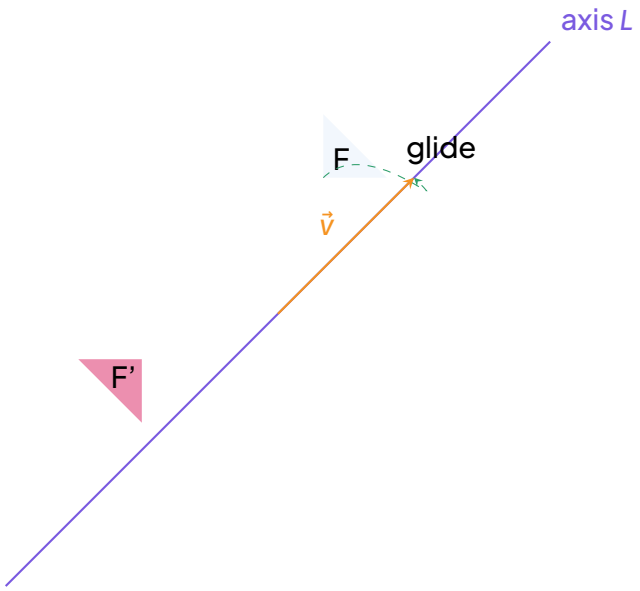
PROPERTY **Characterisation of Direct Isometries**

A direct isometry with a fixed point is a rotation. Indeed, if $T(z) = e^{i\theta}z + b$ and $T(z_0) = z_0$, then $b = (1 - e^{i\theta})z_0$, so $T(z) = e^{i\theta}(z - z_0) + z_0$, a rotation about z_0 by angle θ.

PROPERTY **Characterisation of Indirect Isometries**

An indirect isometry $T(z) = e^{i\theta}\overline{z} + b$:

- If b is orthogonal to the direction $e^{i\theta/2}$ (i.e. $\Re(be^{-i\theta/2}) = 0$), then T is a reflection in a line.
- Otherwise T is a glide reflection: a reflection in a line followed by a translation parallel to that line.



Glide reflection: reflection in axis L followed by translation $\vec{v}$ parallel to L. The flag changes orientation (F → F').

KEY POINTS **Composition Rules for Isometries**

- Direct ∘ Direct = Direct (translation or rotation).
- Direct ∘ Indirect = Indirect (reflection or glide reflection).
- Indirect ∘ Direct = Indirect.
- Indirect ∘ Indirect = Direct.

7.5 Worked Example: Identifying an Isometry

WORKED EXAMPLE 15 **Identify the Isometry**

Determine the type of the transformation $T(z) = i\bar{z} + 2 - i$.

Solution.

The map has the form $e^{i\theta}\bar{z} + b$ with $e^{i\theta} = i$ ($\theta = \pi/2$) and $b = 2 - i$. It is indirect. The pure reflection part is $z \mapsto i\bar{z}$, which is reflection in the line through the origin making an angle $\theta/2 = \pi/4$ with the real axis (the line $y = x$). The direction vector of this axis is $e^{i\pi/4} = \frac{1}{\sqrt{2}}(1 + i)$.

Decompose $b = 2 - i$ into components parallel and perpendicular to the axis:

$$b_{\parallel} = ℜ(be^{-i\pi/4})e^{i\pi/4}, \qquad b_{\perp} = b - b_{\parallel}.$$

Compute $be^{-i\pi/4} = (2 - i)\frac{1}{\sqrt{2}}(1 - i) = \frac{1}{\sqrt{2}}(2 - 2i - i + i^2) = \frac{1}{\sqrt{2}}(1 - 3i)$. Hence $ℜ(be^{-i\pi/4}) = 1/\sqrt{2} \neq 0$, so there is a non-zero parallel component. Therefore T is a **glide reflection**.

The glide vector is $b_{\parallel} = \frac{1}{\sqrt{2}}e^{i\pi/4} = \frac{1}{2} + \frac{1}{2}i$. The axis of the glide reflection is the line parallel to $e^{i\pi/4}$ that is mapped onto itself. Solving $T(z) \in L$ for $z \in L$ with $L : y = x + c$ gives $c = -3/2$. Thus the axis is the line $y = x - \frac{3}{2}$ (or $ℜ(z) = ℜ(z) - \frac{3}{2}$).

7.6 Common Mistakes and Examination Tips

WARNING **Pitfalls in Modulus Inequalities**

- Forgetting to test a point when the boundary is a line (the origin may lie on the line).
- Confusing $|z - a| < r$ (open disc) with $|z - a| \leq r$ (closed disc) — use dashed vs solid circles.
- Misapplying the triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$ gives an upper bound; the reverse inequality gives a lower bound for $|z_1 - z_2|$.

WARNING **Pitfalls in Rigid Motions**

- Writing a rotation as $z \mapsto e^{i\theta}z$ without the translation part when the centre is not the origin.
- Confusing the condition for a glide reflection: the translation vector must have a component parallel to the reflection axis (direction $e^{i\theta/2}$), not just any non-zero translation.
- Forgetting that the composition of two reflections in parallel lines is a translation, while in intersecting lines it is a rotation.

DID YOU KNOW? **Did You Know?**

The classification of plane isometries into four types (translation, rotation, reflection, glide reflection) is a classic result of Euclidean geometry. In crystallography, the symmetry groups of wallpaper patterns are built from these four isometries combined with discrete translations. The famous 17 wallpaper groups are exactly the distinct ways to tile the plane with a repeating motif using only rigid motions.

EXERCISE 7 **Practice**

1. Solve and sketch: $|z - 2i| < 3$ and $|z - 2i| \geq |z + 2|$.
2. Show that the transformation $T(z) = -\overline{z} + 3 + 4i$ is a glide reflection. Find its axis and glide vector.
3. Prove that the composition of a rotation by α about a and a rotation by β about b is a rotation by $\alpha + \beta$ about some point c (unless $\alpha + \beta \equiv 0 \pmod{2\pi}$, in which case it is a translation).
4. A GPS receiver in Yaoundé gives a position $z_0 = 3 + 4i$ (in km) with an error radius of 0.5 km. Write the inequality describing the possible true positions and sketch the region.

SOLUTION **Exercise 1**

1. $|z - 2i| < 3$ is the open disc centre $2i$ (point $(0, 2)$), radius 3. Boundary: dashed circle.
2. $|z - 2i| \geq |z + 2|$: the boundary is the perpendicular bisector of $2i$ and -2. Points: $2i = (0, 2)$, $-2 = (-2, 0)$. Midpoint $= (-1, 1)$. Slope of segment $= 1$, so perpendicular slope $= -1$. Equation: $y - 1 = -(x + 1) \implies y = -x$. This is the line $\Re(z) = -\Im(z)$ (or $\arg z = -\pi/4, 3\pi/4$). Test $z = 0$: $|0 - 2i| = 2$, $|0 + 2| = 2$, so 0 satisfies the inequality (equality). The region is the closed half-plane containing the origin, i.e. the side of the line $y = -x$ where $y \geq -x$ (or $\Re(z) + \Im(z) \geq 0$).
3. The intersection is the part of the open disc $|z - 2i| < 3$ that lies on or above the line $y = -x$.

SOLUTION **Exercise 2**

$T(z) = -\overline{z} + 3 + 4i = e^{i\pi}\overline{z} + 3 + 4i$. Here $\theta = \pi$, so the pure reflection is in the line through the origin with direction $e^{i\pi/2} = i$ (the imaginary axis, $x = 0$). The translation vector is $b = 3 + 4i$.

Decompose b relative to the axis direction i (vertical): Parallel component (vertical): $4i$.
Perpendicular component (horizontal): 3.
Since the parallel component $4i \neq 0$, T is a glide reflection.
The axis is a vertical line $x = c$ mapped onto itself. For $z = c + iy$, $T(z) = -c + 3 + i(y + 4)$. This lies on $x = c$ iff $-c + 3 = c \implies c = 1.5$. So the axis is the line $x = 1.5$ (or $\Re(z) = 1.5$).
The glide vector is the parallel component $4i$ (translation of 4 units upward along the axis).

SOLUTION **Exercise 3**

Let $R_a(z) = e^{i\alpha}(z - a) + a$ and $R_b(z) = e^{i\beta}(z - b) + b$. Then $R_b(R_a(z)) = e^{i\beta}\left(e^{i\alpha}(z - a) + a - b\right) + b = e^{i(\alpha+\beta)}z + \left[e^{i\beta}(a - b) - e^{i\alpha}a + b\right]$. The linear part is $e^{i(\alpha+\beta)}$.

- If $\alpha + \beta \not\equiv 0 \pmod{2\pi}$, then $e^{i(\alpha+\beta)} \neq 1$. The transformation has a unique fixed point c satisfying $c = e^{i(\alpha+\beta)}c + K$, where K is the constant bracket. Solving gives $c = \frac{K}{1-e^{i(\alpha+\beta)}}$. Hence it is a rotation by $\alpha + \beta$ about c.
- If $\alpha + \beta \equiv 0 \pmod{2\pi}$, then $e^{i(\alpha+\beta)} = 1$ and the transformation reduces to $z \mapsto z + K$, a translation by vector K.

SOLUTION **Exercise 4**

The true position z satisfies $|z - (3 + 4i)| \leq 0.5$. This is a closed disc centred at $(3, 4)$ with radius 0.5 km.

8 INTEGRATION ACTIVITY

8.1 Integration Activity: Camtel-Sud Relay Network

Aims of the activity

This integration activity brings together the whole chapter in a single modelling task. By the end of it, you should be able to:

- convert complex numbers between **Cartesian**, **polar** and **exponential** forms and plot affixes in an Argand diagram;
- use **De Moivre's theorem** to find all the n-th roots of a complex number and interpret them geometrically;
- describe and shade **loci** defined by modulus conditions (circles, annuli, perpendicular bisectors);
- recognise a **direct similitude** from its complex form $z' = az + b$, identify its ratio, angle and centre, and write its **matrix**;
- apply **modulus inequalities** (triangle inequality) to justify bounds in a concrete situation;
- classify a **rigid motion** as orientation-preserving or orientation-reversing;
- communicate a complete mathematical argument, as required at the GCE Advanced Level.

KEY POINTS **Key skills assessed**

- De Moivre's theorem: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ and n-th roots.
- Loci: $|z-a| = r$ (circle), $|z-a| \le r$ (disc), $|z-a| = |z-b|$ (perpendicular bisector).
- Direct similitude $z' = az + b$: ratio $|a|$, angle $\arg a$, centre $\frac{b}{1-a}$ (if $a \neq 1$).
- Matrix of $z \mapsto az + \beta\bar{z}$; for $z \mapsto az$ (direct) it is $\begin{pmatrix} \Re a & -\Im a \\ \Im a & \Re a \end{pmatrix}$.
- Rigid motion: isometry $\Leftrightarrow |a| = 1$; orientation-preserving $\Leftrightarrow \det > 0$ (direct), orientation-reversing $\Leftrightarrow \det < 0$ (opposite).

Situation

A telecommunications company, Camtel-Sud, installs a relay mast at a point Ω on a hill in Yaoundé. The surrounding region is modelled by the complex plane (the **Argand diagram**), with Ω at the origin; distances are measured in kilometres, the positive real axis pointing East and the positive imaginary axis pointing North.

Three relay stations A, B and C are placed so that their affixes z_A, z_B, z_C are exactly the three **cube roots of** 8, i.e. the solutions of $z^3 = 8$.

The mast carries a **rotating directional antenna**. Each time the control system sends a pulse, the beam rotates and slightly contracts: a signal point at affix z is moved to the affix

$$z' = \frac{1}{\sqrt{2}}\, e^{i\pi/6}\, z.$$

The engineers apply this transformation repeatedly to track the successive positions $z_0, z_1, z_2, \dots$ of a test signal initially sent to $z_0 = 2\sqrt{2}$ (a point $2\sqrt{2}$ km East of the mast).

The **coverage zone** D guaranteed by the company is the set of points $M(z)$ satisfying simultaneously

$$1 \le |z| \le 3 \qquad \text{and} \qquad |z - 1| \ge |z + 1|.$$

A client K lives at the point of affix $z_K = 2e^{i\pi/4}$.

The company wants a complete mathematical report: positions of the relays, behaviour of the rotating beam, shape of the coverage zone, and a decision on whether the client K is covered.

EXPERIMENT **Guided investigation**

Work through the following tasks step by step. Use a separate sheet for your Argand diagram (scale: 1 cm for 1 km).

Tasks

1. **Relays via De Moivre.** Write 8 in exponential form. Using De Moivre's theorem, determine the three cube roots of 8 in exponential form, then in Cartesian form. Plot O, A, B, C on an Argand diagram (scale: 1 cm for 1 km). What is the nature of triangle ABC? Justify.

2. **The rotating beam.** Write $a = \frac{1}{\sqrt{2}} e^{i\pi/6}$ in the form $a = r(\cos\theta + i\sin\theta)$.

   (a) Identify the **ratio** and the **angle** of the similitude $S : z' = az$. State its centre and its type.

   (b) Compute $z_1 = S(z_0)$, $z_2 = S(z_1)$ and $z_3 = S(z_2)$ where $z_0 = 2\sqrt{2}$. Give each in exponential form.

   (c) Show that $z_n = 2\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)^n e^{in\pi/6}$, and find the smallest n for which the signal point enters the disc $|z| < 1$.

   (d) Describe geometrically the path of the successive signal points.

3. **The coverage zone.**

   (a) Show that the condition $1 \le |z| \le 3$ describes a closed annulus; give its boundary circles, centres and radii.

   (b) Show that $|z - 1| \ge |z + 1|$ is equivalent to $\mathrm{Re}(z) \le 0$. Which line is the boundary, and how is it related to the points of affixes 1 and −1?

   (c) Shade the coverage zone D on your Argand diagram.

4. **Modulus inequalities.** Using the triangle inequality, prove that every point of D satisfies $|z - 1| \le 4$. Is the bound 4 attained? If yes, at which point(s) of D?
5. **The client.** Compute $|z_K|$ and $\mathrm{Re}(z_K)$ for $z_K = 2e^{i\pi/4}$. Decide, with justification, whether the client K is covered.
6. **Matrix and rigid motion.**
   (a) Write the matrix of the linear map $z \mapsto az$ (rotation part combined with the contraction) in the canonical basis.
   (b) Consider now the pure rotation $R : z' = e^{i\pi/6}z$. Write its matrix, compute its determinant, and classify R as a rigid motion: is it orientation-preserving or orientation-reversing? Justify.
7. **Synthesis.** Prepare a group poster: the annotated Argand diagram (relays, beam positions, shaded coverage zone, client K), and a five-line conclusion for the company.

WARNING **Common pitfalls**

- When using De Moivre for roots, remember to add $2k\pi$ before dividing by n: $z_k = \sqrt[n]{r}\, e^{i(\theta+2k\pi)/n}$.
- The condition $|z-1| \ge |z+1|$ means "distance to 1 is at least distance to -1"; the boundary is the perpendicular bisector of $[-1, 1]$, i.e. the imaginary axis. Do not confuse with $|z - 1| \le |z + 1|$.
- Triangle inequality: $|z_1 + z_2| \le |z_1| + |z_2|$. Equality holds iff z_1 and z_2 have the same argument (or one is zero).
- A direct similitude $z' = az + b$ with $a \neq 1$ has centre $\frac{b}{1-a}$. If $b = 0$, the centre is the origin.
- For a rigid motion $z' = e^{i\theta}z$, the matrix is a rotation matrix with determinant $+1$ (orientation-preserving). A reflection would have determinant -1.

Model Answer

WORKED EXAMPLE 16 **Complete solution**

1. Cube roots of 8. $8 = 8e^{i\cdot 0}$. By De Moivre's theorem, the cube roots are

$$w_k = 8^{1/3}e^{i(0+2k\pi)/3} = 2e^{2ik\pi/3}, \qquad k = 0, 1, 2.$$

Explicitly:

$$z_A = 2e^0 = 2, \quad z_B = 2e^{2i\pi/3} = -1 + i\sqrt{3}, \quad z_C = 2e^{4i\pi/3} = -1 - i\sqrt{3}.$$

The three points lie on the circle of centre Ω and radius 2, separated by angles $\frac{2\pi}{3}$. They form an equilateral triangle inscribed in that circle (side length $|z_A - z_B| = |3 - i\sqrt{3}| = \sqrt{9 + 3} = 2\sqrt{3}$).

2. The rotating beam. (a) $a = \dfrac{1}{\sqrt{2}}\left(\cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}\right)$. The similitude S has **ratio** $r = |a| = \dfrac{1}{\sqrt{2}}$, **angle** $\theta = \arg a = \dfrac{\pi}{6}$, and **centre** Ω (since $b = 0$). As $r \neq 1$, S is a direct similitude that is not an isometry: the composite of the rotation of angle $\pi/6$ about Ω and the enlargement of ratio $1/\sqrt{2}$ centred at Ω.
(b) With $z_0 = 2\sqrt{2} = 2\sqrt{2}\,e^{i\cdot 0}$:

$$z_1 = \tfrac{1}{\sqrt{2}}e^{i\pi/6}\cdot 2\sqrt{2} = 2e^{i\pi/6}, \quad z_2 = \sqrt{2}\,e^{i\pi/3}, \quad z_3 = e^{i\pi/2} = i.$$

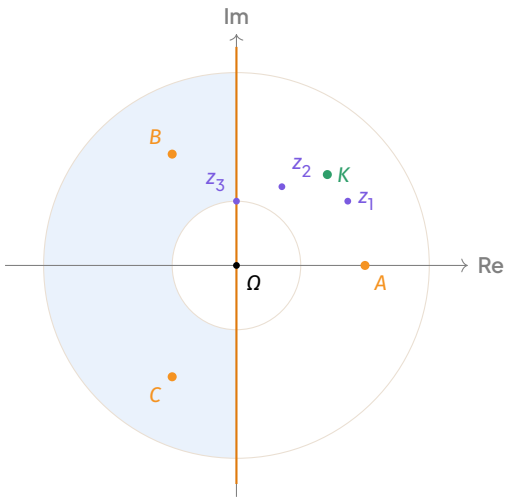
(c) By induction, $z_n = a^n z_0$, and $a^n = \left(\tfrac{1}{\sqrt{2}}\right)^n e^{in\pi/6}$ by De Moivre, hence

$$z_n = 2\sqrt{2}\left(\tfrac{1}{\sqrt{2}}\right)^n e^{in\pi/6}.$$

Then $|z_n| = 2\sqrt{2}\left(\tfrac{1}{\sqrt{2}}\right)^n = 2^{(3-n)/2}$, and $|z_n| < 1 \iff \dfrac{3-n}{2} < 0 \iff n > 3$. The smallest such integer is $\boxed{n = 4}$ (indeed $|z_4| = \tfrac{1}{\sqrt{2}} < 1$).
(d) The points spiral towards Ω: each step rotates by 30° and multiplies the distance to Ω by $\dfrac{1}{\sqrt{2}}$; they lie on an equiangular (logarithmic-type) spiral sampled at equal angles.
3. Coverage zone. (a) $|z| = 1$ and $|z| = 3$ are the circles of centre Ω and radii 1 and 3; $1 \le |z| \le 3$ is the closed annulus between them (boundaries included).
(b) Writing $z = x + iy$:

$$|z-1|^2 \ge |z+1|^2 \iff (x-1)^2 + y^2 \ge (x+1)^2 + y^2 \iff -4x \ge 0 \iff x \le 0.$$

The boundary $x = 0$ (the imaginary axis) is the perpendicular bisector of the segment joining the points of affixes 1 and -1; the condition selects the half-plane closer to -1 than to 1.
(c) D is the left half-annulus: the part of the annulus $1 \le |z| \le 3$ lying in the half-plane $\mathrm{Re}(z) \le 0$, boundaries included.



Coverage zone D (shaded), relays A, B, C, signal points z_1, z_2, z_3, client K.

4. Modulus inequality. For $z \in D$, $|z| \le 3$, so by the triangle inequality

$$|z - 1| \le |z| + |-1| = |z| + 1 \le 3 + 1 = 4.$$

Equality in $|z - 1| \le |z| + 1$ holds iff z and -1 have the same direction, i.e. z is a negative real number. Combined with $|z| = 3$ and $\mathrm{Re}(z) \le 0$, this gives $z = -3$. Since $-3 \in D$ (as $|-3| = 3 \in [1, 3]$ and $\mathrm{Re}(-3) = -3 \le 0$), the bound 4 is attained at $z = -3$.

5. The client. $z_K = 2e^{i\pi/4}$, so $|z_K| = 2 \in [1, 3]$: the first condition holds. But

$$\mathrm{Re}(z_K) = 2\cos\frac{\pi}{4} = \sqrt{2} > 0,$$

so the second condition fails. Conclusion: the client K is NOT covered by the guaranteed zone (he is in the Eastern half-plane).

6. Matrix and rigid motion. (a) $a = \dfrac{1}{\sqrt{2}}\left(\dfrac{\sqrt{3}}{2} + i\dfrac{1}{2}\right) = \dfrac{\sqrt{6}}{4} + i\dfrac{\sqrt{2}}{4}$. The matrix of $z \mapsto az$ in the canonical basis is

$$M = \begin{pmatrix} \frac{\sqrt{6}}{4} & -\frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{6}}{4} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} \\ \sin\frac{\pi}{6} & \cos\frac{\pi}{6} \end{pmatrix}.$$

(b) For $R : z' = e^{i\pi/6}z$, the matrix is the rotation matrix

$$M_R = \begin{pmatrix} \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} \\ \sin\frac{\pi}{6} & \cos\frac{\pi}{6} \end{pmatrix}, \qquad \det M_R = \cos^2\frac{\pi}{6} + \sin^2\frac{\pi}{6} = 1.$$

Since $|e^{i\pi/6}| = 1$, R preserves distances: it is a rigid motion (an isometry). Since $\det M_R = +1 > 0$, R is orientation-preserving: it is a positive (direct) rigid motion, namely the rotation of centre Ω and angle $\dfrac{\pi}{6}$.

7. Synthesis (sample conclusion). The three relays form an equilateral triangle of side $2\sqrt{3}$ km around the mast. The beam describes a contracting spiral (angle 30°, ratio $\frac{1}{\sqrt{2}}$) and enters the 1 km dead-zone after 4 pulses. The guaranteed coverage is the half-annulus $\{1 \le |z| \le 3,\ \mathrm{Re}(z) \le 0\}$; the client K lies outside it and requires an additional relay.

DID YOU KNOW? **Did you know?**

The spiral described by the signal points is a discrete version of the logarithmic spiral $r = r_0 e^{k\theta}$, which appears in nature (nautilus shells, galaxies) and in engineering (antenna patterns). The transformation $z \mapsto az$ with $|a| < 1$ and $\arg a \neq 0$ is a spiral similarity; its matrix is a scaled rotation matrix. In the GCE exam, you may be asked to identify the centre of a similitude given by $z' = az + b$ ($a \neq 1$): it is the unique fixed point $z = \frac{b}{1-a}$.

9 METHODS AND WORKED EXAMPLES

9.1 Methods and Worked Examples

METHOD **Converting Forms and Applying De Moivre's Theorem for Powers and Roots**

Goal: Switch fluently between Cartesian ($z = x + iy$), polar ($z = r(\cos\theta + i\sin\theta)$) and exponential ($z = re^{i\theta}$) forms, and compute integer powers or n-th roots using De Moivre's Theorem.
Steps:

1. Identify the given form. If Cartesian, compute the modulus $r = \sqrt{x^2 + y^2}$ and the argument $\theta = \arg(z)$ (fix the quadrant using $\tan\theta = y/x$ and the signs of x and y). Principal argument: $-\pi < \theta \leq \pi$.
2. Write in polar/exponential form: $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$.
3. For powers z^n ($n \in \mathbb{Z}$): apply De Moivre: $z^n = r^n(\cos n\theta + i\sin n\theta)$. Convert back to Cartesian form if required.
4. For n-th roots $z^{1/n}$: the n distinct roots are
$$w_k = r^{1/n}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right], \quad k = 0, 1, \ldots, n-1.$$
5. Geometric interpretation: the roots lie on the circle of radius $r^{1/n}$ centred at the origin, at the vertices of a regular n-gon. One root (the principal root) has argument θ/n.

Key reflexes:

- Always reduce the argument of the final answer to the principal range $(-\pi, \pi]$.
- For negative powers: $z^{-n} = \dfrac{1}{z^n} = r^{-n}(\cos n\theta - i\sin n\theta)$.
- Check: the sum of the n-th roots of a non-zero complex number is 0; their product is $(-1)^{n-1}z$ (Vieta's formulas applied to $w^n - z = 0$).

WORKED EXAMPLE 17 **Powers, Roots and Geometric Properties**

Statement: Let $z = -\sqrt{3} + i$.

1. Express z in exponential form $re^{i\theta}$ with $r > 0$ and $-\pi < \theta \leq \pi$.
2. Find z^6 in Cartesian form $a + ib$.
3. Determine the three cube roots of z in exponential form. Plot them on an Argand diagram.

4. Show that the sum of these three cube roots is zero and find their product.

SOLUTION

Detailed Solution:

1. Exponential form of z. Modulus: $r = |z| = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$. Argument: $\tan\theta = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}$. The point $(-\sqrt{3}, 1)$ is in the second quadrant. Reference angle: $\frac{\pi}{6}$. Principal argument: $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

$$z = 2e^{i\frac{5\pi}{6}} \quad \left(\text{or } 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)\right).$$

2. Compute z^6. By De Moivre's Theorem:

$$z^6 = 2^6 e^{i\left(6\times\frac{5\pi}{6}\right)} = 64e^{i5\pi}.$$

Reduce the argument: $5\pi \equiv 5\pi - 2(2\pi) = \pi \pmod{2\pi}$.

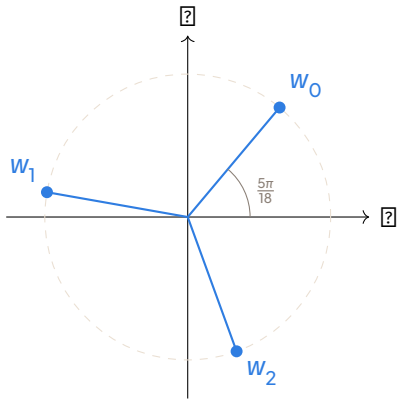
$$z^6 = 64e^{i\pi} = 64(\cos\pi + i\sin\pi) = 64(-1+0i) = \boxed{-64}.$$

3. Cube roots of z. We seek w_k such that $w_k^3 = z = 2e^{i\frac{5\pi}{6}}$. Modulus of the roots: $\sqrt[3]{2} = 2^{1/3}$. Arguments: $\frac{\frac{5\pi}{6} + 2k\pi}{3} = \frac{5\pi}{18} + \frac{2k\pi}{3}$, $k = 0, 1, 2$.

$$k = 0: \quad w_0 = 2^{1/3}e^{i\frac{5\pi}{18}}$$
$$k = 1: \quad w_1 = 2^{1/3}e^{i\left(\frac{5\pi}{18}+\frac{2\pi}{3}\right)} = 2^{1/3}e^{i\frac{17\pi}{18}}$$
$$k = 2: \quad w_2 = 2^{1/3}e^{i\left(\frac{5\pi}{18}+\frac{4\pi}{3}\right)} = 2^{1/3}e^{i\frac{29\pi}{18}}.$$

Reduce the argument of w_2 to the principal range: $\frac{29\pi}{18} - 2\pi = \frac{29\pi}{18} - \frac{36\pi}{18} = -\frac{7\pi}{18}$. So $w_2 = 2^{1/3}e^{-i\frac{7\pi}{18}}$.

Argand diagram (note $\frac{5\pi}{18} = 50^\circ$, $\frac{17\pi}{18} = 170^\circ$, $-\frac{7\pi}{18} = -70^\circ$, and $2^{1/3} \approx 1.26$):



Cube roots of z: vertices of an equilateral triangle on the circle of radius $\sqrt[3]{2}$.

4. Sum and product of the roots. The roots w_0, w_1, w_2 are the solutions of $w^3 - z = 0$, i.e. $w^3 + 0w^2 + 0w - z = 0$. By Vieta's formulas: Sum $= w_0 + w_1 + w_2 = -\dfrac{\text{coefficient of } w^2}{\text{coefficient of } w^3} = 0$. Product $= w_0 w_1 w_2 = -\dfrac{\text{constant term}}{\text{leading coefficient}} = -(-z) = z = \boxed{-\sqrt{3} + i}$. Check: sum of arguments: $\frac{5\pi}{18} + \frac{17\pi}{18} - \frac{7\pi}{18} = \frac{15\pi}{18} = \frac{5\pi}{6}$; product of moduli: $(2^{1/3})^3 = 2$. This matches $z = 2e^{i5\pi/6}$.

METHOD **Deriving Trigonometric Identities via De Moivre's Theorem**

Goal: express $\cos n\theta$, $\sin n\theta$ in powers of $\cos\theta$, $\sin\theta$; or express $\cos^n\theta$, $\sin^n\theta$ in terms of cosines/sines of multiple angles.
Steps (Powers → Multiple angles):

1. Write $2\cos\theta = z + z^{-1}$ and $2i\sin\theta = z - z^{-1}$, where $z = e^{i\theta}$.
2. Raise to the power n: $(2\cos\theta)^n = (z + z^{-1})^n$, $(2i\sin\theta)^n = (z - z^{-1})^n$.
3. Expand using the Binomial Theorem: $(z + z^{-1})^n = \sum_{k=0}^{n} \binom{n}{k} z^{n-2k}$.
4. Group the symmetric terms using $z^m + z^{-m} = 2\cos m\theta$ and $z^m - z^{-m} = 2i\sin m\theta$.
5. Simplify to obtain a linear combination of $\cos m\theta$ (or $\sin m\theta$).

Steps (Multiple angles → Powers):

1. Use De Moivre: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
2. Expand the left-hand side using the Binomial Theorem.
3. Equate real parts → $\cos n\theta$; equate imaginary parts → $\sin n\theta$.
4. Use $\sin^2\theta = 1 - \cos^2\theta$ (or vice versa) to express the result purely in powers of one function.

Key reflexes:

$$\cos n\theta = \Re\left(e^{in\theta}\right), \qquad \sin n\theta = \Im\left(e^{in\theta}\right), \qquad z^m + z^{-m} = 2\cos m\theta, \qquad z^m - z^{-m} = 2i\sin m\theta.$$

WORKED EXAMPLE 18 **Trigonometric Expansions and Integration Application**

Statement:

1. Use De Moivre's theorem to show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.
2. Hence express $\cos^5\theta$ in terms of cosines of multiple angles.

3. Evaluate the exact value of $\int_0^{\pi/2} \cos^5\theta\, d\theta$.

SOLUTION

Detailed Solution:

1. Expansion of $\cos 5\theta$. Let $z = \cos\theta + i\sin\theta = e^{i\theta}$. By De Moivre: $z^5 = \cos 5\theta + i\sin 5\theta$. Binomial expansion of $(\cos\theta + i\sin\theta)^5$:

$$z^5 = \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta + 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta$$
$$= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta.$$

Equate real parts:

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta.$$

Replace $\sin^2\theta = 1 - \cos^2\theta$:

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)^2\\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)\\ &= 11\cos^5\theta - 10\cos^3\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta\\ &= \boxed{16\cos^5\theta - 20\cos^3\theta + 5\cos\theta}.\end{aligned}$$

2. Express $\cos^5\theta$ in multiple angles. Rearrange the identity:

$$16\cos^5\theta = \cos 5\theta + 20\cos^3\theta - 5\cos\theta.$$

We also need $\cos^3\theta$. The same method (or the standard identity) gives $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, hence $\cos^3\theta = \frac{1}{4}(\cos 3\theta + 3\cos\theta)$. Substitute:

$$\begin{aligned}16\cos^5\theta &= \cos 5\theta + 20\left[\frac{1}{4}(\cos 3\theta + 3\cos\theta)\right] - 5\cos\theta\\ &= \cos 5\theta + 5\cos 3\theta + 15\cos\theta - 5\cos\theta\\ &= \cos 5\theta + 5\cos 3\theta + 10\cos\theta.\end{aligned}$$

$$\boxed{\cos^5\theta = \frac{1}{16}(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)}.$$

3. Evaluate the integral.

$$I = \int_0^{\pi/2} \cos^5\theta\, d\theta = \frac{1}{16}\int_0^{\pi/2} (\cos 5\theta + 5\cos 3\theta + 10\cos\theta)\, d\theta.$$

Integrate term by term using $\int \cos k\theta\, d\theta = \frac{1}{k}\sin k\theta$:

$$I = \frac{1}{16}\left[\frac{1}{5}\sin 5\theta + \frac{5}{3}\sin 3\theta + 10\sin\theta\right]_0^{\pi/2}.$$

At $\theta = \pi/2$: $\sin(5\pi/2) = 1$; $\sin(3\pi/2) = -1$; $\sin(\pi/2) = 1$. At $\theta = 0$: all terms vanish.

$$I = \frac{1}{16}\left(\frac{1}{5} - \frac{5}{3} + 10\right) = \frac{1}{16} \times \frac{3 - 25 + 150}{15} = \frac{1}{16} \times \frac{128}{15} = \boxed{\frac{8}{15}}.$$

METHOD **Sketching Loci in the Argand Diagram**

Goal: interpret and sketch loci defined by equations or inequalities in z (involving $|z|$, $\arg z$, $\Re(z)$, $\Im(z)$).
Standard forms and shapes:

Equation / Inequality	Geometric Description	Boundary
$\|z - a\| = r$	Circle, centre a, radius r	Solid circle
$\|z - a\| < r$ / $> r$	Interior / exterior of the circle	Dashed circle
$\arg(z - a) = \alpha$	Half-line from a (a excluded) at angle α	Solid ray, open dot at a
$\alpha < \arg(z - a) < \beta$	Angular sector, vertex a excluded	Dashed rays
$\|z - a\| = \|z - b\|$	Perpendicular bisector of segment AB	Solid line
$\|z - a\| = k\|z - b\|$ $(k > 0, k \neq 1)$	Circle of Apollonius	Solid circle
$\Re(z) = c$ or $\Im(z) = d$	Vertical / horizontal line	Solid line

Steps for sketching:

1. Identify the type (circle, line, ray, region).
2. Find key points: centre and radius for circles; intercepts for lines; vertex for rays.
3. Determine the boundary style: solid for $\leq, \geq, =$; dashed for $<, >$.
4. Test a point (usually the origin $z = 0$, if not on the boundary) to shade the correct region for inequalities.
5. Label clearly: axes, scale, centre coordinates, radius, angles.

Traps:

- $\arg(z - a) = \alpha$ excludes the point a itself (open dot), because arg is undefined there.
- $|z - a| = k|z - b|$ with $k = 1$ is a line (perpendicular bisector), not a circle.
- For $\arg(z) = \alpha$, the half-line starts at the origin, which is excluded.

WORKED EXAMPLE 19 **Compound Locus and Extremal Moduli**

Statement: On a single Argand diagram, sketch the region R defined by the simultaneous inequalities

$$|z - 3i| \leq 3 \quad \text{and} \quad \frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{2}.$$

Determine the greatest value of $|z|$ for $z \in R$, and discuss whether $|z|$ attains a least value

on R.

SOLUTION

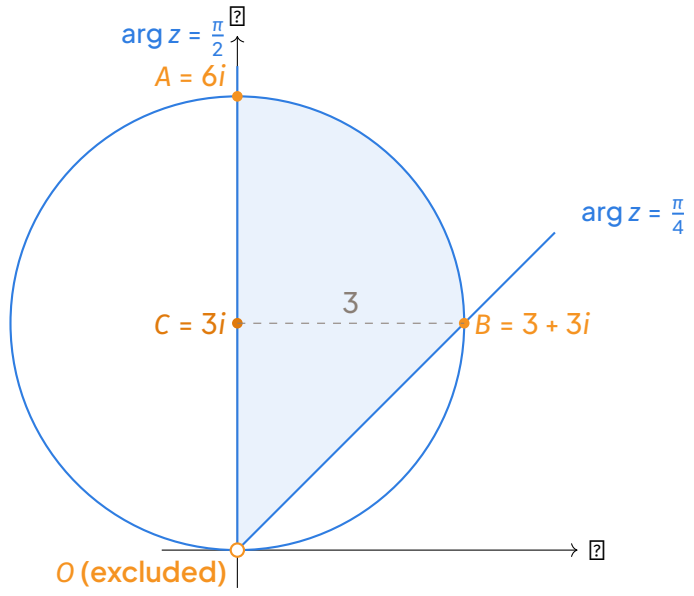
Detailed Solution:
Analysis of the inequalities:

1. $|z-3i| \le 3$: closed disc, centre $C(0,3)$ (affix $3i$), radius 3. Boundary: solid circle, interior included. Note that $|0-3i| = 3$, so the circle passes through the origin.
2. $\frac{\pi}{4} \le \arg(z) \le \frac{\pi}{2}$: closed sector between the ray $\arg(z) = \pi/4$ (the line $y = x$, $x \ge 0$) and the ray $\arg(z) = \pi/2$ (the positive imaginary axis). Boundaries: solid rays. Since $\arg(0)$ is undefined, the origin is not in R.

Key boundary points. The ray $\arg z = \pi/2$ meets the circle where $z = iy$, $|iy-3i| = |y-3| = 3$, giving $y = 0$ (origin, excluded) or $y = 6$: point $A = 6i$. The ray $\arg z = \pi/4$ meets the circle where $z = x + ix$:

$$|x + i(x-3)|^2 = x^2 + (x-3)^2 = 9 \implies 2x^2 - 6x = 0 \implies x = 0 \text{ or } x = 3,$$

giving the origin (excluded) and the point $B = 3 + 3i$.
Sketch:



Region R: intersection of the disc $|z - 3i| \le 3$ with the sector $\pi/4 \le \arg z \le \pi/2$. The origin is excluded.

Greatest modulus. $|z|$ is the distance from O. The farthest point of the closed disc from O lies on the line through O and the centre C, i.e. on the imaginary axis: this is $A = 6i$, which lies in R (it is on the ray $\arg z = \pi/2$). Hence

$$|z|_{\max} = OA = \boxed{6}, \quad \text{attained at } z = 6i.$$

Least modulus — discussion. Points of the ray $\arg z = \pi/4$ satisfy $z = x + ix$, and

$$|z - 3i|^2 = 2x^2 - 6x + 9 \le 9 \iff 2x(x-3) \le 0 \iff 0 \le x \le 3.$$

So the whole segment from O to B (excluding O) lies in R, and $|z| = x\sqrt{2}$ can be made arbitrarily close to 0. Since $O \notin R$ (argument undefined), $|z|$ has no least value on R; its greatest lower bound (infimum) is 0, not attained.

WARNING **A classic trap**

One might be tempted to answer $|z|_{\min} = OB = 3\sqrt{2}$ at $B = 3 + 3i$. This is wrong: B is merely the closest point to O on the circular arc, but the region also contains the segment OB of the ray $\arg z = \pi/4$, along which $|z|$ decreases towards 0. Always check whether the region accumulates at the origin when the boundary circle passes through O.

METHOD **Transformations from the z-plane to the w-plane**

Goal: find the image (locus of w) of a given locus of z under a transformation $w = f(z)$, or find a pre-image.
General algebraic strategy:

1. Write $w = u + iv$ and $z = x + iy$.
2. Express z in terms of w (invert the mapping): $z = f^{-1}(w)$.
3. Substitute $z = f^{-1}(w)$ into the equation of the z-locus (e.g. $|z - a| = r$, $\arg(z - a) = \alpha$, or a Cartesian equation).
4. Simplify to obtain a Cartesian equation in u, v (or a condition on $|w|$, $\arg w$).
5. Identify the locus in the w-plane (line, circle, ray, etc.) and mark any excluded points.

Key standard transformations:

- Translation: $w = z + b$: shift by the vector affix b.
- Rotation–enlargement: $w = az$ ($a \in \mathbb{C}^*$): rotation by $\arg a$ about O, enlargement factor $|a|$.
- Reciprocal: $w = \dfrac{1}{z}$:
  - circle through the origin → line not through the origin;
  - line not through the origin → circle through the origin;
  - line through the origin → line through the origin;

  - circle not through the origin → circle not through the origin.
- Möbius (bilinear) transformation: $w = \dfrac{az+b}{cz+d}$ $(ad - bc \neq 0)$: maps generalized circles (lines and circles) to generalized circles.

Useful formula for $w = 1/z$. If the z-locus is the circle $|z - a| = r$ with $|a| \neq r$ (not through the origin), its image is the circle of centre $w_c = \dfrac{\bar{a}}{|a|^2 - r^2}$ and radius $R = \dfrac{r}{\left||a|^2 - r^2\right|}$.

WORKED EXAMPLE 20 **Image of a Line under a Möbius Transformation**

Statement: The transformation T from the z-plane to the w-plane is defined by

$$w = \frac{z - i}{z + i}, \quad z \neq -i.$$

The line L in the z-plane has equation $\Re(z) = 1$ (i.e. $y = 1$).

1. Find the Cartesian equation of the image of L under T in the w-plane.
2. Sketch the image locus, indicating clearly any excluded points.
3. Find the image of the point $z = 1 + i$.

SOLUTION

Detailed Solution:
1. Equation of the image. Invert the mapping:

$$w(z + i) = z - i \implies wz + wi = z - i \implies z(w - 1) = -i(1 + w) \implies z = i\,\frac{1 + w}{1 - w}.$$

Let $w = u + iv$. Compute z in terms of u, v:

$$z = i\,\frac{(1 + u) + iv}{(1 - u) - iv} = i\,\frac{[(1 + u) + iv][(1 - u) + iv]}{(1 - u)^2 + v^2} = i\,\frac{(1 - u^2 - v^2) + 2iv}{(1 - u)^2 + v^2}.$$

Hence

$$\Re(z) = \frac{-2v}{(1 - u)^2 + v^2}, \qquad \Re(z) = \frac{1 - u^2 - v^2}{(1 - u)^2 + v^2}.$$

The condition $\Re(z) = 1$ gives

$$1 - u^2 - v^2 = (1 - u)^2 + v^2 = 1 - 2u + u^2 + v^2 \implies 2u^2 + 2v^2 - 2u = 0 \implies u^2 + v^2 - u = 0.$$

Completing the square:

$$\boxed{\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}},$$

a circle of centre $\left(\frac{1}{2}, 0\right)$ and radius $\frac{1}{2}$.

2. Sketch and excluded points. The pole $z = -i$ does not lie on L (since $\square(-i) = -1 \neq 1$), so no point of L is sent to infinity: the image is a genuine circle. However, as $z \to \infty$ along L, $w = \frac{z-i}{z+i} \to 1$; since no finite point of L maps to $w = 1$, the point $w = 1$ is excluded (open dot). Also $z = i \in L$ maps to $w = 0$, so the circle passes through the origin.

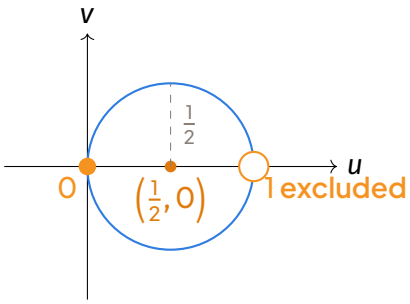


Image of $\square(z) = 1$: circle of centre $(1/2, 0)$, radius $1/2$; the point $w = 1$ is excluded.

3. Image of $z = 1 + i$.

$$w = \frac{(1+i)-i}{(1+i)+i} = \frac{1}{1+2i} = \frac{1-2i}{(1+2i)(1-2i)} = \frac{1-2i}{5} = \boxed{\frac{1}{5} - \frac{2}{5}i}.$$

Check on the circle: $u = 0.2, v = -0.4$: $(0.2 - 0.5)^2 + (-0.4)^2 = 0.09 + 0.16 = 0.25 = (0.5)^2$. ✓

METHOD **Matrix Representation of Plane Transformations**

Goal: represent geometric transformations (rotation, reflection, enlargement, stretch, shear, and their composites) as 2×2 matrices acting on column vectors $\begin{pmatrix} x \\ y \end{pmatrix}$.

Standard matrices (with respect to the standard basis):

Transformation	Matrix **M**	det **M**
Rotation by θ about O (anticlockwise)	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	+1
Reflection in the line $y = (\tan\theta)x$	$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$	−1
Enlargement, factor k, centre O	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	k^2
Stretch parallel to the x-axis, factor a	$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$	a
Stretch parallel to the y-axis, factor b	$\begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}$	b
Shear parallel to the x-axis, factor k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	1
Shear parallel to the y-axis, factor k	$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$	1
Reflection in the x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	−1
Reflection in the y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	−1
Reflection in the line $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	−1
Reflection in the line $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	−1

Composition: "T_1 then T_2" has matrix $\mathbf{M_2M_1}$ (right-to-left order). Inverse: $\mathbf{M}^{-1}$ exists iff $\det\mathbf{M} \neq 0$. Invariant points: solve $\mathbf{Mx} = \mathbf{x}$. Invariant lines through O: solve $\mathbf{Mx} = \lambda\mathbf{x}$ (eigenvectors).

WORKED EXAMPLE 21 **Composite Transformation: Rotation followed by Reflection**

Statement: R is the rotation of 60° anticlockwise about the origin; S is the reflection in the line $y = \sqrt{3}\,x$. Let $T = S \circ R$ (apply R first, then S).

1. Write down the matrices **R** and **S**.
2. Find the matrix **T** representing T.
3. Identify the single geometric transformation represented by **T**, describing it fully.

4. Find the equation of the line of invariant points of T.

SOLUTION

Detailed Solution:

1. Matrices **R** and **S**. $\theta = 60^\circ$: $\cos 60^\circ = \frac{1}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$.

$$\mathbf{R} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

The mirror line $y = \sqrt{3}x$ has $\tan\theta = \sqrt{3}$, so $\theta = 60^\circ$ and $2\theta = 120^\circ$: $\cos 120^\circ = -\frac{1}{2}$, $\sin 120^\circ = \frac{\sqrt{3}}{2}$.

$$\mathbf{S} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

2. Matrix $\mathbf{T} = \mathbf{SR}$.

$$\mathbf{T} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

Entries: $T_{11} = -\frac{1}{4} + \frac{3}{4} = \frac{1}{2}$; $T_{12} = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$; $T_{21} = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$; $T_{22} = -\frac{3}{4} + \frac{1}{4} = -\frac{1}{2}$.

$$\mathbf{T} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

3. Identification. $\det \mathbf{T} = \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)^2 = -\frac{1}{4} - \frac{3}{4} = -1$, so T reverses orientation. The matrix matches the reflection pattern $\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$ with

$$\cos 2\alpha = \frac{1}{2}, \quad \sin 2\alpha = \frac{\sqrt{3}}{2} \implies 2\alpha = 60^\circ \implies \alpha = 30^\circ.$$

Conclusion: T is the reflection in the line $y = (\tan 30^\circ)\,x = \dfrac{x}{\sqrt{3}}$.

4. Line of invariant points. Solve $\mathbf{Tx} = \mathbf{x}$:

$$\frac{1}{2}x + \frac{\sqrt{3}}{2}y = x \implies x = \sqrt{3}\,y; \qquad \frac{\sqrt{3}}{2}x - \frac{1}{2}y = y \implies x = \sqrt{3}\,y \quad \text{(consistent)}.$$

Equation of the invariant line: $\boxed{y = \dfrac{x}{\sqrt{3}}}$ (equivalently $x - \sqrt{3}\,y = 0$) — exactly the mirror line found above, as expected for a reflection.

METHOD **Similitudes: Complex Form, Matrix Form and Properties**

Goal: recognise, analyse and apply similitudes (direct and opposite) in both complex and matrix forms.
Definitions. A similitude is a transformation that preserves angles (hence shapes) and multiplies all distances by a fixed ratio $k > 0$.

- Direct similitude (orientation-preserving): $z' = az + b$, $a, b \in \mathbb{C}$, $a \neq 0$. Ratio $k = |a|$, angle of rotation $\theta = \arg a$. Matrix (affine) form:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \Re(a) & -\Im(a) \\ \Im(a) & \Re(a) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \Re(b) \\ \Im(b) \end{pmatrix}.$$

- Opposite similitude (orientation-reversing): $z' = a\bar{z} + b$, $a \neq 0$. Ratio $k = |a|$. Matrix (affine) form:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \Re(a) & \Im(a) \\ \Im(a) & -\Re(a) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \Re(b) \\ \Im(b) \end{pmatrix}.$$

Key properties:

1. Fixed point (centre): a direct similitude $z' = az + b$ with $a \neq 1$ has a unique fixed point $z_0 = \dfrac{b}{1-a}$: it is the centre of the rotation–enlargement.
2. Composition: direct ∘ direct = direct (ratios multiply, angles add); opposite ∘ opposite = direct; direct ∘ opposite = opposite.
3. Inverse: the inverse of a direct similitude is direct; the inverse of an opposite similitude is opposite.

Determining a direct similitude from two pairs (z_1, z_1'), (z_2, z_2'):

$$a = \frac{z_1' - z_2'}{z_1 - z_2}, \qquad b = z_1' - az_1, \qquad z_0 = \frac{b}{1-a}.$$

WORKED EXAMPLE 22 **Analysing a Direct Similitude from Point Pairs**

Statement: A direct similitude S maps $z_1 = 1 + i$ to $z_1' = 4 + 2i$ and $z_2 = 2 - i$ to $z_2' = 5 - 4i$.

1. Determine the complex form $z' = az + b$ of S.
2. Find the ratio of similitude k and the angle of rotation θ.
3. Find the centre (fixed point) z_0 of S.
4. Write the matrix (affine) representation of S.
5. Determine the Cartesian equation of the image of the line $\Re(z) = 1$.

SOLUTION

Detailed Solution:

1. Find a and b.

$$a = \frac{z_1' - z_2'}{z_1 - z_2} = \frac{(4+2i)-(5-4i)}{(1+i)-(2-i)} = \frac{-1+6i}{-1+2i} = \frac{(-1+6i)(-1-2i)}{(-1)^2+2^2} = \frac{13-4i}{5} = 2.6-0.8i.$$

Then, using z_1:

$$az_1 = (2.6-0.8i)(1+i) = 2.6+2.6i-0.8i-0.8i^2 = 3.4+1.8i,$$

$$b = z_1' - az_1 = (4+2i)-(3.4+1.8i) = 0.6+0.2i.$$

$$\boxed{z' = (2.6-0.8i)\,z + (0.6+0.2i)} \qquad \left(\text{equivalently } z' = \tfrac{1}{5}\left[(13-4i)z + (3+i)\right]\right).$$

Check with z_2: $az_2 = (2.6-0.8i)(2-i) = 5.2-2.6i-1.6i+0.8i^2 = 4.4-4.2i$; then $az_2+b = 5-4i = z_2'$. ✓

2. Ratio k and angle θ.

$$k = |a| = \frac{|13-4i|}{5} = \frac{\sqrt{169+16}}{5} = \boxed{\frac{\sqrt{185}}{5}} \approx 2.72.$$

$\tan\theta = \dfrac{-4}{13}$ with $\Re(a) > 0$, $\Im(a) < 0$ (fourth quadrant):

$$\theta = -\arctan\frac{4}{13} \approx -17.1^\circ.$$

3. Centre z_0.

$$1-a = \frac{5-(13-4i)}{5} = \frac{-8+4i}{5}, \qquad z_0 = \frac{b}{1-a} = \frac{3+i}{-8+4i} = \frac{(3+i)(-8-4i)}{64+16} = \frac{-20-20i}{80} = \boxed{-\frac{1}{4}-\frac{1}{4}i}.$$

Check: $az_0 = (2.6-0.8i)(-0.25-0.25i) = -0.85-0.45i$; $az_0 + b = -0.25-0.25i = z_0$. ✓

4. Matrix representation. With $\Re(a) = 2.6$, $\Im(a) = -0.8$:

$$\boxed{\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2.6 & 0.8 \\ -0.8 & 2.6 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0.6 \\ 0.2 \end{pmatrix}}.$$

5. Image of the line $x = 1$. Write $a = \alpha + i\beta$ with $\alpha = 2.6$, $\beta = -0.8$, and $b = u_0 + iv_0$ with $u_0 = 0.6$, $v_0 = 0.2$. For $z = x + iy$:

$$u = \alpha x - \beta y + u_0, \qquad v = \beta x + \alpha y + v_0.$$

Setting $x = 1$:

$$u = 2.6 + 0.8y + 0.6 = 3.2 + 0.8y, \qquad v = -0.8 + 2.6y + 0.2 = -0.6 + 2.6y.$$

Eliminate y: from the first equation $y = \dfrac{u-3.2}{0.8}$; substituting,

$$v = -0.6 + 2.6\cdot\frac{u-3.2}{0.8} = -0.6 + 3.25(u-3.2) = \boxed{3.25\,u - 11}.$$

The image is a straight line (as expected: a similitude maps lines to lines). Consistency check: the original line is vertical (direction angle 90°); after rotation by $\theta \approx -17.1^\circ$ the direction angle is $\approx 72.9^\circ$, and $\tan 72.9^\circ \approx 3.25$ — matching the slope found.

METHOD **Solving Modulus Inequalities and Classifying Rigid Motions**

Goal: solve inequalities involving $|z-a| \lessgtr |z-b|$ and interpret them geometrically; classify rigid motions (isometries) of the plane.
Modulus inequalities — algebraic method. For $|z-a| \lessgtr |z-b|$, write $z = x + iy$, $a = x_a + iy_a$, $b = x_b + iy_b$ and square both sides (valid since both sides are ≥ 0):

$$(x-x_a)^2 + (y-y_a)^2 \lessgtr (x-x_b)^2 + (y-y_b)^2.$$

The terms x^2, y^2 cancel, leaving the linear inequality

$$2(x_b-x_a)\,x + 2(y_b-y_a)\,y \lessgtr |b|^2 - |a|^2,$$

a half-plane bounded by the perpendicular bisector of the segment AB.
Rigid motions (isometries): transformations preserving distances, $|z_1' - z_2'| = |z_1 - z_2|$.

- Direct (orientation-preserving, det = +1):
  - Translation $z' = z + b$ ($b \neq 0$): no fixed point.
  - Rotation $z' - c = e^{i\theta}(z-c)$ ($\theta \neq 2k\pi$): unique fixed point, the centre c.
- Opposite (orientation-reversing, det = −1):
  - Reflection: a line of fixed points (the mirror).
  - Glide reflection (reflection composed with a translation parallel to the mirror): no fixed point.

Classification algorithm for $\mathbf{x}' = \mathbf{M}\mathbf{x} + \mathbf{t}$ with $\mathbf{M}$ orthogonal ($\mathbf{M}^T\mathbf{M} = \mathbf{I}$):

1. Compute $\det \mathbf{M}$.
2. If $\det \mathbf{M} = +1$ (direct): if $\mathbf{M} = \mathbf{I}$, it is a translation by $\mathbf{t}$; otherwise it is a rotation, with centre $\mathbf{c} = (\mathbf{I} - \mathbf{M})^{-1}\mathbf{t}$ and angle read from $\mathbf{M}$.
3. If $\det \mathbf{M} = -1$ (opposite): solve $(\mathbf{I} - \mathbf{M})\mathbf{x} = \mathbf{t}$. If there is a line of solutions, it is a reflection (the solution line is the mirror); if there is no solution, it is a glide reflection.

WORKED EXAMPLE 23 **Modulus Inequality Region and Rigid Motion Classification**

Statement:

1. Solve the inequality $|z-3| < |z-i|$ and shade the region on an Argand diagram.
2. A transformation T is defined by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

   (a) Show that T is an isometry (rigid motion).

(b) Classify T fully (type and mirror line).

(c) Find the image of the circle $|z| = 2$ under T.

SOLUTION

Detailed Solution:

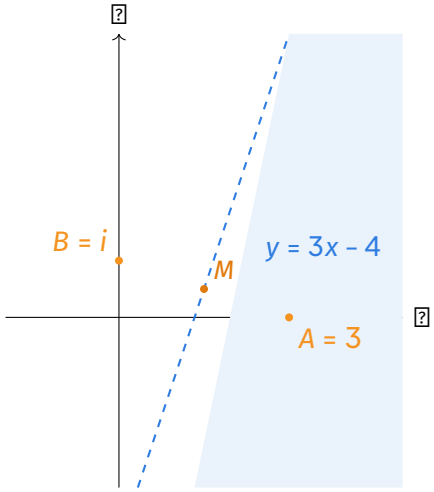
1. Inequality $|z - 3| < |z - i|$. Geometric reading: points closer to $A(3, 0)$ than to $B(0, 1)$; the boundary is the perpendicular bisector of AB. Midpoint of AB: $M(1.5,\ 0.5)$. Gradient of AB: $\frac{1-0}{0-3} = -\frac{1}{3}$, so the bisector has gradient 3:

$$y - 0.5 = 3(x - 1.5) \implies y = 3x - 4.$$

Algebraic check with $z = x + iy$:

$$(x-3)^2 + y^2 < x^2 + (y-1)^2 \implies -6x + 9 < -2y + 1 \implies \boxed{y < 3x - 4}.$$

Test the origin: $|0 - 3| = 3 > |0 - i| = 1$, so O is not in the region; the region is the half-plane below the line (the side containing $A(3, 0)$). Boundary dashed (strict inequality).



Region $|z - 3| < |z - i|$: open half-plane below the bisector $y = 3x - 4$ (dashed).

2(a). Isometry check.

$$\mathbf{M}^T\mathbf{M} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I},$$

so $\mathbf{M}$ is orthogonal: T preserves distances and is a rigid motion.

2(b). Classification. $\det \mathbf{M} = (0)(0) - (-1)(-1) = -1$: T is an opposite isometry. Fixed points: solve $(\mathbf{I} - \mathbf{M})\mathbf{x} = \mathbf{t}$:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \implies x + y = 2,$$

a whole line of fixed points. Hence T is a reflection in the mirror line

$$\boxed{x + y = 2}.$$

Consistency: $\mathbf{M}$ alone is the reflection in $y = -x$; the translation $(2, 2)$ is perpendicular to that line, and composing gives the reflection in the parallel line through $(1, 1)$, i.e. $x + y = 2$. ✓

2(c). Image of the circle $|z| = 2$. An isometry maps circles to circles of the same radius. The centre $O(0, 0)$ maps to

$$\mathbf{M}\begin{pmatrix}0\\0\end{pmatrix} + \begin{pmatrix}2\\2\end{pmatrix} = \begin{pmatrix}2\\2\end{pmatrix}.$$

Hence the image is the circle of centre $(2, 2)$ and radius 2:

$$\boxed{|z - (2 + 2i)| = 2} \qquad \left(\text{i.e. } (x - 2)^2 + (y - 2)^2 = 4\right).$$

10 EXERCISES

10.1 I check my knowledge

EXERCISE 8 **MCQ: Forms and De Moivre**

Which of the following is the exponential form of $z = 1 + i\sqrt{3}$?

1. $2e^{i\pi/3}$
2. $2e^{i\pi/6}$
3. $4e^{i\pi/3}$
4. $2e^{-i\pi/3}$

EXERCISE 9 **True/False with justification**

State whether each statement is true or false. Justify your answer.

1. $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ for all non-zero $z_1, z_2 \in \mathbb{C}$.
2. The locus $|z - 2| = |z + 2i|$ is a straight line.
3. The transformation $w = \frac{1}{z}$ maps the line $\mathrm{Re}(z) = 2$ to a circle.
4. A similitude $z' = az + b$ with $|a| = 1$ and $a \neq 1$ is a rotation.

EXERCISE 10 **Definitions and direct calculations**

1. Write the complex number $z = -2 + 2i$ in trigonometric form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with $r > 0$ and $\theta \in (-\pi, \pi]$.
2. Use De Moivre's theorem to compute $(1 - i)^8$ exactly.
3. Find all the cube roots of $-8i$, giving them in exponential form.

EXERCISE 11 **Quick computation**

Given $z_1 = 3e^{i\pi/4}$ and $z_2 = 2e^{-i\pi/6}$, compute exactly:

1. $z_1 z_2$ and $\frac{z_1}{z_2}$ in exponential form.

2. $\mathrm{Re}(z_1\overline{z_2})$ and $\mathrm{Im}(z_1\overline{z_2})$.

10.2 I practise

EXERCISE 12 **Trigonometric and exponential forms**

Let $z = 1 + i\sqrt{3}$.

1. Write z in trigonometric form and in exponential form.
2. Compute z^{12} exactly.
3. Find all the fourth roots of $-4z$. Plot them on an Argand diagram.

EXERCISE 13 **De Moivre and** $\cos 5\theta$

1. Use De Moivre's theorem to show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.
2. Deduce the exact value of $\cos(\pi/10)$ as a surd.

EXERCISE 14 **Loci in the Argand diagram**

On separate Argand diagrams, sketch the following loci:

1. $|z - 2 + i| = 2$.
2. $|z - 2| = |z + 2i|$.
3. $\arg(z - 1) = \pi/4$.

On a single diagram, shade the region satisfying all three conditions simultaneously.

EXERCISE 15 **Transformation** $w = 1/z$

Find the image of the line $\mathrm{Re}(z) = 2$ under the transformation $w = 1/z$. Give the centre and radius of the resulting circle.

10.3 I prepare for the GCE

EXERCISE 16 **Composite transformation and matrix] [12 marks**

A transformation T_1 is represented by the matrix $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and a transformation T_2 by $N = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

1. Find the single matrix representing the composite transformation $T_2 \circ T_1$.
2. State the geometric effect of this composite transformation.
3. Determine its area scale factor.
4. Find the inverse transformation and its matrix.

EXERCISE 17 **Similitude analysis] [13 marks**

A similitude is given by $z' = (1 - i)z + 3i$.

1. Determine its ratio k and its angle of rotation θ.
2. Find its centre c.
3. Classify the similitude (rotation, homothety, translation, etc.).
4. Find the image of the triangle with vertices $0, 1, i$.

EXERCISE 18 **Modulus inequality and rigid motions] [15 marks**

1. Prove the triangle inequality $|z_1 + z_2| \le |z_1| + |z_2|$ for all $z_1, z_2 \in \mathbb{C}$.
2. Solve in $\mathbb{C}$ the inequality $|z - 3i| \le |z + 3|$. Shade the solution set on an Argand diagram.
3. Classify the rigid motions:

   (a) $z' = iz + 2 - i$,

   (b) $z' = i\overline{z} + 1$.

   For each, state whether it preserves or reverses orientation and give its characteristic elements (centre and angle, or axis and vector).

11 SOLUTIONS TO THE EXERCISES

12 WORKED EXERCISES AND EXAMINATION PRACTICE

SOLUTION **Exercise 1 — MCQ: Forms and De Moivre**

The correct answer is a) $2e^{i\pi/3}$.
Justification. For $z = 1 + i\sqrt{3}$:

- Modulus: $|z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$.
- Argument: $\cos\theta = \frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{\pi}{3}$ (principal value).

Hence $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 2e^{i\pi/3}$.
Options b) and d) have the wrong argument, and c) has the wrong modulus ($4 = |z|^2$, a classic trap).

SOLUTION **Exercise 2 — True/False with justification**

a) TRUE (modulo 2π). If $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$, then $z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}$, so $\arg(z_1z_2) = \arg z_1 + \arg z_2 \pmod{2\pi}$. The statement is true provided we accept equality up to a multiple of 2π; if principal values in $(-\pi, \pi]$ are required, the sum may need adjusting by $\pm 2\pi$.
b) TRUE. $|z-2| = |z+2i| = |z-(-2i)|$ says that $P(z)$ is equidistant from $A(2,0)$ and $B(0,-2)$. The locus of points equidistant from two fixed points is the **perpendicular bisector** of $[AB]$, which is a straight line. Algebraically, with $z = x+iy$: $(x-2)^2+y^2 = x^2+(y+2)^2$ gives $-4x = 4y$, i.e. $y = -x$.
c) TRUE. The line $\mathrm{Re}(z) = 2$ does not pass through the origin, and inversion $w = 1/z$ maps any line not through O to a circle through O. Explicitly (see Exercise 8), the image is the circle $\left|w - \frac{1}{4}\right| = \frac{1}{4}$.
d) TRUE. $z' = az + b$ with $|a| = 1$, $a \neq 1$, has a unique fixed point $c = \frac{b}{1-a}$, and writing $z' - c = a(z-c)$ shows it is the rotation of centre c and angle $\arg(a)$. (If $a = 1$ it would be a translation.)

SOLUTION **Exercise 3 — Definitions and direct calculations**

a) $z = -2 + 2i$: $|z| = \sqrt{4+4} = 2\sqrt{2}$. Then $z = 2\sqrt{2}\left(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)$, so $\cos\theta = -\frac{\sqrt{2}}{2}$, $\sin\theta = \frac{\sqrt{2}}{2}$, giving

$\theta = \dfrac{3\pi}{4}$ (in $(-\pi, \pi]$).

$$z = 2\sqrt{2}\left(\cos\tfrac{3\pi}{4} + i\sin\tfrac{3\pi}{4}\right) = 2\sqrt{2}\,e^{3i\pi/4}.$$

b) $1 - i = \sqrt{2}\,e^{-i\pi/4}$. By De Moivre:

$$(1-i)^8 = (\sqrt{2})^8 e^{-8i\pi/4} = 16\,e^{-2i\pi} = 16.$$

c) $-8i = 8e^{-i\pi/2} = 8e^{i(-\pi/2+2k\pi)}$. The cube roots are

$$z_k = 2\,e^{i(-\pi/6+2k\pi/3)}, \quad k = 0, 1, 2,$$

i.e. $z_0 = 2e^{-i\pi/6}$, $z_1 = 2e^{i\pi/2} = 2i$, $z_2 = 2e^{7i\pi/6} = 2e^{-5i\pi/6}$.

SOLUTION **Exercise 4 — Quick computation**

a) $z_1 z_2 = 3 \cdot 2\,e^{i(\pi/4-\pi/6)} = 6e^{i\pi/12}$. $\dfrac{z_1}{z_2} = \dfrac{3}{2}e^{i(\pi/4+\pi/6)} = \dfrac{3}{2}e^{5i\pi/12}$.

b) $z_1\overline{z_2} = 3e^{i\pi/4} \cdot 2e^{i\pi/6} = 6e^{5i\pi/12}$. Now $\dfrac{5\pi}{12} = \dfrac{\pi}{4} + \dfrac{\pi}{6}$, so

$$\cos\tfrac{5\pi}{12} = \cos\tfrac{\pi}{4}\cos\tfrac{\pi}{6} - \sin\tfrac{\pi}{4}\sin\tfrac{\pi}{6} = \tfrac{\sqrt{2}}{2}\cdot\tfrac{\sqrt{3}}{2} - \tfrac{\sqrt{2}}{2}\cdot\tfrac{1}{2} = \tfrac{\sqrt{6}-\sqrt{2}}{4},$$

$$\sin\tfrac{5\pi}{12} = \sin\tfrac{\pi}{4}\cos\tfrac{\pi}{6} + \cos\tfrac{\pi}{4}\sin\tfrac{\pi}{6} = \tfrac{\sqrt{6}+\sqrt{2}}{4}.$$

Therefore

$$\operatorname{Re}(z_1\overline{z_2}) = 6 \cdot \tfrac{\sqrt{6}-\sqrt{2}}{4} = \tfrac{3(\sqrt{6}-\sqrt{2})}{2}, \qquad \operatorname{Im}(z_1\overline{z_2}) = \tfrac{3(\sqrt{6}+\sqrt{2})}{2}.$$

SOLUTION **Exercise 5 — Trigonometric and exponential forms**

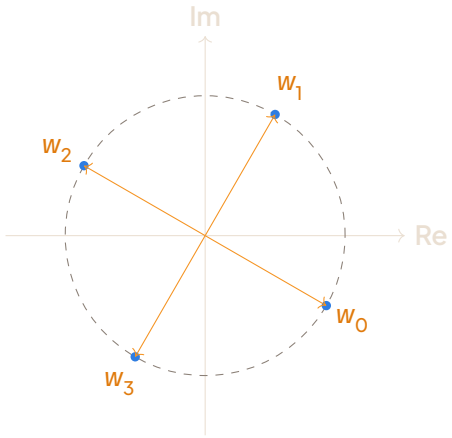
a) From Exercise 1: $z = 1 + i\sqrt{3} = 2\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right) = 2e^{i\pi/3}$.

b) By De Moivre: $z^{12} = 2^{12}e^{12i\pi/3} = 4096\,e^{4i\pi} = 4096$.

c) $-4z = -4(1 + i\sqrt{3})$. Modulus: $|-4z| = 4|z| = 8$. Argument: $-4z$ is z rotated by π (multiplication by the negative real -4), so $\arg(-4z) = \dfrac{\pi}{3} + \pi = \dfrac{4\pi}{3} \equiv -\dfrac{2\pi}{3}$. Hence $-4z = 8e^{-2i\pi/3}$.

The fourth roots are $w_k = 8^{1/4}e^{i(-2\pi/3+2k\pi)/4} = 2^{3/4}e^{i(-\pi/6+k\pi/2)}$, $k = 0, 1, 2, 3$:

$$w_0 = 2^{3/4}e^{-i\pi/6}, \quad w_1 = 2^{3/4}e^{i\pi/3}, \quad w_2 = 2^{3/4}e^{5i\pi/6}, \quad w_3 = 2^{3/4}e^{4i\pi/3} = 2^{3/4}e^{-2i\pi/3}.$$



The four fourth roots of $-4z$: vertices of a square on the circle of radius $2^{3/4}$.

The roots form a square inscribed in the circle of radius $2^{3/4} \approx 1.68$ centred at O, successive roots being separated by an angle $\dfrac{\pi}{2}$.

SOLUTION **Exercise 6 — De Moivre and $\cos 5\theta$**

a) By De Moivre, $\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$. Writing $c = \cos\theta$, $s = \sin\theta$ and expanding by the binomial theorem:

$$(c + is)^5 = c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5.$$

Equating real parts: $\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$. Using $s^2 = 1 - c^2$:

$$\begin{aligned}\cos 5\theta &= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 \\ &= 16c^5 - 20c^3 + 5c.\end{aligned}$$

Hence $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. ■

b) Put $\theta = \dfrac{\pi}{10}$: then $5\theta = \dfrac{\pi}{2}$, so $\cos 5\theta = 0$. With $c = \cos\dfrac{\pi}{10} \neq 0$:

$$16c^5 - 20c^3 + 5c = 0 \implies 16c^4 - 20c^2 + 5 = 0.$$

This is quadratic in c^2: $c^2 = \dfrac{20 \pm \sqrt{400 - 320}}{32} = \dfrac{20 \pm \sqrt{80}}{32} = \dfrac{5 \pm \sqrt{5}}{8}$.

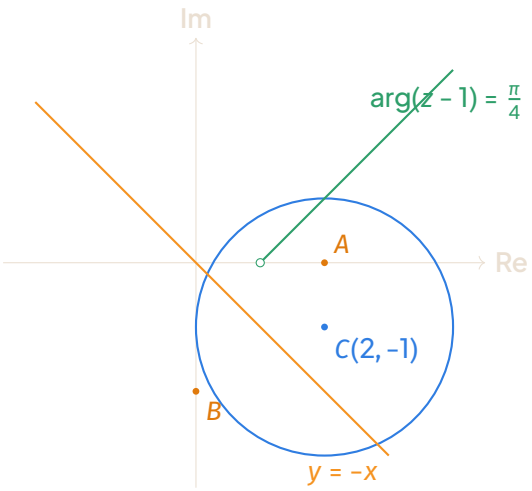
Since $\dfrac{\pi}{10} < \dfrac{\pi}{4}$, we have $\cos\dfrac{\pi}{10} > \cos\dfrac{\pi}{4} = \dfrac{1}{\sqrt{2}}$, so $c^2 > \dfrac{1}{2}$. Check: $\dfrac{5-\sqrt{5}}{8} \approx 0.345 < \dfrac{1}{2}$ and

$\frac{5+\sqrt{5}}{8} \approx 0.905 > \frac{1}{2}$. Hence $c^2 = \frac{5+\sqrt{5}}{8}$, and since $c > 0$:

$$\cos\frac{\pi}{10} = \sqrt{\frac{5+\sqrt{5}}{8}} = \frac{\sqrt{10+2\sqrt{5}}}{4}.$$

SOLUTION **Exercise 7 — Loci in the Argand diagram**

a) $|z-(2-i)| = 2$: circle of centre $C(2,-1)$ and radius 2.
b) $|z-2| = |z+2i|$: perpendicular bisector of $A(2,0)$ and $B(0,-2)$, i.e. the line $y = -x$.
c) $\arg(z-1) = \frac{\pi}{4}$: half-line (ray) from the point $(1,0)$ — excluded — making an angle $\frac{\pi}{4}$ with the positive real axis, i.e. the ray $y = x-1$, $x > 1$.



The three loci: circle, perpendicular bisector, and half-line (open point at $(1,0)$).

Simultaneous region. The three conditions are equalities, so the solution set is the set of points lying on all three loci at once. The ray $y = x-1$ meets the line $y = -x$ where $x-1 = -x$, i.e. $x = \frac{1}{2}$, which does not satisfy $x > 1$: the ray and the bisector do not intersect. Hence no point satisfies all three conditions simultaneously; the region is empty. (If the intended question was the region defined by the corresponding inequalities — inside the circle, the half-plane containing O bounded by $y = -x$, and the sector $\arg(z-1) \in [\frac{\pi}{4}, \frac{\pi}{2}]$ say — one shades the intersection of the disc, the half-plane and the angular sector.)

SOLUTION **Exercise 8 — Transformation $w = 1/z$**

Let $z = x + iy$ on the line $\mathrm{Re}(z) = 2$, so $z = 2 + iy$, and write $w = u + iv = \dfrac{1}{z}$, i.e. $z = \dfrac{1}{w}$.

Then $\mathrm{Re}(z) = \mathrm{Re}\left(\dfrac{1}{w}\right) = \mathrm{Re}\left(\dfrac{\overline{w}}{|w|^2}\right) = \dfrac{u}{u^2+v^2} = 2$.

Hence $u = 2(u^2+v^2)$, i.e. $u^2 + v^2 - \dfrac{u}{2} = 0$. Completing the square:

$$\left(u - \frac{1}{4}\right)^2 + v^2 = \frac{1}{16}, \qquad \text{i.e.} \qquad \left|w - \frac{1}{4}\right| = \frac{1}{4}.$$

The image is the circle of centre $\left(\frac{1}{4}, 0\right)$ and radius $\frac{1}{4}$, with the origin excluded (the point $w = 0$ is never reached since $|w| = 1/|z|$ is finite). The circle passes through O, as expected: a line not through O is mapped by inversion to a circle through O.

SOLUTION **Exercise 9 — Composite transformation and matrix (12 marks)**

a) The composite $T_2 \circ T_1$ (apply T_1 first, then T_2) is represented by

$$NM = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}.$$

[M1 A1]

b) M is the rotation about O through $+\dfrac{\pi}{2}$ (since $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $\theta = \frac{\pi}{2}$), and N is the enlargement (homothety) of centre O and scale factor 2. The composite is a rotation of $\dfrac{\pi}{2}$ about O combined with an enlargement of scale factor 2 — a spiral similarity of ratio 2 and angle $\dfrac{\pi}{2}$. [B1 B1]

c) The area scale factor is $|\det(NM)| = |0 \cdot 0 - (-2)(2)| = 4$. (Equivalently, enlargement factor 2 squares to 4; rotation preserves area.) [M1 A1]

d) The inverse undoes the enlargement then the rotation: scale by $\frac{1}{2}$ and rotate by $-\dfrac{\pi}{2}$. Its matrix:

$$(NM)^{-1} = \frac{1}{4}\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}.$$

Check: $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. [M1 A1 A1]

Examiner expectations: correct order of multiplication (NM, not MN) for the first mark; geometric description must mention both components; area factor must be the determinant, not the linear scale factor; the inverse matrix must be verified or derived from $\frac{1}{\det}$ adj. Common error: giving area factor 2 instead of 4.

SOLUTION **Exercise 10 — Similitude analysis (13 marks)**

Given $z' = (1-i)z + 3i$, a direct similitude $z' = az + b$ with $a = 1-i$, $b = 3i$.
a) Ratio: $k = |a| = |1-i| = \sqrt{2}$. Angle: $\theta = \arg(1-i) = -\frac{\pi}{4}$. [B1 B1]
b) The centre is the unique fixed point (since $a \neq 1$): solve $c = (1-i)c + 3i$:

$$c - (1-i)c = 3i \implies ic = 3i \implies c = 3.$$

Centre: the point $\Omega(3, 0)$. Check: $(1-i)\cdot 3 + 3i = 3 - 3i + 3i = 3$. [M1 A1]
c) Since $k = \sqrt{2} \neq 1$ and $a \notin \mathbb{R}$, this is a spiral similarity (similitude with rotation and enlargement): centre $\Omega(3, 0)$, ratio $\sqrt{2}$, angle $-\frac{\pi}{4}$. It is neither a pure rotation ($k \neq 1$), nor a homothety (a not real), nor a translation ($a \neq 1$). [B1 B1]
d) Images of the vertices:

$$\begin{aligned} 0 &\mapsto 3i, \\ 1 &\mapsto (1-i) + 3i = 1 + 2i, \\ i &\mapsto (1-i)i + 3i = i - i^2 + 3i = 1 + 4i. \end{aligned}$$

The image triangle has vertices $A'(0, 3)$, $B'(1, 2)$, $C'(1, 4)$. [M1 A1 A1]
Check of similarity: original side from 0 to 1 has length 1; image side from $3i$ to $1 + 2i$ has length $|1 - i| = \sqrt{2} = k \cdot 1$. ✓ [B1]
Examiner expectations: quote $k = |a|$ and $\theta = \arg(a)$ explicitly; the fixed-point equation is the standard route to the centre; classification must be justified using the values of k and θ; images computed with correct complex arithmetic. A frequent error is solving $c = ac + b$ as $c = \frac{b}{1-a}$ and mishandling the division — here $\frac{3i}{i} = 3$ directly.

SOLUTION **Exercise 11 — Modulus inequality and rigid motions (15 marks)**

a) Proof of the triangle inequality. For any $z_1, z_2 \in \mathbb{C}$, using $|w|^2 = w\overline{w}$:

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) = |z_1|^2 + z_1\overline{z_2} + \overline{z_1}z_2 + |z_2|^2 \\ &= |z_1|^2 + 2\,\mathrm{Re}(z_1\overline{z_2}) + |z_2|^2. \end{aligned}$$

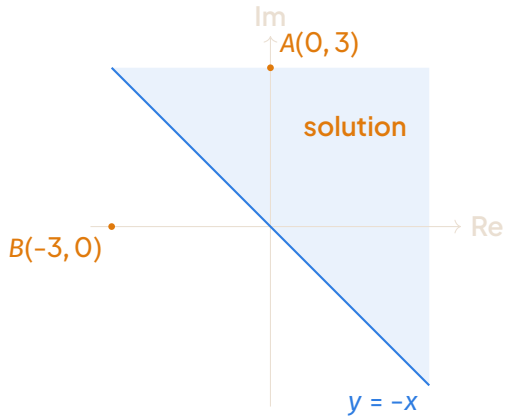
Now $\mathrm{Re}(z_1\overline{z_2}) \le |z_1\overline{z_2}| = |z_1||z_2|$, so

$$|z_1 + z_2|^2 \le |z_1|^2 + 2|z_1||z_2| + |z_2|^2 = (|z_1| + |z_2|)^2.$$

Both sides are non-negative, so taking square roots: $|z_1 + z_2| \le |z_1| + |z_2|$. ■
Equality holds iff $\mathrm{Re}(z_1\overline{z_2}) = |z_1\overline{z_2}|$, i.e. iff $z_1\overline{z_2}$ is a non-negative real, i.e. z_1 and z_2 have the same direction (one is a non-negative real multiple of the other). [M1 A1 A1 B1]
b) $|z - 3i| \le |z + 3|$: the set of points $P(z)$ at least as close to $A(0, 3)$ as to $B(-3, 0)$. The boundary is the perpendicular bisector of $[AB]$. With $z = x + iy$:

$$x^2 + (y-3)^2 \le (x+3)^2 + y^2 \implies -6y + 9 \le 6x + 9 \implies y \ge -x.$$

Solution set: the closed half-plane $\{(x, y) : y \geq -x\}$, bounded by the line $y = -x$ and containing $A(0, 3)$. [M1 A1 A1]



The half-plane $|z - 3i| \leq |z + 3|$, i.e. $y \geq -x$ (boundary included).

c) (i) $z' = iz + 2 - i$: direct isometry ($a = i$, $|a| = 1$, $a \neq 1$), hence a rotation. It preserves orientation. Fixed point: $c = ic + 2 - i \Rightarrow c(1 - i) = 2 - i \Rightarrow c = \dfrac{2 - i}{1 - i} = \dfrac{(2 - i)(1 + i)}{2} = \dfrac{3 + i}{2}$. Angle: $\arg(i) = \dfrac{\pi}{2}$. So: rotation of centre $\left(\frac{3}{2}, \frac{1}{2}\right)$ and angle $+\dfrac{\pi}{2}$. [B1 M1 A1]

(ii) $z' = i\bar{z} + 1$: indirect isometry (conjugate form $z' = a\bar{z} + b$, $|a| = 1$), hence an orientation-reversing rigid motion. Fixed points: $z = i\bar{z} + 1$. With $z = x + iy$: $x + iy = i(x - iy) + 1 = y + 1 + ix$, giving $x = y + 1$ and $y = x$ — impossible ($x = x + 1$). No fixed point, so it is a glide reflection (not a pure reflection).

Analysis of the glide reflection. The transformation is $z' = i\bar{z} + 1$.

- The reflection part $z \mapsto i\bar{z}$ is reflection in the line through the origin with angle $\frac{1}{2}\arg(i) = \frac{\pi}{4}$, i.e. the line $y = x$.
- The translation vector is $b = 1$ (real axis). Decompose b into components parallel and perpendicular to the axis $y = x$ (direction vector $(1, 1)$):

$$b_{\parallel} = \frac{1}{2}(1 + i) \quad (\text{vector } (\tfrac{1}{2}, \tfrac{1}{2})), \qquad b_{\perp} = \frac{1}{2}(1 - i) \quad (\text{vector } (\tfrac{1}{2}, -\tfrac{1}{2})).$$

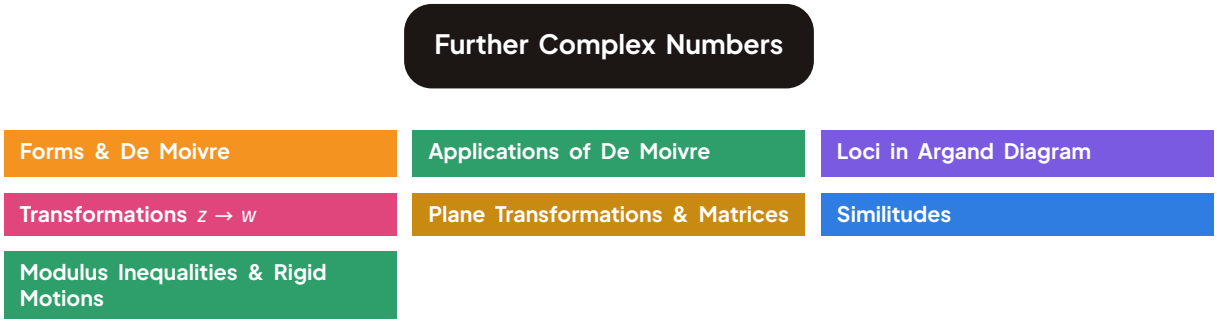
- The glide reflection axis is the line $y = x$ shifted by $\frac{1}{2}b_{\perp} = (\frac{1}{4}, -\frac{1}{4})$, giving the line $y = x - \frac{1}{2}$.
- The glide vector is $b_{\parallel} = (\frac{1}{2}, \frac{1}{2})$ (parallel to the axis, length $\frac{1}{\sqrt{2}}$).

Conclusion: $z' = i\bar{z} + 1$ is the glide reflection with axis $y = x - \frac{1}{2}$ and translation vector $(\frac{1}{2}, \frac{1}{2})$; it reverses orientation. [B1 M1 A1 A1]

Examiner expectations: in (a) the proof must use $|w|^2 = w\bar{w}$ and the bound $\mathrm{Re}(w) \leq |w|$; in (b) the boundary must be identified as a perpendicular bisector and the correct half-plane selected (test a point, e.g. A); in (c) the form $az + b$ vs $a\bar{z} + b$ decides orientation,

and the existence or non-existence of fixed points distinguishes rotation from translation, and reflection from glide reflection.

13 SUMMARY SHEET



Mind map of Chapter FMA02: Further Complex Numbers

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **Complex number**: $z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$.
- **Modulus**: $|z| = r = \sqrt{x^2 + y^2}$.
- **Argument**: $\arg z = \theta$ (principal value in $(-\pi, \pi]$).
- **De Moivre's theorem**: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for $n \in \mathbb{Z}$.
- **Locus**: set of points satisfying a condition on z.
- **Transformation** $w = f(z)$: mapping from z-plane to w-plane.
- **Similitude**: transformation $w = az + b$ or $w = a\bar{z} + b$ with $a \neq 0$.
- **Rigid motion**: isometry preserving distances; orientation preserving (rotation, translation) or reversing (reflection, glide reflection).

Key Formulas & Theorems

- Euler: $e^{i\theta} = \cos\theta + i\sin\theta$.
- $z^n = r^n e^{in\theta}$; $z^{1/n} = r^{1/n} e^{i(\theta+2k\pi)/n}$, $k = 0, \dots, n-1$.
- $\cos n\theta$, $\sin n\theta$ expressed via binomial expansion of $(\cos\theta + i\sin\theta)^n$.
- Loci: $|z - z_0| = r$ (circle), $|z - z_1| = |z - z_2|$ (perpendicular bisector), $\arg(z - z_0) = \alpha$ (half-line), $|z - z_1| + |z - z_2| = k$ (ellipse), $||z - z_1| - |z - z_2|| = k$ (hyperbola).
- Linear transformation: $w = \frac{az+b}{cz+d}$ (Möbius) but here elementary: translation $w = z + b$, rotation $w = e^{i\alpha}z$, enlargement $w = kz$, reflection $w = \bar{z}$, shear $w = z + \alpha\bar{z}$.

- Matrix representation: translation not linear; rotation/enlargement/reflection/shear via 2×2 matrices.
- Similitude matrix: $a + ib$ corresponds to $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ for direct; $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ for opposite.
- Modulus inequalities: $|z - z_1| < |z - z_2|$ defines half-plane; $|z - z_1| + |z - z_2| < k$ interior of ellipse.

Methods

- Convert forms: Cartesian ↔ Polar ↔ Exponential.
- De Moivre: expand $(\cos\theta + i\sin\theta)^n$, equate real/imaginary parts to get $\cos n\theta$, $\sin n\theta$; solve $z^n = w$ by taking nth roots.
- Loci: translate condition into Cartesian (x, y) or geometric description; sketch on Argand diagram.
- Transformations $z \to w$: find image of lines/circles; use $w = f(z)$ and substitute $z = x + iy$; identify type (line, circle, etc.).
- Matrix representation: write transformation as $\begin{pmatrix} x' \\ y' \end{pmatrix} = M \begin{pmatrix} x \\ y \end{pmatrix} + \mathbf{t}$; identify M and $\mathbf{t}$.
- Similitudes: express as $w = az + b$ (direct) or $w = a\bar{z} + b$ (opposite); find a, b from given conditions; matrix form.
- Modulus inequalities: square both sides carefully (preserve inequality direction); convert to Cartesian; identify region (half-plane, disc, ellipse, hyperbola).
- Rigid motions: check orientation by determinant of matrix (+1 preserving, −1 reversing); composition of reflections yields rotation/translation.

Common Mistakes

- Forgetting to add $2k\pi$ when finding arguments of roots.
- Confusing principal argument with general argument.
- Incorrectly squaring modulus inequalities without considering sign (e.g., $|z-1| < |z+1|$ leads to $x > 0$ after squaring).
- Misidentifying transformation type: $w = 1/z$ maps lines not through origin to circles through origin.
- Mixing up direct and opposite similitudes: direct uses z, opposite uses $\bar{z}$.
- Matrix representation: translation is not linear; must use homogeneous coordinates or separate vector.
- Orientation: determinant of 2×2 matrix > 0 preserves, < 0 reverses.

GCE Tips

- Always show the conversion between forms explicitly; examiners award marks for correct modulus/argument.
- For De Moivre expansions, write binomial expansion and separate real/imaginary parts clearly.
- When sketching loci, indicate centre, radius, asymptotes, and shade region for inequalities.
- For $w = f(z)$, find image of a generic point $z = x + iy$; eliminate x, y to get equation in u, v where $w = u + iv$.
- In matrix questions, write the matrix in the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ for rotation+enlargement; for reflection use $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.
- Similitude: if $w = az + b$, then $a = |a|e^{i\alpha}$ gives rotation by α and enlargement by $|a|$; b gives translation.
- Modulus inequalities: square only when both sides are non-negative (always true for moduli); then simplify to Cartesian.
- Rigid motions: composition of two reflections in intersecting lines gives rotation; in parallel lines gives translation.
- Time management: allocate 12 minutes per 10-mark question; practice past papers.

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